

Exponential-Integral Periods across Arithmetic, Padé, Hankel, Reciprocal Geometry, and Frobenius Framing

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Abstract

Write $\text{Ein}(z) = \int_0^z (1 - e^{-t}) dt/t$ and $\delta = eE_1(1)$. We first repair and globalize the multipoint relation theorem

$$\text{trdeg}_{\overline{\mathbb{Q}}}(e^\lambda, \text{Ein}(\lambda) : \lambda \in \Lambda) = \dim_{\mathbb{Q}} \langle \Lambda \rangle_{\mathbb{Q}} + |\Lambda|.$$

It follows that all nonzero algebraic-point Ein-values are jointly algebraically independent over the field generated by every algebraic exponential. Linear projection then determines the exact graphic-matroid relation ideal of definite Ein-periods and yields algebraic independence of the countable family $\{\text{Si}(n), \text{Cin}(n), \text{Shi}(n), \text{Cinh}(n) : n \geq 1\}$, together with the complete first-jet ideal of these four functions. A generic-point transfer principle also gives algebraically independent Hankel determinants and formal Favard parameters, exact Plücker relations, simple algebraically independent symmetric spectra, and realizations of every $\overline{\mathbb{Q}}$ -representable matroid by explicit Ein-periods. Finite differences of $\text{Ein}(b + am)$ yield explicit positive Hausdorff measures whose infinite Hankel matrices are strictly totally positive. Their moments, principal Hankel determinants, Favard coordinates, and every finite Jacobi spectrum are algebraically independent; the associated self-adjoint Jacobi operators have simple purely absolutely continuous spectrum, exact Nevai limits, capacity asymptotics, and arcsine zero distribution.

This finite theorem extends through the classical one-level polyexponential tower

$$\text{Ein}_r(z) = \sum_{n \geq 1} \frac{(-1)^{n-1} z^n}{n^r n!},$$

$$\text{trdeg}_{\overline{\mathbb{Q}}}(\overline{\mathbb{Q}}(e^{\lambda_i}, \text{Ein}_r(\lambda_i) : 1 \leq i \leq s, 1 \leq r \leq R) = d + Rs.$$

Thus all nonzero algebraic-point values in every order are countably algebraically independent over the pure exponential field. A positive Hausdorff–Jacobi realization exists at every order, with its arithmetic coordinates jointly algebraically independent across orders; for a fixed order the spectra of two consecutive truncations form a sharp transcendence basis of size $2N - 1$ for the full finite spectral tower. Splitting Euler’s Gamma integral at one further gives an all-order Gamma-jet/tail decomposition whose resulting combinations are countably algebraically independent.

On the divergent side we prove, for every prime and every nonzero integral parameter, an exact two-branch p -adic radius formula for the classical Euler–Padé remainder, an equivalent local denominator law, a nonzero p -curvature obstruction, and a fixed logarithmic Borel singularity. Geronimus–Markov reduction evaluates an entire Laguerre–Hankel tower as a Kummer U -function for all sizes, while a confluent reciprocal-Gamma determinant becomes a monomer–dimer polynomial whose complex zeros lie in an explicit vertical strip. Bell–Gould moment functionals admit monic common annihilators modulo every prime power and, by coefficientwise CRT, modulo every integer; the $N!$ -annihilator has degree $3N + O(\log N)$. A scalar one-pole Borel–Laguerre lift has exact least

denominator $(M-1)!$. A corrected compact double integral has size $\sim \pi 16^{-n}/(12n \log 2)$, but its integer form has a strictly positive logarithmic coefficient at every prime in $[n+1, 2n+1]$. Positive mixed Hankel determinants $\mathcal{P}_d(\gamma, \delta)$ have full degree, and their first two values recover $\mathbb{Q}(\gamma, \delta)$ through an extension of degree at most four; in particular $\mathcal{P}_1 + i\mathcal{P}_2$ is transcendental.

A common-shift theorem shows that translating an independent family by γ loses at most one transcendence degree: dependence in an explicit Euler residual family forces γ, δ to be algebraically independent, while all anchored residual differences are unconditionally algebraically independent. We also prove pairwise algebraicity exclusion among $E_1, \text{Ei}, \text{Ci}, \text{Chi}$ at every positive algebraic level and construct a positive quadratically sampled E_1 -family with exact e^{-m^2}/m^2 decay and countable algebraic independence. Finite algebraic-point Ein-representations are rigid: the one-variable Euler level has at most one algebraic zero, and the rank-one witness family $n \text{Ein}(z) - \text{Ein}(z^n) - (n-1)\gamma$ has an exact real-zero classification and at most one algebraic zero. The first two nontrivial inverse traces of the divisor of $\text{Ein} - \gamma$ recover γ by a rational formula.

Finally the reciprocal sum descends through the Joukowski map to an entire function H with an exact product for H' . Its values at distinct algebraic points are algebraically independent over their exponential field; hence every preimage of an algebraic level is transcendental. At the Euler level, algebraicity of the unique real reciprocal orbit would force γ and δ to be algebraically independent and would produce an explicit transcendental element of the intersection of the E -value and arithmetic-Gevrey value rings. A positive Bessel–Laplace representation places every zero of H in a strict left half-plane and every reciprocal zero in the open left half-plane. A level-zero de Branges–Clark construction produces algebraically independent scattering values at algebraic sample points and recovers γ from their asymptotic phase. We further obtain an exact Hadamard trace at the Euler level, recover γ nontrivially from the asymptotic phase of the genuinely γ -free reciprocal zero set, and prove that the full primitive clearing polynomial for the symmetric Laurent truncations saturates its defining-height budget: the factorial approximation gain is exactly cancelled by denominator clearing, while the level first appears in the full top norm. On the dyadic subsequence the clearing polynomial is irreducible by a 2-Eisenstein argument, its reciprocal lift is irreducible of degree $2N$, and the same critical exponent persists for the actual minimal Weil height. Rational polynomial capacity pullbacks and unimodular full-determinant normalizations are shown to remain exactly critical. Finally, the rank-two scalar semilinear Frobenius model with $|q| \neq 1$ is proved to split locally, so bare Frobenius data cannot by itself detect the global extension class. The remaining motivic gap is isolated instead as a depth-one numerical-to-motivic comparison/fullness principle, sufficient but presently unproved for the transcendence of γ . These are structural closures and conditional detectors. Nothing here proves the irrationality or transcendence of γ or δ , and no named open problem is claimed solved.

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1 Introduction and scope

Euler’s constant and the Euler–Gompertz constant are

$$\gamma = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{1}{k} - \log n \right), \quad \delta = \int_0^\infty \frac{e^{-t}}{1+t} dt = eE_1(1).$$

Neither number is known to be irrational; see Lagarias’ survey for context [34]. The elementary connection formula

$$\text{Ein}(1) = \gamma + E_1(1) = \gamma + \frac{\delta}{e} \tag{1}$$

places this problem on the interface between Siegel E -functions and arithmetic Gevrey series of order one. Throughout, the pure exponential field is

$$\mathcal{E}_{\text{exp}} := \overline{\mathbb{Q}}(e^\alpha : \alpha \in \overline{\mathbb{Q}}).$$

The purpose of this note is to make several rigorous statements on that interface and, equally importantly, to mark the point at which they stop.

The first result is a multipoint algebraic-independence theorem. Its proof is an application of Kolchin–Ostrowski followed by Beukers’ refined Siegel–Shidlovskii theorem, but a denominator step in the differential field is essential: arbitrary rational functions cannot simply be expanded as finite Laurent sums. We supply the missing Darboux-polynomial argument.

Theorem 1.1 (Multipoint Ein theorem). *For distinct $\lambda_1, \dots, \lambda_s \in \overline{\mathbb{Q}}^\times$, put $d = \dim_{\mathbb{Q}} \langle \lambda_1, \dots, \lambda_s \rangle_{\mathbb{Q}}$. Then*

$$\text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(e^{\lambda_i}, \text{Ein}(\lambda_i) : 1 \leq i \leq s) = d + s.$$

Moreover the $\text{Ein}(\lambda_i)$ are algebraically independent over $\overline{\mathbb{Q}}(e^{\lambda_1}, \dots, e^{\lambda_s})$, and every algebraic relation among the displayed numbers is generated by the multiplicative, or toric, relations among the exponential coordinates.

The most concrete consequence for the classical constants is the following.

Theorem 1.2 (Six constants). *Among*

$$\gamma, \quad \delta, \quad E_1(1), \quad \text{Ei}(1), \quad \text{Ci}(1), \quad \text{Chi}(1)$$

at most one number is algebraic. Consequently at least five are transcendental.

This is a density statement, not an identification statement: it does not say which five numbers are transcendental. In particular it does not settle γ or δ individually.

The same multipoint theorem produces a genuinely positive spectral model, not merely the formal Favard system obtained from the raw moments $\text{Ein}(m+1)$.

Theorem 1.3 (Positive Hausdorff–Jacobi realization). *For algebraic $a, b > 0$ and $r \geq 1$, put*

$$\mu_m = (-1)^{r-1} \mathbf{D}^r \text{Ein}(b + am).$$

Then

$$\mu_m = \int_{e^{-a}}^1 x^m \frac{x^{b/a-1} (1-x)^r}{-\log x} dx.$$

The infinite Hankel matrix $(\mu_{i+j})_{i,j \geq 0}$ is strictly totally positive. The moments, the principal Hankel determinants, and the complete Favard coordinate family are each algebraically independent over $\mathcal{E}_{\text{exp}}$. The associated Jacobi operator has simple purely absolutely continuous spectrum $[e^{-a}, 1]$; every finite principal spectrum consists of jointly algebraically independent real numbers, and the eigenvalue counting measures converge to the arcsine law on that interval.

Theorem 1.4 (Resonant polyexponential closure). *For $q \geq 1$, put*

$$\text{Ein}_q(z) = \sum_{n \geq 1} \frac{(-1)^{n-1} z^n}{n^q n!}.$$

If $\lambda_1, \dots, \lambda_s \in \overline{\mathbb{Q}}^\times$ are distinct and $d = \dim_{\mathbb{Q}} \langle \lambda_1, \dots, \lambda_s \rangle_{\mathbb{Q}}$, then, for every $R \geq 1$,

$$\text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(e^{\lambda_i}, \text{Ein}_q(\lambda_i) : 1 \leq i \leq s, 1 \leq q \leq R) = d + Rs.$$

All the $\text{Ein}_q(\lambda)$, jointly in $q \geq 1$ and $\lambda \in \overline{\mathbb{Q}}^\times$, are algebraically independent over $\mathcal{E}_{\text{exp}}$. Their positive finite-difference moment systems have the same Hausdorff–Jacobi conclusions as Theorem 1.3. For a fixed order, if Σ_j is the spectrum of the j -by- j Jacobi truncation, then

$$\text{trdeg}_{\mathcal{E}_{\text{exp}}} \mathcal{E}_{\text{exp}}(\Sigma_1, \dots, \Sigma_N) = 2N - 1;$$

the two consecutive spectra $\Sigma_{N-1} \cup \Sigma_N$ are jointly algebraically independent and form a sharp transcendence basis.

These are classical one-level polyexponentials, not newly defined special functions. The contribution here is the explicit resonant integral-shift proof, its multipoint toric relation ideal, and the arithmetic spectral applications; exact historical priority is treated conservatively below. Splitting the Euler Gamma integral at one also yields

$$g_k - T_k = (-1)^{k+1} \text{Ein}_{k+1}(1),$$

where $\Gamma(s) = s^{-1} + \sum_{k \geq 0} g_k s^k$ and $T_k = k!^{-1} \int_1^\infty e^{-u} (\log u)^k du / u$. Hence the entire family $\{g_k - T_k : k \geq 0\}$ is algebraically independent over $\mathcal{E}_{\text{exp}}$.

A second linear-algebra consequence is a precise defect-one principle. A common translation of r independent coordinates can lose at most one transcendence degree. Applied to explicit residual combinations of $E_1(2^N)$ and $\text{Ei}(n)$, it gives an unconditional countable family of algebraically independent anchored differences; any relation among the unanchored residuals would force γ and δ to be algebraically independent over $\mathcal{E}_{\text{exp}}$.

Our second theme is zero removal. An E -function is stable under division by $z - \alpha$ when it vanishes at a nonzero algebraic point α . The analogous assertion for arithmetic Gevrey series is conjectural in the form relevant to Euler’s factorial series. We prove an exact obstruction.

Theorem 1.5 (Denominator obstruction). *Let $0 \neq \xi \in \mathbb{Z}$, $\alpha \in \overline{\mathbb{Q}}$, and*

$$\mathcal{E}_\xi(z) = \sum_{n \geq 0} (-1)^n n! \xi^n z^n.$$

Write

$$\frac{\mathcal{E}_\xi(z) - \alpha}{z - 1} = \sum_{n \geq 0} n! b_n z^n.$$

If K is a number field containing α , and Δ_N is the least positive integer such that $\Delta_N b_0, \dots, \Delta_N b_N \in \mathcal{O}_K$, then

$$\lim_{N \rightarrow \infty} \frac{\log \Delta_N}{N} = +\infty.$$

Thus the quotient is not an arithmetic Gevrey series of order one and, in particular, does not belong to the arithmetic-Gevrey **D**-class.

For $\xi = 1$, the same Padé construction has a sharp local signature at every prime.

Theorem 1.6 (Padé radius transition). *Let*

$$q_n = \sum_{k=0}^n k! \binom{n}{k}^2,$$

let $s_n = P_n(1)$ be the numerator in the classical Padé approximation $Q_n(z)\mathcal{E}_1(z) - P_n(z) = R_n(z)$, and put

$$F_\alpha(z) = \sum_{n \geq 0} (q_n \alpha - s_n) \frac{z^n}{n!}.$$

For every prime p and $\alpha \in \mathbb{Q}_p$, with $\delta_p = \sum_{m \geq 0} (-1)^m m! \in \mathbb{Z}_p$,

$$R_p(F_\alpha) = \begin{cases} p^{1/(p-1)}, & \alpha = \delta_p, \\ p^{-1/(p-1)}, & \alpha \neq \delta_p. \end{cases}$$

Moreover $F_\alpha \in \mathbb{Z}_p[[z]]$ if and only if $\alpha = \delta_p$.

The local arithmetic can also be organized directly on the Bell and Gould moment functionals. If $M = \prod p^{r_p}$, there is a monic polynomial $C_M \in \mathbb{Z}[X]$, of degree

$$\max_{p|M} (m_{p,r_p} + pr_p), \quad m_{p,r} = \min\{m : v_p(m!) \geq r\},$$

which annihilates both functionals modulo M after multiplication by every polynomial in $\mathbb{Z}[X]$. For $M = N!$ this degree is $3N + O(\log N)$. This is an exact common-divisibility theorem; it does not assert that the resulting δ -linear forms are small. A paired logarithmic integral does, however, give an unconditional nonzero $\mathcal{E}_{\text{exp}}$ -transcendental mixed value with coefficient tending to infinity.

The third theme imports the capacity viewpoint. The relevant determinant is not merely small; it sits exactly on the critical constant.

Theorem 1.7 (Shifted harmonic Hankel determinant). *Let $H_0 = 0$, $H_n = \sum_{k=1}^n 1/k$, and fix $x \in \mathbb{C}$. Then*

$$\det(H_{i+j} - x)_{0 \leq i, j \leq n} = (-1)^n \frac{2H_n - x}{n! \prod_{j=1}^n \binom{2j}{j} \binom{2j-1}{j}}. \quad (2)$$

Consequently

$$\lim_{n \rightarrow \infty} |\det(H_{i+j} - x)_{0 \leq i, j \leq n}|^{1/n^2} = \frac{1}{4} = \text{cap}([0, 1]).$$

The shift by x , including $x = \gamma$, changes only the subquadratic part of the determinant asymptotic.

The mixed positive-moment construction gives a complementary positive result. It produces nonzero polynomials $\mathcal{P}_d(\gamma, \delta) > 0$ of exact total degree $2d$, with

$$[\gamma^d \delta^d] \mathcal{P}_d = (-1)^{d(d-1)/2} ((d-1)!)^d.$$

More unexpectedly, $\mathbb{Q}(\gamma, \delta)$ is algebraic of degree at most four over $\mathbb{Q}(\mathcal{P}_1, \mathcal{P}_2)$. Thus the first two positive determinants preserve the full transcendence degree of the original pair; at least one is transcendental over $\mathcal{E}_{\text{exp}}$, and $\mathcal{P}_1 + i\mathcal{P}_2$ is unconditionally transcendental.

The exact mixed family suggested by (1) is analysed in [section 23](#). The conclusion is deliberately negative but sharp: the proposed one-family zero-removal implication is equivalent to the transcendence of γ , rather than a smaller lemma from which that transcendence follows cheaply.

The reciprocal quotient supplies the final theme. The entire function H is characterized on $\mathbb{C}^\times$ by

$$H(z + z^{-1}) = \text{Ein}(z) + \text{Ein}(z^{-1})$$

has an explicit critical lattice and no algebraic value at an algebraic point. A positive Bessel–Laplace representation proves that the unique real point in every real fibre is its unique rightmost point. At level zero this puts all zeros of H , and all reciprocal zeros of $H(z + z^{-1})$, in the left half-plane and proves their simplicity and transcendence. Rotation and centering then give a Cartwright Hermite–Biehler function and a zero-mean-type meromorphic inner quotient. Its values at nonzero algebraic real points are algebraically independent over the pure exponential field, while their asymptotic phase recovers γ with an explicit first correction.

The divisor of $H - 2\gamma$ has an exact second Hadamard trace, whereas the upper exterior zeros of the genuinely γ -free level $H = 0$ recover γ through a second-order phase shift. The algebraic zeros of the Laurent truncations approximate a fixed reciprocal zero with exponent one relative to the full primitive clearing polynomial, but integerization makes those auxiliary polynomial values grow. On $N = 2^r$, a 2-Eisenstein argument and a norm-sign obstruction prove that the reciprocal clearing polynomial is itself irreducible of degree $2N$; hence exponent one persists for the actual minimal Weil height. Vieta’s formulas separately show that the level is invisible to every proper exterior power and first appears in the factorially weighted top norm.

Status and priority. The one-point exponential-integral phenomenon is already present in Mahler’s work [35]. The finite multipoint toric formulation here packages a direct use of classical Kolchin–Ostrowski and Siegel–Shidlovskii machinery [33, 5], with an explicit Laurent-denominator repair. The higher-order functions used here are Hardy’s classical one-level polyexponentials [46, 43]; Shidlovskii’s classical theorem covers the nearby nonintegral rational-shift family [49, Chap. 7, Sec. 3], and recent analytic context is given by Aminov–Arnaudo [42]. We record the regularized resonant integral-shift subsystem considered here, its multipoint toric closure, and its Jacobi consequences, without claiming priority for the underlying all-order phenomenon. The p -adic Padé formulas are due to Matala-aho and Zudilin [36]. The case $x = 0$ of (2) appears in Cigler [12]; the rank-one shift and the capacity interpretation are recorded here. The logarithmic E -value corollary in section 25 is an application of a recent theorem of Fischler and Rivoal [23]. Related recent independence results for generalized constants appear in Powers [41]. No claim of historical priority is made for the particular syntheses in this draft. Most importantly, nothing in this paper proves that γ or δ is irrational or transcendental.

2 Preliminaries

We use

$$\text{Ein}(z) = \sum_{n \geq 1} \frac{(-1)^{n-1} z^n}{n n!}, \quad \text{Ein}'(z) = \frac{1 - e^{-z}}{z}.$$

This is a strict E -function. We need two standard transfer principles. In contrast, $E_1(z) = \text{Ein}(z) - \gamma - \text{Log } z$ has a logarithmic singularity at the origin and is not an E -function. None of the effective E -value arguments below is applied directly to E_1 . First, the primitive form of the Kolchin–Ostrowski theorem [33] says that if a differential field K of characteristic zero has algebraically closed constant field C , the ambient differential extension has the same constant field, and $y'_i \in K$, then the primitives y_i are algebraically dependent over K only if $\sum c_i y'_i = Dg$ for some $c_i \in C$, not all zero, and $g \in K$. Second, Beukers’ refinement of the Siegel–Shidlovskii theorem [5] lifts every algebraic relation among values of a system of E -functions at a nonzero algebraic ordinary point to a functional algebraic relation.

For the divergent side, call a formal series

$$G(z) = \sum_{n \geq 0} n! a_n z^n, \quad a_n \in \overline{\mathbb{Q}},$$

an *arithmetic Gevrey series of order one* if it is holonomic and there is $C > 0$ such that, for every N , the conjugates of $a_0, \dots, a_N$ have size at most C^{N+1} and a common integer denominator at most C^{N+1} . This normalization is sufficient for every nonmembership statement below.

3 The Laurent denominator lemma

The key differential-field point is isolated here. Let $\nu_1, \dots, \nu_d \in \overline{\mathbb{Q}}$ be $\mathbb{Q}$ -linearly independent, put

$$R = \overline{\mathbb{Q}}(z)[X_1^{\pm 1}, \dots, X_d^{\pm 1}], \quad K = \text{Frac}(R),$$

and extend $D = d/dz$ by $DX_j = -\nu_j X_j$. Thus the specialization $X_j = e^{-\nu_j z}$ respects the derivation.

Lemma 3.1 (No nonmonomial Darboux denominator). *If an irreducible $P \in R$ satisfies $P \mid DP$, then P is a unit of the Laurent ring.*

Proof. Write $DP = AP$ with $A \in R$. If $A = 0$, Laurent-coefficient comparison shows that every nonzero coefficient $a_u(z)$ would satisfy $a'_u/a_u = u \cdot \nu$; rationality forces $u = 0$, and then a_0 is constant. Thus P is already a unit. We may therefore assume $A \neq 0$. The Newton polytope of DP is contained in that of P , while $\text{NP}(AP) = \text{NP}(A) + \text{NP}(P)$. Hence $\text{NP}(A) = \{0\}$, so $A \in \overline{\mathbb{Q}}(z)$. Write $P = \sum_{u \in S} a_u(z) X^u$. For $u, v \in S$, comparison of coefficients gives

$$\frac{d}{dz} \log \frac{a_u}{a_v} = (u - v) \cdot \nu.$$

A nonzero rational function cannot have logarithmic derivative equal to a nonzero constant: its logarithmic derivative tends to zero at infinity (and has only simple polar principal parts). Since the ν_j are $\mathbb{Q}$ -linearly independent, $(u - v) \cdot \nu = 0$ forces $u = v$. Thus S has one element and P is a Laurent monomial times an element of $\overline{\mathbb{Q}}(z)^\times$, hence a unit. $\square$

Lemma 3.2 (Denominator removal). *If $g \in K$ and $Dg \in R$, then $g \in R$.*

Proof. Write $g = A/B$ in lowest terms. If an irreducible P occurs in B , let $m = -\text{ord}_P(g) > 0$. Unless $P \mid DP$, differentiation lowers the P -valuation from $-m$ to $-m - 1$, contradicting $Dg \in R$. Thus $P \mid DP$, impossible by [lemma 3.1](#) because a denominator factor is not a unit. Hence B is a unit. $\square$

Lemma 3.3 (Exact-derivative obstruction). *Suppose $c_u \in \overline{\mathbb{Q}}$ are finitely supported and*

$$Dg = \frac{1}{z} \sum_{u \in \mathbb{Z}^d} c_u X^u, \quad g \in K.$$

Then every c_u is zero.

Proof. By [lemma 3.2](#), write $g = \sum a_u(z) X^u$ with $a_u \in \overline{\mathbb{Q}}(z)$. Coefficient comparison gives

$$a'_u(z) - (u \cdot \nu) a_u(z) = \frac{c_u}{z}. \tag{3}$$

If $u = 0$, a derivative of a rational function has zero residue at every finite pole, so (3) forces $c_0 = 0$. If $u \neq 0$, then $u \cdot \nu \neq 0$. A finite pole of a_u away from zero would produce an uncanceled pole of one order higher in a'_u . A pole at zero produces the same contradiction, whereas a function regular at zero makes the left-hand side regular there. Thus a nonzero right-hand side c_u/z is impossible. $\square$

The differential constants of K are $\overline{\mathbb{Q}}$. One way to see this is to apply the same denominator argument and then compare Laurent coefficients; the equation $a'_u - (u \cdot \nu)a_u = 0$ has no nonzero rational solution for $u \neq 0$, and $a'_0 = 0$.

4 Multipoint algebraic independence

Theorem 4.1. *Let $\Lambda = \{\lambda_1, \dots, \lambda_s\} \subset \overline{\mathbb{Q}}^\times$ consist of distinct elements, and put $d = \dim_{\mathbb{Q}} \langle \Lambda \rangle_{\mathbb{Q}}$. Then*

$$\text{trdeg}_{\overline{\mathbb{Q}}}(\overline{\mathbb{Q}}(e^{\lambda_i}, \text{Ein}(\lambda_i) : 1 \leq i \leq s)) = d + s.$$

If

$$L_\Lambda = \{m \in \mathbb{Z}^s : \sum_i m_i \lambda_i = 0\},$$

the kernel of

$$\overline{\mathbb{Q}}[U_1^{\pm 1}, \dots, U_s^{\pm 1}, V_1, \dots, V_s] \longrightarrow \mathbb{C}, \quad U_i \mapsto e^{\lambda_i}, \quad V_i \mapsto \text{Ein}(\lambda_i),$$

is generated by $U^{m_+} - U^{m_-}$, $m \in L_\Lambda$.

Proof. The additive group generated by Λ is free of rank d . Choose a $\mathbb{Z}$ -basis $\nu_1, \dots, \nu_d$, so that $\lambda_i = m_i \cdot \nu$ with $m_i \in \mathbb{Z}^d$. Equivalently, one may first choose a $\mathbb{Q}$ -basis and divide it by a common denominator; this lattice clearing is necessary for the Laurent-ring argument.

Set $X_j = e^{-\nu_j z}$ formally and let K be the differential field of the previous section. Distinct Laurent monomials X^u are linearly independent over $\overline{\mathbb{Q}}(z)$; after specialization this is the standard linear independence of exponentials with distinct exponents. Hence the exponential torus has transcendence degree d .

Put $Y_i = \text{Ein}(\lambda_i z)$. Then

$$DY_i = \frac{1 - X^{m_i}}{z} \in K.$$

Under $X_j = e^{-\nu_j z}$, the field generated by K and the Y_i lies in $\overline{\mathbb{Q}}((z))$, whose differential constants are exactly $\overline{\mathbb{Q}}$. Thus the no-new-constants hypothesis in Kolchin–Ostrowski is satisfied. If the Y_i were algebraically dependent over K , the primitive Kolchin–Ostrowski theorem would give constants $c_i \in \overline{\mathbb{Q}}$, not all zero, and $g \in K$ such that

$$Dg = \sum_{i=1}^s c_i DY_i = \frac{\sum_i c_i - \sum_i c_i X^{m_i}}{z}.$$

Because the m_i are distinct, lemma 3.3 forces every $c_i = 0$, a contradiction. Thus the Y_i are algebraically independent over K , and the functional transcendence degree is $d + s$.

The same degree holds over $\mathbb{C}(z)$: after clearing denominators, a relation there would give a finite homogeneous linear system with $\overline{\mathbb{Q}}$ -valued Taylor coefficients, so scalar-extension invariance of rank would produce a relation over $\overline{\mathbb{Q}}(z)$.

For specialization, include in the differential system the constant function 1, the exponentials $e^{\pm \nu_j z}$ and $e^{-\lambda_i z}$, and all Y_i . The additional characters $e^{-\lambda_i z}$ ensure that the displayed first-order system is closed even when some λ_i is a nontrivial integral combination of the ν_j . They are E -functions, the system has coefficients in $\overline{\mathbb{Q}}(z)$, and its only finite pole is at

$z = 0$; hence $z = 1$ is ordinary. Beukers' refined Siegel–Shidlovskii theorem transfers the functional transcendence degree to the values at 1.

Finally, since the V_i remain algebraically independent over the exponential field, the full relation ideal is the relation ideal of the exponential torus. Its character lattice is exactly L_Λ , which gives the stated binomial generators. $\square$

Remark 4.2. The exclusion of 0 from Λ is substantive only for notation: $e^0 = 1$ and $\text{Ein}(0) = 0$ contribute fixed algebraic coordinates and should simply be deleted before applying the theorem.

5 Pure exponential base change and special-function periods

The toric relation theorem has a useful global form. Define the pure exponential field

$$\mathcal{E}_{\text{exp}} := \overline{\mathbb{Q}}(e^\alpha : \alpha \in \overline{\mathbb{Q}}) \subset \mathbb{C}. \quad (4)$$

Although this is an infinitely generated field, every coefficient of a polynomial relation belongs to a subfield generated by only finitely many algebraic exponentials. This elementary observation removes the toric coordinates altogether.

Theorem 5.1 (Pure exponential base change). *The family*

$$\{\text{Ein}(\lambda) : \lambda \in \overline{\mathbb{Q}}^\times\}$$

is algebraically independent over $\mathcal{E}_{\text{exp}}$, in the finite-subset sense. Equivalently, for any distinct $\lambda_1, \dots, \lambda_s \in \overline{\mathbb{Q}}^\times$,

$$\text{trdeg}_{\mathcal{E}_{\text{exp}}}(\text{Ein}(\lambda_1), \dots, \text{Ein}(\lambda_s)) = s.$$

Proof. Suppose that a polynomial relation exists. Its finitely many coefficients belong to $\overline{\mathbb{Q}}(e^\omega : \omega \in \Omega)$ for some finite $\Omega \subset \overline{\mathbb{Q}}$. Clear coefficient denominators and apply Theorem 4.1 to

$$\Sigma = \{\lambda_1, \dots, \lambda_s\} \cup (\Omega \setminus \{0\}).$$

That theorem makes all $\text{Ein}(\sigma)$, $\sigma \in \Sigma$, algebraically independent over $\overline{\mathbb{Q}}(e^\sigma : \sigma \in \Sigma)$. In particular the selected $\text{Ein}(\lambda_j)$ are independent over the coefficient field, a contradiction. $\square$

We shall also use the following elementary linear-image consequence. It is recorded because it identifies full relation ideals, rather than only their transcendence degrees.

Lemma 5.2 (Linear images of the Ein-coordinates). *Let $Y = (\text{Ein}(\lambda_1), \dots, \text{Ein}(\lambda_s))^t$, where the λ_j are distinct nonzero algebraic numbers, and let $B \in M_{r,s}(\overline{\mathbb{Q}})$. Put $W = BY$. The kernel of*

$$\mathcal{E}_{\text{exp}}[T_1, \dots, T_r] \longrightarrow \mathbb{C}, \quad T_j \longmapsto W_j,$$

is the linear ideal

$$\left(\sum_{j=1}^r c_j T_j : c \in \overline{\mathbb{Q}}^r, \ c^t B = 0 \right).$$

In particular, the W_j are algebraically independent over $\mathcal{E}_{\text{exp}}$ when B has full row rank.

Proof. By Theorem 5.1, the entries of Y are independent indeterminates over $\mathcal{E}_{\text{exp}}$. After invertible row and column operations over $\overline{\mathbb{Q}}$, the map $Y \mapsto BY$ is a coordinate projection of rank $\text{rank } B$. Its polynomial kernel is therefore exactly the displayed linear ideal. $\square$

5.1 A graphic matroid of definite periods

Let $G = (V, E)$ be a finite oriented graph, let $c(G)$ be its number of connected components, and attach pairwise distinct labels $\lambda_v \in \overline{\mathbb{Q}}^\times$ to its vertices. For an edge e , write $s(e)$ and $t(e)$ for its source and target and set

$$P_e := \int_{\lambda_{s(e)}}^{\lambda_{t(e)}} \frac{1 - e^{-z}}{z} dz = \text{Ein}(\lambda_{t(e)}) - \text{Ein}(\lambda_{s(e)}). \quad (5)$$

The integrand is entire after filling its removable singularity at zero, so the integral is path independent.

Let $\partial : \overline{\mathbb{Q}}^E \rightarrow \overline{\mathbb{Q}}^V$ be the oriented incidence map, $\partial e = t(e) - s(e)$. Its kernel is the cycle space of G .

Theorem 5.3 (Graphic-period relation ideal). *The kernel of*

$$\mathcal{E}_{\text{exp}}[X_e : e \in E] \longrightarrow \mathbb{C}, \quad X_e \longmapsto P_e,$$

is

$$\left(\sum_{e \in E} c_e X_e : c \in \ker \partial \right). \quad (6)$$

Thus every algebraic relation among the periods P_e is generated by the signed cycle sums. In particular,

$$\text{trdeg}_{\mathcal{E}_{\text{exp}}} \mathcal{E}_{\text{exp}}(P_e : e \in E) = |V| - c(G),$$

and the periods attached to any spanning forest are algebraically independent over $\mathcal{E}_{\text{exp}}$.

Proof. Put $Y_v = \text{Ein}(\lambda_v)$. In vector notation, $(P_e)_{e \in E} = \partial^t(Y_v)_{v \in V}$. Theorem 5.1 makes the Y_v algebraically independent over $\mathcal{E}_{\text{exp}}$, so Lemma 5.2 identifies the kernel with the linear relations among the rows of ∂^t , namely $\ker \partial$. The incidence matrix has rank $|V| - c(G)$, and its cycle space is spanned by signed circuit vectors. This proves every claim. $\square$

5.2 The fourth-root orbit

Define the four entire integral functions

$$\begin{aligned} \text{Si}(z) &= \int_0^z \frac{\sin t}{t} dt, & \text{Cin}(z) &= \int_0^z \frac{1 - \cos t}{t} dt, \\ \text{Shi}(z) &= \int_0^z \frac{\sinh t}{t} dt, & \text{Cinh}(z) &= \int_0^z \frac{\cosh t - 1}{t} dt. \end{aligned} \quad (7)$$

Here $\text{Cinh}(z) = \text{Chi}(z) - \gamma - \text{Log } z$ on a compatible logarithmic branch; the integral in (7) is its entire normalization. Termwise comparison gives

$$\begin{aligned} \text{Ein}(z) &= \text{Shi}(z) - \text{Cinh}(z), & \text{Ein}(-z) &= -\text{Shi}(z) - \text{Cinh}(z), \\ \text{Ein}(iz) &= \text{Cin}(z) + i \text{Si}(z), & \text{Ein}(-iz) &= \text{Cin}(z) - i \text{Si}(z). \end{aligned} \quad (8)$$

Theorem 5.4 (Fourth-root orbit). *For every $a \in \overline{\mathbb{Q}}^\times$, the six numbers*

$$e^a, \quad e^{ia}, \quad \text{Si}(a), \quad \text{Cin}(a), \quad \text{Shi}(a), \quad \text{Cinh}(a) \quad (9)$$

are algebraically independent over $\overline{\mathbb{Q}}$. More strongly, the last four numbers are algebraically independent over $\mathcal{E}_{\text{exp}}$.

Proof. The four arguments $a, -a, ia, -ia$ are distinct. Their $\mathbb{Q}$ -span has dimension two, with basis a, ia . Theorem 4.1 therefore gives transcendence degree six for the corresponding exponential and Ein-coordinates. The four-by-four transformation (8) is invertible over $\overline{\mathbb{Q}}$, proving the first assertion. The stronger assertion follows directly from Theorem 5.1 and the same invertible transformation. $\square$

Corollary 5.5 (A countable independent family). *The countable family*

$$\{\text{Si}(n), \text{Cin}(n), \text{Shi}(n), \text{Cinh}(n) : n \geq 1\} \quad (10)$$

is algebraically independent over $\mathcal{E}_{\text{exp}}$.

Proof. It suffices to treat a finite set of positive integers. The fourth-root orbits $\{n, -n, in, -in\}$ are pairwise disjoint as n varies. Apply Theorem 5.1 to their union and perform the invertible transformation (8) separately in every block. $\square$

Remark 5.6 (Relation to recent sine-integral results). Delaygue proved that, when $a_1^2, b_1^2, \dots, a_m^2, b_m^2$ are pairwise distinct algebraic numbers, the integrals $\int_{a_j}^{b_j} (\sin t/t) dt$ are linearly independent over $\overline{\mathbb{Q}}$; see [15, Proposition 4.4]. Theorem 5.1 and Lemma 5.2 upgrade this, under the same nonzero square-distinctness hypothesis, to algebraic independence over $\mathcal{E}_{\text{exp}}$, because

$$\int_a^b \frac{\sin t}{t} dt = \frac{\text{Ein}(ib) - \text{Ein}(-ib) - \text{Ein}(ia) + \text{Ein}(-ia)}{2i}.$$

This is a strict strengthening of the stated conclusion of Delaygue's proposition, but a separate priority search in the older Mahler–Shidlovskii literature is required before presenting the explicit upgrade as historically new.

5.3 The exact first-jet ideal

The preceding theorem also determines every algebraic relation involving the first derivatives of the four functions. This is stronger than a transcendence-degree count.

Theorem 5.7 (Exact first-jet ideal). *Fix $a \in \overline{\mathbb{Q}}^\times$, and abbreviate*

$$S = \text{Si}(a), \quad C = \text{Cin}(a), \quad H = \text{Shi}(a), \quad K = \text{Cinh}(a)$$

and

$$S_1 = \text{Si}'(a), \quad C_1 = \text{Cin}'(a), \quad H_1 = \text{Shi}'(a), \quad K_1 = \text{Cinh}'(a).$$

The kernel of the evaluation homomorphism

$$\overline{\mathbb{Q}}[S, C, H, K, S_1, C_1, H_1, K_1] \longrightarrow \mathbb{C}$$

at $(S, C, H, K, S_1, C_1, H_1, K_1)$ is generated by the two polynomials

$$(aS_1)^2 + (1 - aC_1)^2 - 1, \quad (11)$$

$$(1 + aK_1)^2 - (aH_1)^2 - 1. \quad (12)$$

In particular, no nonzero polynomial relation mixes the four zeroth-order values with the derivative coordinates.

Proof. Put $x = e^{ia}$ and $y = e^a$. Differentiating (7) gives

$$\begin{aligned} aS_1 &= \frac{x - x^{-1}}{2i}, & 1 - aC_1 &= \frac{x + x^{-1}}{2}, \\ aH_1 &= \frac{y - y^{-1}}{2}, & 1 + aK_1 &= \frac{y + y^{-1}}{2}. \end{aligned} \tag{13}$$

Conversely,

$$x = 1 - aC_1 + iaS_1, \quad y = 1 + aK_1 + aH_1,$$

and the conjugate signs in these formulas give x^{-1} and y^{-1} . Thus the derivative coordinate ring is exactly $\overline{\mathbb{Q}}[x^{\pm 1}, y^{\pm 1}]$. Its presentation in the four derivative variables has precisely the two conic relations (11)–(12); the inverse formulas show that there can be no further kernel. Finally, Theorem 5.4 makes S, C, H, K algebraically independent over the larger field $\mathcal{E}_{\text{exp}}$, hence over $\overline{\mathbb{Q}}(x, y)$. Adjoining them is therefore a polynomial extension of the derivative coordinate ring, and introduces no additional relations. $\square$

Remark 5.8 (All positive-order jets). Repeated differentiation of the four integrands in (7) shows that every derivative of positive order at a belongs to $\overline{\mathbb{Q}}(e^a, e^{ia})$. Conversely the first derivatives recover e^a, e^{ia} by (13). Hence Theorem 5.7 also says that the four values are algebraically independent over the field generated by all of their positive-order jets.

5.4 A quantitative consequence

The results above are qualitative. A recent general theorem supplies Liouville-type lower bounds, but the exponent is controlled by the differential transcendence degree, not merely by the number of displayed integral values.

Proposition 5.9 (Quantitative orbit bound). *Let $A = \{a_1, \dots, a_N\}$ be distinct positive real algebraic numbers, and put*

$$r = \dim_{\mathbb{Q}} \text{span}_{\mathbb{Q}}\{a_1, \dots, a_N\}, \quad \tau = 4N + 2r.$$

Let ξ_A be the $4N$ -tuple consisting of

$$\text{Si}(a_j), \text{Cin}(a_j), \text{Shi}(a_j), \text{Cinh}(a_j) \quad (1 \leq j \leq N).$$

There exist a constant $C_2 > 0$ and positive constants $C_1(D) > 0$, depending on A and on the degree bound D , such that every nonzero polynomial $P \in \mathbb{Z}[\mathbf{X}]$ satisfies

$$|P(\xi_A)| \geq C_1(\deg P) H(P)^{-C_2(\deg P)^\tau}. \tag{14}$$

In particular the alternative “ $P(\xi_A) = 0$ ” in the general measure theorem never occurs here.

Proof. Consider the $4N$ E-functions obtained from (7) by replacing their argument by $a_j z$, and evaluate them at $z = 1$. The functional proof of Theorem 4.1, followed by the blockwise transformation (8), shows that these functions are algebraically independent over the exponential differential field generated by $e^{\pm a_j z}$ and $e^{\pm ia_j z}$. Their derivatives recover that field, whose torus rank is $2r$. Consequently the differential field generated by the functions and all their derivatives has transcendence degree $4N + 2r = \tau$ over $\overline{\mathbb{Q}}(z)$.

Apply the E-function case of [1, Theorem 1.3] at the regular point $z = 1$. It gives the alternative between vanishing and the lower bound (14), with exponent $(\deg P)^\tau$. Theorem 5.1 rules out vanishing for every nonzero P . $\square$

Remark 5.10 (Scope of the quantitative import). Proposition 5.9 is an application of a general algebraic-independence measure, not a new measure theorem. The constants in (14) are those furnished by Adamczewski–Faverjon; their paper discusses effectivity separately. The same procedure applies to any fixed full-rank linear family supplied by Lemma 5.2, but its exponent must be computed from the associated differential field before a numerical value is stated. No such quantitative estimate bears on the arithmetic of γ or δ , which are connection constants outside the finite-point E-function field.

6 Generic-point transfer and concrete algebraic models

Theorem 5.1 does more than certify isolated special values. It provides a concrete transcendence basis on which any finite algebraic construction may be evaluated without acquiring accidental relations. We record this elementary transfer principle before applying it to Hankel determinants, formal orthogonal polynomials, Grassmannians, symmetric spectra, and representable matroids.

Throughout this section put $K = \mathcal{E}_{\text{exp}}$.

Proposition 6.1 (Generic-point transfer). *Let $\lambda_1, \dots, \lambda_s \in \overline{\mathbb{Q}}^\times$ be distinct and put $y_i = \text{Ein}(\lambda_i)$. For polynomials $F_1, \dots, F_r \in K[X_1, \dots, X_s]$, the two homomorphisms*

$$K[T_1, \dots, T_r] \longrightarrow \mathbb{C}, \quad T_j \longmapsto F_j(y_1, \dots, y_s),$$

and

$$K[T_1, \dots, T_r] \longrightarrow K[X_1, \dots, X_s], \quad T_j \longmapsto F_j(X_1, \dots, X_s),$$

have the same kernel. The analogous assertion holds for rational maps after localizing at denominators which do not vanish identically.

Proof. Theorem 5.1 says precisely that the evaluation map $K[X_1, \dots, X_s] \rightarrow \mathbb{C}$, $X_i \mapsto y_i$, is injective. Composing the second displayed homomorphism with this injection proves the assertion. Localization gives the rational version. $\square$

Thus the point $y = (y_1, \dots, y_s)$ is a generic point of affine s -space over K . The proposition is classical generic-point algebra; the arithmetic content is the explicit realization of the coordinates as finite-point E -values.

6.1 A Hankel–Favard transcendence basis

Put

$$\mu_m = \text{Ein}(m+1), \quad \Delta_n = \det(\mu_{i+j})_{0 \leq i, j \leq n}, \quad \Delta_{-1} = 1. \quad (15)$$

Theorem 6.2 (Hankel determinants and formal Jacobi parameters). *The countable family*

$$\Delta_0, \Delta_1, \Delta_2, \dots$$

is algebraically independent over K , in the finite-subset sense.

Moreover the functional $\mathcal{L} \in \mathbb{C}[x]^*$ defined by $\mathcal{L}(x^m) = \mu_m$ is quasi-definite. Let its monic formal orthogonal polynomials be normalized by

$$p_{n+1}(x) = (x - a_n)p_n(x) - b_n p_{n-1}(x), \quad p_{-1} = 0, \quad p_0 = 1. \quad (16)$$

Then

$$\mu_0, a_0, a_1, \dots, b_1, b_2, \dots$$

is algebraically independent over K .

Proof. Expansion of Δ_n at its lower-right entry gives

$$\Delta_n = \Delta_{n-1}\mu_{2n} + R_n(\mu_0, \dots, \mu_{2n-1}) \quad (17)$$

for a polynomial R_n . The coefficient Δ_{n-1} is a nonzero polynomial in the preceding independent moments. Hence Δ_n is transcendental over $K(\Delta_0, \dots, \Delta_{n-1})$, and induction proves the first assertion. In particular every Δ_n is nonzero, which is the quasi-definiteness criterion.

Finite Gram–Schmidt elimination expresses

$$(\mu_0, \dots, \mu_{2N+1}) \mapsto (\mu_0, a_0, \dots, a_N, b_1, \dots, b_N)$$

rationally. Conversely, the matrix identity $\mu_m = \mu_0(J^m)_{00}$ for the finite Jacobi truncation, equivalently the weighted Motzkin-path expansion, expresses the moments rationally (indeed polynomially after adjoining μ_0) in the displayed Favard parameters. The two lists both contain $2N+2$ entries and are birational. Since the moment list is algebraically independent, so is the parameter list. Letting N vary proves the countable statement. $\square$

The system in Theorem 6.2 is formal rather than positive-definite: for example $\Delta_1 < 0$. It therefore does not define a self-adjoint Jacobi operator. Also the Hankel and Favard families must not be joined into one independent family, since

$$b_n \Delta_{n-1}^2 = \Delta_n \Delta_{n-2}. \quad (18)$$

6.2 Plücker coordinates, spectra, and matroids

Corollary 6.3 (Explicit generic Plücker data). *Let $1 \leq r < n$, choose pairwise distinct nonzero algebraic numbers λ_{ij} , and form the $r \times n$ matrix $M = (\text{Ein}(\lambda_{ij}))$. If p_I denotes its maximal minors, then the complete relation ideal of the p_I over K is the classical Plücker ideal. Consequently*

$$\text{trdeg}_K K(p_I) = r(n-r) + 1.$$

The corresponding projective Plücker ratios have transcendence degree $r(n-r)$.

Proof. Apply Proposition 6.1 to the maximal-minor map on a generic $r \times n$ matrix. Its kernel is the Plücker ideal and its image is the affine cone over $\text{Gr}(r, n)$; see, for example, [8]. The dimension statements are the dimensions of that cone and its projectivization. $\square$

Theorem 6.4 (A simple real algebraically independent spectrum). *For $1 \leq i \leq j \leq n$, choose pairwise distinct positive algebraic λ_{ij} and define the real symmetric matrix*

$$A_{ij} = A_{ji} = \text{Ein}(\lambda_{ij}).$$

Then A has n distinct real eigenvalues $\rho_1, \dots, \rho_n$, and these eigenvalues are algebraically independent over K .

Proof. The characteristic coefficients of a generic symmetric matrix are algebraically independent: restriction to diagonal matrices gives the elementary symmetric polynomials in n independent variables. The discriminant is likewise a nonzero polynomial, as diagonal matrices with distinct entries show. Proposition 6.1 therefore makes the characteristic coefficients of A independent and its discriminant nonzero. Symmetry makes all roots real, while the nonzero discriminant makes them distinct. The splitting field is finite over the field generated by the n characteristic coefficients, so

$$\text{trdeg}_K K(\rho_1, \dots, \rho_n) = n.$$

As there are n eigenvalues, they are jointly algebraically independent. $\square$

Corollary 6.5 (Representable matroids as Ein-period matroids). *Every finite matroid representable over $\overline{\mathbb{Q}}$ occurs as the algebraic matroid over K of explicit algebraic linear combinations of Ein-values. More precisely, if a rank- m representation has columns $b_e = (b_{1e}, \dots, b_{me})^t$ and*

$$W_e = \sum_{i=1}^m b_{ie} \text{Ein}(\lambda_i)$$

for distinct $\lambda_i \in \overline{\mathbb{Q}}^\times$, then for every subset S of the ground set

$$\text{trdeg}_K K(W_e : e \in S) = \text{rank}(b_e : e \in S).$$

The complete relation ideal is generated by the linear forms $\sum_e c_e T_e$ with $\sum_e c_e b_e = 0$.

Proof. This is Lemma 5.2 applied to the representing matrix and to each column subfamily. $\square$

These corollaries do not introduce new Grassmannian, spectral, or matroid machinery: each ambient construction is classical. They give exact arithmetic realizations by special values and solve no named external open problem. The historical priority of these particular Ein realizations should be checked separately before publication.

7 Positive Ein moment systems and algebraically independent Jacobi spectra

The generic Hankel system built directly from the moments $\text{Ein}(m+1)$ is quasi-definite but not positive-definite. A finite difference removes this defect and produces a genuine Hausdorff moment problem without sacrificing any algebraic independence.

Fix $a, b \in \overline{\mathbb{Q}}_{>0}$ and an integer $r \geq 1$. For a sequence $f = (f_m)_{m \geq 0}$ write

$$(\text{D}f)_m = f_{m+1} - f_m,$$

and put

$$f_m = \text{Ein}(b + am), \quad \mu_m^{(a,b,r)} = (-1)^{r-1} (\text{D}^r f)_m \quad (m \geq 0). \quad (19)$$

When there is no danger of confusion, write simply μ_m .

Theorem 7.1 (Positive finite-difference Ein moment theorem). *Let $K = \mathcal{E}_{\text{exp}}$. The moments in (19) have the exact representation*

$$\mu_m = \int_0^1 e^{-(b+am)t} \frac{(1 - e^{-at})^r}{t} dt \quad (20)$$

$$= \int_{e^{-a}}^1 x^m w_{a,b,r}(x) dx, \quad w_{a,b,r}(x) = \frac{x^{b/a-1} (1-x)^r}{-\log x}. \quad (21)$$

In particular $(\mu_m)_{m \geq 0}$ is a strictly positive-definite Hausdorff moment sequence. Moreover:

1. *The countable family $\mu_0, \mu_1, \dots$ is algebraically independent over K , in the finite-subset sense. The infinite Hankel matrix $(\mu_{i+j})_{i,j \geq 0}$ is strictly totally positive: every one of its finite minors is positive.*

2. *With*

$$\mathcal{H}_n = \det(\mu_{i+j})_{0 \leq i,j \leq n}, \quad \mathcal{H}_{-1} = 1, \quad (22)$$

one has

$$\mathcal{H}_n = \frac{1}{(n+1)!} \int_{[e^{-a}, 1]^{n+1}} \prod_{0 \leq i < j \leq n} (x_i - x_j)^2 \prod_{j=0}^n w_{a,b,r}(x_j) dx_j > 0. \quad (23)$$

The family $\mathcal{H}_0, \mathcal{H}_1, \dots$ is algebraically independent over K , and its exact capacity scale is

$$\lim_{n \rightarrow \infty} \mathcal{H}_n^{1/n^2} = \text{cap}([e^{-a}, 1]) = \frac{1 - e^{-a}}{4}. \quad (24)$$

3. Let $p_{-1} = 0, p_0 = 1$ be the monic orthogonal polynomials for the functional $\mathcal{L}(x^m) = \mu_m$, and normalize the Favard recurrence by

$$p_{n+1}(x) = (x - \alpha_n)p_n(x) - \beta_n p_{n-1}(x). \quad (25)$$

Then $\beta_n > 0$ for every $n \geq 1$, and

$$\mu_0, \alpha_0, \alpha_1, \dots, \beta_1, \beta_2, \dots$$

is algebraically independent over K .

4. The Jacobi matrix

$$J = \begin{pmatrix} \alpha_0 & \sqrt{\beta_1} & & \\ \sqrt{\beta_1} & \alpha_1 & \sqrt{\beta_2} & \\ & \sqrt{\beta_2} & \alpha_2 & \ddots \\ & & \ddots & \ddots \end{pmatrix} \quad (26)$$

defines a bounded self-adjoint operator on $\ell^2(\mathbb{N}_0)$. It has simple purely absolutely continuous spectrum

$$\sigma(J) = [e^{-a}, 1]$$

of multiplicity one. Its monic recurrence coefficients lie in the Nevai class of this interval:

$$\alpha_n \rightarrow \frac{1 + e^{-a}}{2}, \quad \beta_n \rightarrow \frac{(1 - e^{-a})^2}{16}. \quad (27)$$

5. For every $N \geq 1$, the N eigenvalues of the leading $N \times N$ principal truncation J_N are distinct, belong to $(e^{-a}, 1)$, and are jointly algebraically independent over K . If they are denoted in increasing order by $\lambda_{N,1}, \dots, \lambda_{N,N}$, then their normalized counting measures satisfy

$$\frac{1}{N} \sum_{j=1}^N \delta_{\lambda_{N,j}} \xrightarrow[N \rightarrow \infty]{\text{weak}} \frac{\mathbf{1}_{(e^{-a}, 1)}(x)}{\pi \sqrt{(x - e^{-a})(1 - x)}} dx. \quad (28)$$

Proof. The standard integral

$$\text{Ein}(s) = \int_0^1 \frac{1 - e^{-st}}{t} dt \quad (s > 0)$$

shows, after taking the r th forward difference in m , that

$$(-1)^{r-1} (D^r f)_m = \int_0^1 e^{-(b+am)t} \frac{(1 - e^{-at})^r}{t} dt.$$

The substitution $x = e^{-at}$ gives (21). The density is positive on $(e^{-a}, 1)$. At $x = 1$ it is asymptotic to $x^{b/a-1}(1-x)^{r-1}$, while the other endpoint is strictly positive; hence the

density is integrable and has essential support $[e^{-a}, 1]$. The Gram determinant formula now gives (23); its integrand is positive off the diagonals, so every $\mathcal{H}_n$ is strictly positive.

More generally, choose strictly increasing nonnegative integer lists $i_1 < \dots < i_k$ and $j_1 < \dots < j_k$. Andreief's identity writes the corresponding Hankel minor as

$$\frac{1}{k!} \int_{[e^{-a}, 1]^k} \det(x_\ell^{i_p})_{p,\ell=1}^k \det(x_\ell^{j_p})_{p,\ell=1}^k \prod_{\ell=1}^k w_{a,b,r}(x_\ell) dx_\ell.$$

On the ordered simplex $e^{-a} < x_1 < \dots < x_k < 1$, each generalized Vandermonde determinant is a positive Vandermonde times a Schur polynomial with positive values. Their product is positive on a set of positive measure, proving strict total positivity.

For algebraic independence, let

$$y_m = \text{Ein}(b + am).$$

The points $b + am$ are distinct nonzero algebraic numbers, so the multipoint Ein theorem makes every finite subfamily of the y_m algebraically independent over K . For fixed M , the $M + 1$ linear forms $\mu_0, \dots, \mu_M$ in $y_0, \dots, y_{M+r}$ are linearly independent: the coefficient of y_{M+r} occurs only in μ_M , and descending induction finishes the argument. Extending these coefficient vectors to a basis gives an invertible linear change of variables. Thus $\mu_0, \dots, \mu_M$ are algebraically independent over K .

Expansion of the determinant at its lower-right entry gives

$$\mathcal{H}_n = \mathcal{H}_{n-1} \mu_{2n} + R_n(\mu_0, \dots, \mu_{2n-1}). \quad (29)$$

Induction in n , viewing a putative relation as a polynomial in μ_{2n} , proves the algebraic independence of the $\mathcal{H}_n$.

Strict positivity of all Gram determinants gives the monic Favard recurrence, with

$$\beta_n = \frac{\mathcal{H}_n \mathcal{H}_{n-2}}{\mathcal{H}_{n-1}^2} > 0. \quad (30)$$

For each $N \geq 0$, finite Gram–Schmidt elimination gives a rational map

$$(\mu_0, \dots, \mu_{2N+1}) \mapsto (\mu_0, \alpha_0, \dots, \alpha_N, \beta_1, \dots, \beta_N).$$

Conversely, the weighted Motzkin-path expansion, equivalently $\mu_m = \mu_0(J^m)_{00}$, recovers the moments through order $2N + 1$ polynomially from the displayed Favard variables. The two lists have $2N + 2$ entries and the transformations are inverse on the open set of nonzero Hankel determinants. They are therefore birational. Algebraic independence of the moments proves that of the Favard variables.

Normalize the measure in (21) by its mass μ_0 . Polynomials are dense in its L^2 space, and the orthonormal polynomial transform identifies J with multiplication by x on

$$L^2([e^{-a}, 1], \mu_0^{-1} w_{a,b,r}(x) dx).$$

This proves bounded self-adjointness. Since the density is positive almost everywhere on the whole interval, multiplication by x has purely absolutely continuous spectrum $[e^{-a}, 1]$. The constant function is cyclic, so the spectral multiplicity is one. The Denisov–Rakhmanov theorem [44], affinely rescaled from $[-2, 2]$ to $[e^{-a}, 1]$, gives

$$\alpha_n \rightarrow \frac{1 + e^{-a}}{2}, \quad \sqrt{\beta_n} \rightarrow \frac{1 - e^{-a}}{4},$$

which is exactly (27) in the monic normalization (25). If $h_n = \mathcal{L}(p_n^2)$, then $h_n/h_{n-1} = \beta_n$ and $\mathcal{H}_n = \prod_{j=0}^n h_j$. Consequently

$$\mathcal{H}_n = \mu_0^{n+1} \prod_{k=1}^n \beta_k^{n+1-k}.$$

Taking logarithms and using (27) proves (24).

It remains to prove the arithmetic assertion about finite truncations. The positivity of the off-diagonal entries makes every J_N an irreducible real symmetric tridiagonal matrix. Hence its eigenvalues are simple; the standard zero theorem for orthogonal polynomials places them in $(e^{-a}, 1)$. Write the characteristic polynomial as

$$\det(TI_N - J_N) = T^N + c_1 T^{N-1} + \cdots + c_N.$$

Each c_j is a polynomial over $\mathbb{Q}$ in $\alpha_0, \dots, \alpha_{N-1}, \beta_1, \dots, \beta_{N-1}$. These N polynomials are algebraically independent: after formally specializing all β_i to zero, they become the signed elementary symmetric polynomials in the independent diagonal variables $\alpha_0, \dots, \alpha_{N-1}$. The algebraic independence of the actual Favard variables makes their evaluation map injective, so the actual $c_1, \dots, c_N$ are algebraically independent over K .

If $\lambda_{N,1}, \dots, \lambda_{N,N}$ are the eigenvalues, then the c_j are their elementary symmetric functions and the splitting field is finite over $K(c_1, \dots, c_N)$. Consequently

$$\text{trdeg}_K K(\lambda_{N,1}, \dots, \lambda_{N,N}) = N.$$

There are exactly N eigenvalues, so they are jointly algebraically independent over K .

Finally, these eigenvalues are the zeros of p_N . Formula (27) gives their limiting distribution directly. Indeed, put

$$c = \frac{1 + e^{-a}}{2}, \quad d = \frac{1 - e^{-a}}{4}.$$

For every fixed integer $k \geq 0$, a closed-walk expansion of the trace, with the $O(k)$ boundary rows discarded, gives

$$\lim_{N \rightarrow \infty} \frac{1}{N} \text{tr}(J_N^k) = \frac{1}{2\pi} \int_0^{2\pi} (c + 2d \cos \theta)^k d\theta.$$

The matrices J_N are uniformly bounded, so polynomial approximation extends this convergence from moments to every continuous test function. The push-forward of $d\theta/(2\pi)$ under $\theta \mapsto c + 2d \cos \theta$ is precisely the probability measure on the right side of (28). This proves the weak convergence. Equivalently, it is the classical zero-counting theorem for a regular measure on a single interval. $\square$

Remark 7.2 (What is new and what is imported). The Hausdorff moment representation and the spectral consequences are classical once (21) is known. The arithmetic content is that the explicit moments, all Hankel determinants, all Favard coordinates, and every finite Jacobi spectrum acquire the stated algebraic independence from the multipoint Ein theorem. Historical priority for this exact arithmetic realization should be audited before publication.

8 Multipoint one-level polyexponentials of arbitrary weight

For $r \geq 1$, define the order- r complementary exponential integral by

$$\text{Ein}_r(z) := \sum_{n \geq 1} \frac{(-1)^{n-1} z^n}{n^r n!}. \quad (31)$$

We call r the *order* (or *weight*), not the level: these are classical one-level polyexponentials, not the multiple-index polyexponentials of level greater than one [46, 43, 42]. Thus $\text{Ein}_1 = \text{Ein}$, and

$$z \text{Ein}'_1(z) = 1 - e^{-z}, \quad z \text{Ein}'_r(z) = \text{Ein}_{r-1}(z) \quad (r \geq 2). \quad (32)$$

We use the harmless notation $\text{Ein}_0(z) = 1 - e^{-z}$ only when writing the chain compactly. For each fixed r , Ein_r is an E -function. Indeed, its coefficient sequence is hypergeometric, hence holonomic. Moreover, for every N , the single integer $\text{lcm}(1, \dots, N)^r$ clears simultaneously all coefficients $n! [z^n] \text{Ein}_r(z)$, $0 \leq n \leq N$; it grows at most exponentially in N . The coefficients are rational of absolute value at most one, so the conjugate-growth condition is automatic.

We first record the rational-denominator step needed in the iterated primitive argument. It is useful beyond the present family.

Lemma 8.1 (Affine form of a primitive rational function). *Let F be a differential field of characteristic zero whose constant field C is algebraically closed. Let $a_1, \dots, a_s \in F$ have the property*

$$\sum_{i=1}^s c_i a_i \in D(F), \quad c_i \in C \quad \implies \quad c_1 = \dots = c_s = 0. \quad (33)$$

Adjoin algebraically independent primitives $u_1, \dots, u_s$, with $Du_i = a_i$, and put $L = F(u_1, \dots, u_s)$. If $g \in L$ and $Dg \in F$, then

$$g = f + \sum_{i=1}^s b_i u_i \quad (f \in F, b_i \in C). \quad (34)$$

In particular, the constant field of L is again C .

Proof. Work first in $F[u_1, \dots, u_s]$. There is no nonconstant irreducible Darboux polynomial in this ring. Indeed, suppose that an irreducible nonconstant P divides DP . Degree in the u_i shows that $DP = hP$ for some $h \in F$. Fix a monomial order refining total degree and divide P by the coefficient of its leading monomial. For the resulting monic polynomial Q , one has $DQ = (h - Dp/p)Q$, where p was that leading coefficient. Since the derivation lowers the u -degree of the monomial part and the leading coefficient of Q is one, comparison of the leading monomial forces $h - Dp/p = 0$; hence $DQ = 0$. Write Q_n for its highest nonzero homogeneous part. The degree- n part of $DQ = 0$ shows that every coefficient of Q_n belongs to C . The degree- $(n-1)$ part is

$$DQ_{n-1} + \sum_{i=1}^s a_i \frac{\partial Q_n}{\partial u_i} = 0.$$

Since Q_n is nonconstant, at least one coefficient comparison in this identity writes a nonzero constant linear combination of the a_i as an element of $D(F)$, contrary to (33).

Now write $g = A/B$ in lowest terms. If a nonconstant irreducible factor P occurs in B , put $m = -v_P(g) > 0$. If $P \nmid DP$, the usual quotient-rule computation in the discrete valuation v_P gives $v_P(Dg) = -m - 1$: the term obtained by differentiating P^{-m} has that valuation and cannot cancel with terms of valuation at least $-m$. But every nonzero element of F has P -valuation zero. Thus $Dg \in F$ forces $P \mid DP$, which the preceding paragraph rules out. Hence g is a polynomial in the u_i .

Let g_n be its highest homogeneous part. Since $Dg \in F$, the degree- n coefficient comparison makes the coefficients of g_n constant. If $n \geq 2$, the degree- $(n-1)$ comparison again makes a nonzero constant linear combination of the a_i exact in F , a contradiction. Thus $n \leq 1$, and (34) follows. If $Dg = 0$, then $Df + \sum_i b_i a_i = 0$; (33) gives $b_i = 0$, and then $f \in C$. Thus L has no new constants. $\square$

We now apply this lemma to the exponential torus used in Lemma 3.3. Let $\lambda_1, \dots, \lambda_s \in \overline{\mathbb{Q}}^\times$ be distinct, let $\nu_1, \dots, \nu_d$ be a $\mathbb{Z}$ -basis of the additive group generated by the λ_i , and write $\lambda_i = m_i \cdot \nu$, with $m_i \in \mathbb{Z}^d$. Put

$$K = \text{Frac}(\overline{\mathbb{Q}}(z)[X_1^{\pm 1}, \dots, X_d^{\pm 1}]), \quad DX_j = -\nu_j X_j.$$

Because the ν_j form a basis of the group generated by the λ_i , each ν_j is also an integral combination of the λ_i . Thus passing from λ to ν changes neither the generated exponential field nor its character lattice. The vectors m_i are pairwise distinct and nonzero. The constant field of K is $\overline{\mathbb{Q}}$. Lemma 3.3 implies both

$$\frac{1}{z} \notin D(K) \quad (35)$$

and

$$\sum_{i=1}^s c_i \frac{1 - X^{m_i}}{z} \in D(K), \quad c_i \in \overline{\mathbb{Q}} \quad \implies \quad c_1 = \dots = c_s = 0. \quad (36)$$

Indeed, after collecting the distinct Laurent characters, each assertion is an immediate instance of that lemma.

All the functions below have Taylor coefficients in $\overline{\mathbb{Q}}$, and the field they generate at the origin embeds in $\overline{\mathbb{Q}}((z))$. Since the constant field of $\overline{\mathbb{Q}}((z))$ under d/dz is $\overline{\mathbb{Q}}$, the ambient extension used in every invocation of Kolchin–Ostrowski has no new constants. The affine lemma below also verifies this internally after each primitive layer is adjoined.

For $q \geq 1$, set

$$Y_{i,q}(z) = \text{Ein}_q(\lambda_i z), \quad K_q = K(Y_{i,j} : 1 \leq i \leq s, 1 \leq j \leq q).$$

Then

$$DY_{i,1} = \frac{1 - X^{m_i}}{z}, \quad DY_{i,q} = \frac{Y_{i,q-1}}{z} \quad (q \geq 2). \quad (37)$$

Lemma 8.2 (Nonexactness throughout the tower). *For every $q \geq 0$, with $K_0 = K$, the constant field of K_q is $\overline{\mathbb{Q}}$, and*

$$\frac{1}{z} \notin D(K_q). \quad (38)$$

Moreover, at each level the variables $Y_{1,q}, \dots, Y_{s,q}$ are algebraically independent over K_{q-1} .

Proof. The primitive form of the Kolchin–Ostrowski theorem [33] says that primitives with derivatives in a differential field with algebraically closed constants are algebraically dependent only if a nonzero constant linear combination of their derivatives is exact in the base field. Equation (36) therefore makes the first layer algebraically independent over K . Applying Lemma 8.1 also shows that K_1 has no new constants.

Suppose $Dg = 1/z$ with $g \in K_1$. Since $Dg \in K$, Lemma 8.1 gives $g = k + \sum_i b_i Y_{i,1}$, where $k \in K$ and $b_i \in \overline{\mathbb{Q}}$. Hence

$$Dk = \frac{1 - \sum_i b_i + \sum_i b_i X^{m_i}}{z},$$

contradicting Lemma 3.3. This proves (38) for $q = 1$.

Assume the assertions through level $q \geq 1$. The extension K_q/K_{q-1} admits all translations

$$\tau_h(Y_{i,q}) = Y_{i,q} + h_i, \quad h = (h_1, \dots, h_s) \in \overline{\mathbb{Q}}^s,$$

as differential automorphisms. If, for some $g \in K_q$,

$$Dg = \sum_{i=1}^s c_i \frac{Y_{i,q}}{z}$$

with $c \neq 0$, choose h with $c \cdot h \neq 0$. Then

$$D(\tau_h g - g) = \frac{c \cdot h}{z},$$

contradicting the inductive nonexactness of $1/z$ in K_q . Thus no nonzero constant linear combination of the derivatives of the next layer is exact. Kolchin–Ostrowski gives the algebraic independence of $Y_{1,q+1}, \dots, Y_{s,q+1}$ over K_q , and Lemma 8.1 shows that the constants remain $\overline{\mathbb{Q}}$.

Finally, suppose $Dg = 1/z$ with $g \in K_{q+1}$. The affine lemma gives

$$g = k + \sum_i b_i Y_{i,q+1}, \quad k \in K_q, \quad b_i \in \overline{\mathbb{Q}},$$

and therefore

$$Dk = \frac{1 - \sum_i b_i Y_{i,q}}{z}.$$

If $b \neq 0$, translate the q -th layer by an h with $b \cdot h \neq 0$; the difference makes a nonzero multiple of $1/z$ exact in K_q , a contradiction. If $b = 0$, the displayed identity directly contradicts the inductive hypothesis. This completes the induction. $\square$

For clarity, the first new case can also be checked without the induction. For one nonzero algebraic λ , put $Y = \text{Ein}(\lambda z)$ and $Z = \text{Ein}_2(\lambda z)$. Exactness in K first shows that Y is transcendental over K ; the affine lemma then gives constants $\overline{\mathbb{Q}}$ in $K(Y)$, and its displayed calculation proves $1/z \notin D(K(Y))$. If Z were algebraic over $K(Y)$, the one-primitive Kolchin–Ostrowski theorem would give $Dg = Y/z$ for some $g \in K(Y)$, after rescaling a nonzero constant. Because Y is transcendental, $\tau_h(Y) = Y + h$ is a differential automorphism of $K(Y)/K$. Applying it with $h \neq 0$ gives $D(\tau_h g - g) = h/z$, the required contradiction. Thus $e^{\lambda z}, Y, Z$ have functional transcendence degree three when $s = 1$, and the order-two claim is independent of any higher-order induction.

Theorem 8.3 (Multipoint resonant polyexponential theorem). *Let $\lambda_1, \dots, \lambda_s \in \overline{\mathbb{Q}}^\times$ be distinct, let*

$$d = \dim_{\mathbb{Q}} \text{span}_{\mathbb{Q}}\{\lambda_1, \dots, \lambda_s\},$$

and let $R \geq 1$. Then

$$\text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(e^{\lambda_i}, \text{Ein}_r(\lambda_i) : 1 \leq i \leq s, 1 \leq r \leq R) = d + Rs. \quad (39)$$

In particular, the Rs polyexponential values are algebraically independent over $\overline{\mathbb{Q}}(e^{\lambda_1}, \dots, e^{\lambda_s})$. All algebraic relations among the displayed exponential and polyexponential coordinates are therefore the toric relations among the exponential coordinates.

Proof. Lemma 8.2 gives functional transcendence degree $d + Rs$ over $\overline{\mathbb{Q}}(z)$: the exponential torus contributes d , and each of the R primitive layers contributes s .

The same functional transcendence degree holds over $\mathbb{C}(z)$. Indeed, all Taylor coefficients are algebraic; after clearing rational-function denominators, any finite relation over $\mathbb{C}(z)$ would be a finite linear dependence over $\mathbb{C}$ among series with coefficients in $\overline{\mathbb{Q}}$, and rank invariance under the scalar extension $\overline{\mathbb{Q}} \subset \mathbb{C}$ would produce a relation over $\overline{\mathbb{Q}}(z)$.

For specialization, include in one vector the constant function 1, the finitely many exponentials $e^{\pm\lambda_i z}$, and all $\text{Ein}_r(\lambda_i z)$, $1 \leq r \leq R$. The exponentials are E -functions. If $q_i \in \mathbb{Z}_{>0}$ clears the algebraic denominator of λ_i , then for every N the integer

$$q_i^N \text{lcm}(1, \dots, N)^r$$

clears simultaneously the first N normalized coefficients of $\text{Ein}_r(\lambda_i z)$; all of their conjugates have size at most C_i^N for a fixed C_i . Thus these scaled order functions satisfy the denominator and conjugate-growth axioms for E -functions. By (32), the displayed vector satisfies a first-order linear system over $\overline{\mathbb{Q}}(z)$, with diagonal constant entries for the exponentials and only $1/z$ entries in the order chains. Consequently its coefficient matrix has no finite pole except possibly $z = 0$, so the nonzero algebraic point $z = 1$ is ordinary. Beukers' refined Siegel–Shidlovskii theorem [5] transfers the functional transcendence degree, and all relations, to the values at $z = 1$. This proves (39); the final statement follows from the algebraic independence of all polyexponential coordinates over the exponential field. $\square$

Corollary 8.4 (Pure exponential base change in every order). *The countable family*

$$\{\text{Ein}_r(\lambda) : \lambda \in \overline{\mathbb{Q}}^\times, r \geq 1\}$$

is algebraically independent over $\mathcal{E}_{\text{exp}} = \overline{\mathbb{Q}}(e^\alpha : \alpha \in \overline{\mathbb{Q}})$, in the finite-subset sense.

Proof. Any proposed polynomial relation uses finitely many polyexponential values and finitely many coefficients from $\mathcal{E}_{\text{exp}}$. Enlarge the finite set of algebraic points to include the exponents occurring in those coefficients, take the largest order appearing, clear coefficient denominators, and apply Theorem 8.3. The selected order values remain algebraically independent over the enlarged exponential field. $\square$

Corollary 8.5 (Hausdorff moments and consecutive finite spectra). *Fix $a, b \in \overline{\mathbb{Q}} \cap \mathbb{R}_{>0}$ and an integer $\ell \geq 1$, and write $\Delta_a f(s) = f(s+a) - f(s)$. For $q \geq 1$ and $m \geq 0$, put*

$$\mu_m^{(q)} := (-1)^{\ell-1} (\Delta_a^\ell \text{Ein}_q)(b + am). \quad (40)$$

Then $(\mu_m^{(q)})_{m \geq 0}$ is a strict Hausdorff moment sequence on $[e^{-a}, 1]$, with

$$\mu_m^{(q)} = \int_{e^{-a}}^1 x^m w_q(x) dx, \quad w_q(x) = \frac{x^{b/a-1} (1-x)^\ell}{(q-1)! (-\log x)} \left(\log \frac{a}{-\log x} \right)^{q-1}. \quad (41)$$

The entire countable family

$$\{\mu_m^{(q)} : q \geq 1, m \geq 0\} \quad (42)$$

is algebraically independent over $\mathcal{E}_{\text{exp}}$.

For each q , let

$$\mathcal{H}_n^{(q)} = \det(\mu_{i+j}^{(q)})_{0 \leq i, j \leq n}, \quad \mathcal{H}_{-1}^{(q)} = 1.$$

All these Hankel determinants, jointly for $q \geq 1$ and $n \geq 0$, are algebraically independent over $\mathcal{E}_{\text{exp}}$, and they are strictly positive. Let the monic orthogonal polynomials for $w_q(x) dx$ satisfy

$$P_{n+1}^{(q)}(x) = (x - \alpha_n^{(q)}) P_n^{(q)}(x) - \beta_n^{(q)} P_{n-1}^{(q)}(x), \quad \beta_n^{(q)} > 0 \quad (n \geq 1). \quad (43)$$

Then the family

$$\{\mu_0^{(q)}, \alpha_n^{(q)}, \beta_{n+1}^{(q)} : q \geq 1, n \geq 0\} \quad (44)$$

is algebraically independent over $\mathcal{E}_{\text{exp}}$.

For every fixed q , the infinite Hankel matrix $(\mu_{i+j}^{(q)})_{i,j \geq 0}$ is strictly totally positive. The bounded Jacobi operator associated with (43) has simple, purely absolutely continuous spectrum $[e^{-a}, 1]$ of multiplicity one, and

$$\alpha_n^{(q)} \longrightarrow \frac{1 + e^{-a}}{2}, \quad \beta_n^{(q)} \longrightarrow \frac{(1 - e^{-a})^2}{16}. \quad (45)$$

Consequently

$$\lim_{n \rightarrow \infty} (\mathcal{H}_n^{(q)})^{1/n^2} = \frac{1 - e^{-a}}{4}, \quad (46)$$

and the normalized zero-counting measures of the monic orthogonal polynomials $P_N^{(q)}$ converge weakly to

$$\frac{\mathbf{1}_{(e^{-a}, 1)}(x)}{\pi \sqrt{(x - e^{-a})(1 - x)}} dx. \quad (47)$$

Finally, let $J_N^{(q)}$ be the N -by- N Jacobi truncation attached to (43). For every fixed q and $N \geq 2$, the $2N - 1$ eigenvalues of $J_N^{(q)}$ and $J_{N-1}^{(q)}$, taken together, are algebraically independent over $\mathcal{E}_{\text{exp}}$. The same joint assertion holds for finitely many distinct orders q , with one chosen consecutive pair at each order. Each spectrum is simple and the two consecutive spectra strictly interlace. The count $2N - 1$ is sharp: if $\Sigma_j^{(q)}$ denotes the full spectrum of $J_j^{(q)}$, then

$$\text{trdeg}_{\mathcal{E}_{\text{exp}}}(\Sigma_1^{(q)}, \dots, \Sigma_N^{(q)}) = 2N - 1. \quad (48)$$

Thus the consecutive pair $\Sigma_{N-1}^{(q)} \cup \Sigma_N^{(q)}$ is a transcendence basis for the full finite spectral tower up to algebraic extension.

Proof. The Mellin identity

$$\frac{1}{n^q} = \frac{1}{(q-1)!} \int_0^1 t^{n-1} (-\log t)^{q-1} dt$$

and termwise integration give, first near the origin and then everywhere by analyticity,

$$\text{Ein}_q(s) = \frac{1}{(q-1)!} \int_0^1 (1 - e^{-st}) (-\log t)^{q-1} \frac{dt}{t}. \quad (49)$$

Since

$$(-1)^{\ell-1} \Delta_a^\ell (1 - e^{-st}) = e^{-st} (1 - e^{-at})^\ell,$$

substitution of $s = b + am$, followed by $x = e^{-at}$, gives exactly (41). Its density is positive on $(e^{-a}, 1)$. At $x = e^{-a}$ the logarithmic factor vanishes when $q > 1$ (and is interpreted as one when $q = 1$); as $x \uparrow 1$,

$$w_q(x) = O\left((1-x)^{\ell-1} |\log(1-x)|^{q-1}\right).$$

Thus the density is integrable at both endpoints and has infinite support. This proves the strict Hausdorff assertion.

For fixed q , every finite set of the $\mu_m^{(q)}$ is obtained from the values $\text{Ein}_q(b + ak)$ by a matrix whose row indexed by m has last nonzero entry, with nonzero coefficient, in column $m + \ell$. Distinct rows therefore have distinct pivots. For several orders the matrix is block diagonal in q . Corollary 8.4 then proves (42).

Strict positivity of the measure gives $\mathcal{H}_n^{(q)} > 0$. Moreover,

$$\frac{\partial \mathcal{H}_n^{(q)}}{\partial \mu_{2n}^{(q)}} = \mathcal{H}_{n-1}^{(q)}. \quad (50)$$

The Jacobian with respect to $\mu_0^{(q)}, \mu_2^{(q)}, \dots, \mu_{2n}^{(q)}$ is triangular with nonzero diagonal; together with the independence of the moments, this proves the assertion about all Hankel determinants.

The finite Stieltjes–Favard algorithm gives, for every $N \geq 1$, the birational equality

$$\mathcal{E}_{\text{exp}}(\mu_0^{(q)}, \dots, \mu_{2N-1}^{(q)}) = \mathcal{E}_{\text{exp}}(\mu_0^{(q)}, \alpha_0^{(q)}, \dots, \alpha_{N-1}^{(q)}, \beta_1^{(q)}, \dots, \beta_{N-1}^{(q)}). \quad (51)$$

Both sides have $2N$ displayed generators. Hence (44) follows, jointly in all orders.

Strict total positivity follows from Andreief’s identity: every minor is an integral of the product of two generalized Vandermonde determinants against the positive density in (41). After ordering the integration variables, both determinants have the same strict sign. Normalizing the measure by its mass identifies the infinite Jacobi operator with multiplication by x on its L^2 -space. Since w_q is positive almost everywhere on the whole interval, this gives simple purely absolutely continuous spectrum $[e^{-a}, 1]$. The Denisov–Rakhmanov theorem [44], affinely rescaled to that interval, gives (45). The identity

$$\mathcal{H}_n^{(q)} = (\mu_0^{(q)})^{n+1} \prod_{k=1}^n (\beta_k^{(q)})^{n+1-k}$$

then proves (46); the standard fixed-walk trace argument for Jacobi truncations with convergent coefficients gives the arcsine law (47).

The characteristic polynomial of $J_N^{(q)}$ is $P_N^{(q)}$. Given the two monic polynomials $P_N^{(q)}$ and $P_{N-1}^{(q)}$, successive Euclidean divisions in (43) recover $\alpha_0^{(q)}, \dots, \alpha_{N-1}^{(q)}$ and $\beta_1^{(q)}, \dots, \beta_{N-1}^{(q)}$ rationally; conversely the recurrence constructs the two polynomials. Their $2N-1$ nonleading coefficients are therefore algebraically independent. Adjoining all roots is an algebraic extension and introduces exactly $2N-1$ displayed roots, so those roots are algebraically independent as claimed. Positivity of the $\beta_n^{(q)}$ gives simplicity and strict interlacing. The same Euclidean divisions recover every earlier monic polynomial $P_j^{(q)}$, $j < N-1$, from the coefficients of $P_N^{(q)}$ and $P_{N-1}^{(q)}$. All earlier eigenvalues are therefore algebraic over the field generated by the consecutive pair, which proves (48). $\square$

Corollary 8.6 (The order-two Basel split). *Put*

$$\eta = \text{Ein}(1) = \gamma + \frac{\delta}{e}, \quad T = \int_1^\infty \frac{e^{-u} \log u}{u} du.$$

Then each of the following triples is algebraically independent over $\overline{\mathbb{Q}}$:

$$(e, \eta, \text{Ein}_2(1)), \quad \left(e, \gamma + \frac{\delta}{e}, \gamma^2 + \frac{\pi^2}{6} - 2T \right). \quad (52)$$

Consequently

$$\text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(e, \gamma, \delta, \text{Ein}_2(1)) \geq 3. \quad (53)$$

Proof. The first triple is the case $s = 1$, $\lambda_1 = 1$, and $R = 2$ of Theorem 8.3. The exact identity

$$2\text{Ein}_2(1) + 2T = \gamma^2 + \frac{\pi^2}{6}$$

turns the first triple into the second by replacing the third coordinate with twice itself. Finally $\eta = \gamma + \delta/e$, so the first triple is contained in the field on the left of (53). No individual irrationality or transcendence conclusion for γ or δ is asserted. $\square$

Remark 8.7 (Classical lineage and scope). The functions in (31) belong to Hardy's classical polyexponential lineage. In the notation

$$\mathfrak{e}_p(x, a) = \sum_{n \geq 0} \frac{x^n}{n!(n+a)^p},$$

one has $\text{Ein}_r(z) = z\mathfrak{e}_{r+1}(-z, 1)$, whereas $\mathfrak{e}_1(x, 1) = (e^x - 1)/x$. Thus $a = 1$ is a resonant integral shift whose first coordinate is already elementary. Shidlovskii proved an all-order algebraic-independence theorem for a nearby shifted family with a nonintegral rational shift; see [49, Sec. 7.3.3] and the account [43, Sec. 9]. The argument above treats the regularized integral-shift endpoint needed for Ein , simultaneously at several algebraic dilation points, and identifies the complete toric relation ideal. It does not cover multiple-index polyexponentials, whose shuffle-type relations lie outside the present one-level tower. We make no broader historical-priority claim.

Corollary 8.8 (All-order gamma-jet split). *Write the Laurent expansion at the origin as*

$$\Gamma(s) = \frac{1}{s} + \sum_{k \geq 0} g_k s^k$$

and put

$$T_k = \frac{1}{k!} \int_1^\infty \frac{e^{-u}(\log u)^k}{u} du \quad (k \geq 0).$$

Then, for every $k \geq 0$,

$$g_k = (-1)^{k+1} \text{Ein}_{k+1}(1) + T_k. \quad (54)$$

Consequently the countable family

$$\{g_k - T_k : k \geq 0\}$$

is algebraically independent over $\mathcal{E}_{\text{exp}}$. In particular, for every $R \geq 1$ the $R+1$ numbers

$$e, \quad g_0 - T_0, \quad g_1 - T_1, \quad \dots, \quad g_{R-1} - T_{R-1}$$

are algebraically independent over $\overline{\mathbb{Q}}$.

The coefficients g_k are explicit polynomials in the classical gamma jet. Namely, if Y_n denotes the complete exponential Bell polynomial, then

$$g_k = \frac{1}{(k+1)!} Y_{k+1}(x_1, \dots, x_{k+1}), \quad x_1 = -\gamma, \quad x_n = (-1)^n (n-1)! \zeta(n) \quad (n \geq 2). \quad (55)$$

Thus (54) gives, at every order, an explicit algebraically independent combination of $\gamma, \zeta(2), \dots, \zeta(k+1)$ and the tail T_k . For $k = 1$ it is precisely

$$\text{Ein}_2(1) + \int_1^\infty \frac{e^{-u} \log u}{u} du = \frac{\gamma^2}{2} + \frac{\pi^2}{12}.$$

No individual arithmetic conclusion for a zeta value or a tail integral is asserted.

Proof. For $\Re s > 0$, split Euler's integral at 1:

$$\Gamma(s) = \int_0^1 e^{-u} u^{s-1} du + \int_1^\infty e^{-u} u^{s-1} du.$$

After isolating the pole $1/s$, the resulting identities extend meromorphically to a neighborhood of $s = 0$, where termwise integration of the first part gives

$$\begin{aligned} \int_0^1 e^{-u} u^{s-1} du &= \frac{1}{s} + \sum_{n \geq 1} \frac{(-1)^n}{n!(n+s)} \\ &= \frac{1}{s} + \sum_{k \geq 0} (-1)^{k+1} \text{Ein}_{k+1}(1) s^k. \end{aligned}$$

Expansion under the exponentially convergent second integral gives

$$\int_1^\infty e^{-u} u^{s-1} du = \sum_{k \geq 0} T_k s^k.$$

Coefficient comparison proves (54). The algebraic-independence assertions are Theorem 8.3 and Corollary 8.4 at the point 1.

Finally,

$$\Gamma(1+s) = \exp \left(-\gamma s + \sum_{n \geq 2} \frac{(-1)^n \zeta(n)}{n} s^n \right).$$

The defining generating identity for the complete exponential Bell polynomials gives (55). $\square$

9 Gamma-free curvature, logarithms, and a probabilistic transfer

The pure-exponential base change is stable under every full-rank linear projection. This produces explicit gamma-free families among classical logarithmic special functions and transfers the same algebraic matroid to a probability law.

9.1 A geometric-grid curvature family

Fix an integer $q \geq 2$. For $n \geq 0$, define

$$\begin{aligned} A_n &= E_1(q^{n+2}) - 2E_1(q^{n+1}) + E_1(q^n), \\ B_n &= \text{Ei}(q^{n+2}) - 2\text{Ei}(q^{n+1}) + \text{Ei}(q^n), \\ C_n &= \text{Ci}(q^{n+2}) - 2\text{Ci}(q^{n+1}) + \text{Ci}(q^n), \\ S_n &= \text{Si}(q^{n+2}) - 2\text{Si}(q^{n+1}) + \text{Si}(q^n). \end{aligned}$$

Theorem 9.1 (Gamma-free geometric curvature). *The countable family*

$$\{A_n, B_n, C_n, S_n : n \geq 0\} \tag{56}$$

is algebraically independent over $\mathcal{E}_{\text{exp}}$, in the finite-subset sense.

Proof. For positive x , the connection formulas are

$$\text{Ein}(x) = \gamma + \log x + E_1(x), \quad \text{Ein}(-x) = \gamma + \log x - \text{Ei}(x),$$

and

$$\frac{\text{Ein}(ix) + \text{Ein}(-ix)}{2} = \gamma + \log x - \text{Ci}(x), \quad \frac{\text{Ein}(ix) - \text{Ein}(-ix)}{2i} = \text{Si}(x).$$

The second-difference row $(1, -2, 1)$ kills both the constant γ and the affine sequence $\log(q^n) = n \log q$. Hence every displayed quantity is a linear form in Ein-values on one of the four disjoint orbits $\{q^j\}$, $\{-q^j\}$, $\{iq^j\}$, and $\{-iq^j\}$. Within each block, every finite family of second-difference rows is linearly independent: inspect the largest terminal index in a linear combination. The four blocks have disjoint support. Lemma 5.2 therefore proves joint algebraic independence over $\mathcal{E}_{\text{exp}}$. $\square$

The geometric grid is essential to the uncorrected formula: on a general grid one must add the logarithmic second difference $\log(x_{n+2}x_n/x_{n+1}^2)$. Omitting it would reintroduce the connection term and invalidate the linear-image argument.

9.2 Logarithms of rational period expressions

Proposition 9.2 (Global logarithm principle). *Let $\lambda_1, \dots, \lambda_s \in \overline{\mathbb{Q}}^\times$ be distinct, and let $R \in \mathcal{E}_{\text{exp}}(X_1, \dots, X_s)$ be nonconstant. Suppose*

$$r = R(\text{Ein}(\lambda_1), \dots, \text{Ein}(\lambda_s))$$

is defined and nonzero. Every solution of $e^w = r$, hence every branch of $\log r$, is transcendental.

Proof. If w were algebraic, then $e^w \in \mathcal{E}_{\text{exp}}$. On the other hand, Theorem 5.1 makes the $\text{Ein}(\lambda_j)$ independent indeterminates over $\mathcal{E}_{\text{exp}}$, so a nonconstant rational expression in them cannot belong to $\mathcal{E}_{\text{exp}}$. This contradiction proves the claim. $\square$

The proposition applies after every invertible Fourier transform in (8). For example, every logarithm of each of $\text{Si}(a)$, $\text{Ci}(a)$, $\text{Shi}(a)$, $\text{Cinh}(a)$, with $a \in \overline{\mathbb{Q}}^\times$, is transcendental. It also proves the transcendence of every branch of $\log \text{Ein}(1)$ without an exceptional-value analysis; the non-ultra-Liouville refinement in section 25 still uses the quantitative theorem of Fischler–Rivoal.

More generally, let $G(W, e^{\beta_1 W}, \dots, e^{\beta_m W})$ be a rational expression over $\overline{\mathbb{Q}}$, where the β_j are algebraic. Every solution w of

$$G(w, e^{\beta_1 w}, \dots, e^{\beta_m w}) = r$$

is transcendental whenever both sides are defined: an algebraic w would make the left side an element of $\mathcal{E}_{\text{exp}}$. Logarithms and Lambert W -preimages of r are elementary special cases.

9.3 The Dickman log-Laplace matroid

Let Φ denote the Laplace transform of the normalized Dickman law. The classical de Bruijn formula is

$$\Phi(s) = \exp\left(\int_0^1 \frac{e^{-sx} - 1}{x} dx\right) = e^{-\text{Ein}(s)} \quad (s > 0); \quad (57)$$

see, for example, [39]. Thus the Lévy exponent, rather than the transform value itself, is an Ein-period.

Corollary 9.3 (Dickman log-Laplace periods). *For distinct positive algebraic $s_1, \dots, s_N$, the numbers*

$$-\log \Phi(s_j) = \text{Ein}(s_j) \quad (1 \leq j \leq N)$$

are algebraically independent over $\mathcal{E}_{\text{exp}}$. If these parameters label the vertices of an oriented graph, then the log-ratios

$$\log \frac{\Phi(s_{t(e)})}{\Phi(s_{s(e)})} = -\text{Ein}(s_{t(e)}) + \text{Ein}(s_{s(e)})$$

have precisely the signed cycle relations and no others.

Proof. Use (57), Theorem 5.1, and the incidence-matrix argument of Theorem 5.3. $\square$

This is an exact transfer to the logarithmic Laplace coordinates of a classical smooth-number distribution. It gives no arithmetic conclusion about the values $\Phi(s_j) = e^{-\text{Ein}(s_j)}$ themselves, which lie outside the finite-point Siegel–Shidlovskii mechanism.

10 New classical families from balanced Ein coordinates

The pure-exponential base-change theorem turns every full-rank linear image of distinct Ein-coordinates into an exact algebraic-independence statement. Two consequences are particularly concrete: a pairwise algebraicity exclusion at every positive algebraic level, and a countable family that is simultaneously algebraically independent and sharply small.

10.1 Four classical values at every algebraic level

Theorem 10.1 (Four classical values at an algebraic level). *For every positive algebraic number a , among*

$$E_1(a), \quad \text{Ei}(a), \quad \text{Ci}(a), \quad \text{Chi}(a) \quad (58)$$

at most one is algebraic.

More precisely, each of the following six quantities is transcendental over $\mathcal{E}_{\text{exp}}$:

$$\begin{array}{lll} E_1(a) + \text{Ei}(a), & E_1(a) + \text{Ci}(a), & E_1(a) + \text{Chi}(a), \\ \text{Ei}(a) - \text{Ci}(a), & \text{Ei}(a) - \text{Chi}(a), & \text{Chi}(a) - \text{Ci}(a). \end{array} \quad (59)$$

Proof. Put

$$A = \text{Ein}(a), \quad B = \text{Ein}(-a), \quad C = \frac{\text{Ein}(ia) + \text{Ein}(-ia)}{2}.$$

The connection formulas give, in the order displayed in (59),

$$A - B, \quad A - C, \quad \frac{A - B}{2}, \quad C - B, \quad \frac{A - B}{2}, \quad C - \frac{A + B}{2}. \quad (60)$$

The four values $\text{Ein}(a), \text{Ein}(-a), \text{Ein}(ia), \text{Ein}(-ia)$ are algebraically independent over $\mathcal{E}_{\text{exp}}$. Every expression in (60) is a nonzero linear form in those independent coordinates and is therefore transcendental over $\mathcal{E}_{\text{exp}}$. If any two members of (58) were algebraic, the corresponding witness in (59) would be algebraic, a contradiction. $\square$

10.2 A sharply decaying algebraically independent family

Theorem 10.2 (Quadratic balanced E_1 family). *For $m \geq 1$, put*

$$Q_m = E_1(m^2) - 2E_1(m(m+1)) + E_1((m+1)^2). \quad (61)$$

Then the countable family $\{Q_m : m \geq 1\}$ is algebraically independent over $\mathcal{E}_{\text{exp}}$, in the finite-subset sense. Every Q_m is positive, and

$$Q_m = \frac{e^{-m^2}}{m^2} \left(1 + O(m^{-2}) + O(e^{-m})\right). \quad (62)$$

Consequently

$$\lim_{m \rightarrow \infty} \frac{-\log Q_m}{(m+1)^2} = 1. \quad (63)$$

Proof. The two cancellations

$$1 - 2 + 1 = 0, \quad \log(m^2) - 2\log(m(m+1)) + \log((m+1)^2) = 0$$

give the exact identity

$$Q_m = \text{Ein}(m^2) - 2\text{Ein}(m(m+1)) + \text{Ein}((m+1)^2). \quad (64)$$

For $m \geq 1$, the integer $m(m+1)$ is not a square: since $\gcd(m, m+1) = 1$, a square product would force both factors to be squares, and two consecutive positive squares do not exist. Moreover $m(m+1)$ is strictly increasing with m . Hence the middle coordinate $\text{Ein}(m(m+1))$ occurs in no row of (64) other than the m -th row. In any finite linear combination of these rows, inspection of the middle coordinates forces every row coefficient to vanish. The rows therefore have full rank. The pure-exponential base-change theorem and Lemma 5.2 prove the asserted algebraic independence.

For $x > 0$, integration by parts gives the standard bounds

$$\frac{e^{-x}}{x+1} < E_1(x) < \frac{e^{-x}}{x}. \quad (65)$$

Indeed, the upper bound is immediate, while

$$E_1(x) = \frac{e^{-x}}{x} - \int_x^\infty \frac{e^{-t}}{t^2} dt > \frac{e^{-x}}{x} - \frac{E_1(x)}{x}$$

gives the lower bound. Therefore

$$\frac{2E_1(m^2 + m)}{E_1(m^2)} < 2e^{-m} \frac{m^2 + 1}{m^2 + m} \leq \frac{2}{e} < 1.$$

Thus $E_1(m^2) > 2E_1(m^2 + m)$, and the last positive term in (61) proves $Q_m > 0$.

Finally, the standard expansion

$$E_1(x) = \frac{e^{-x}}{x} \left(1 + O(x^{-1})\right)$$

shows that

$$\frac{E_1(m^2 + m)}{E_1(m^2)} = O(e^{-m}), \quad \frac{E_1((m+1)^2)}{E_1(m^2)} = O(e^{-2m}).$$

This proves (62); taking logarithms gives (63). $\square$

Remark 10.3 (Uniform moving-point boundary). The scale $(m+1)^2$ in (63) is the largest evaluation argument, not a Weil height or a relation-polynomial height. The theorem therefore gives a sharp counterexample to naive lower bounds that are uniform only in the moving archimedean evaluation scale. It does not contradict fixed-point algebraic-independence measures for E -functions.

10.3 A conditional E - D collision

Corollary 10.4 (An explicit conditional E - D collision). *Let $\mathbf{E}$ and $\mathbf{D}$ be the rings of finite E -values and arithmetic-Gevrey values of order one, respectively, in the sense of Fischler–Rivoal [22]. Then*

$$\boxed{\gamma \in \overline{\mathbb{Q}} \implies \delta \in (\mathbf{E} \cap \mathbf{D}) \setminus \overline{\mathbb{Q}}.} \quad (66)$$

Consequently the conjectural equality $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ implies that γ is transcendental.

Proof. Assume $\gamma \in \overline{\mathbb{Q}}$. Then

$$f(z) = e^z(\text{Ein}(z) - \gamma)$$

is an E -function with algebraic Taylor coefficients, and $f(1) = \delta$. Hence $\delta \in \mathbf{E}$. The standard Borel-sum realization of Euler’s factorial series gives $\delta \in \mathbf{D}$. If δ were algebraic, then

$$\text{Ein}(1) = \gamma + e^{-1}\delta$$

would belong to $\mathcal{E}_{\text{exp}}$, contradicting the pure-exponential base-change theorem. Thus $\delta \notin \overline{\mathbb{Q}}$, proving the assertion. $\square$

11 Common-shift defect one and residual families

The following field-theoretic lemma isolates exactly what a common translation can and cannot lose. The independent anchor is essential: merely excluding the possibility that the shift parameter belongs to the base field would not prove the final two-variable independence statement.

Theorem 11.1 (Common-shift defect one). *Let $K \subset \mathbb{C}$ be a field of characteristic zero, and let $z_0, z_1, \dots, z_r$ be algebraically independent over K , where $r \geq 1$. Fix $c, e_0 \in K^\times$, and put*

$$T_j = c(z_j - \gamma) \quad (1 \leq j \leq r), \quad \Delta = e_0(z_0 - \gamma). \quad (67)$$

Then

$$\text{trdeg}_K K(T_1, \dots, T_r) \geq r - 1. \quad (68)$$

More precisely, the $r - 1$ anchored differences

$$T_2 - T_1, \dots, T_r - T_1$$

are algebraically independent over K . Consequently the T_j are algebraically dependent over K if and only if

$$\text{trdeg}_K K(T_1, \dots, T_r) = r - 1.$$

In that case γ and Δ are algebraically independent over K .

Proof. The linear forms $z_j - z_1$, $2 \leq j \leq r$, have row rank $r - 1$ in the independent coordinates $z_1, \dots, z_r$. Hence

$$T_j - T_1 = c(z_j - z_1), \quad 2 \leq j \leq r,$$

are algebraically independent over K , proving (68).

Suppose now that the T_j are algebraically dependent and write $L = K(T_1, \dots, T_r)$. The lower bound gives $\text{trdeg}_K L = r - 1$. Since $z_j = \gamma + c^{-1}T_j$, one has

$$K(z_1, \dots, z_r, \gamma) = L(\gamma). \quad (69)$$

The left side contains the purely transcendental field $K(z_1, \dots, z_r)$, of transcendence degree r , whereas the right side has transcendence degree at most $(r-1) + 1 = r$. Thus both sides of (69) have transcendence degree exactly r . It follows that γ is algebraic over $K(z_1, \dots, z_r)$, and also that γ is transcendental over L , hence over K .

Because $z_0, z_1, \dots, z_r$ are algebraically independent, z_0 is transcendental over the algebraic closure of $K(z_1, \dots, z_r)$. In particular it is transcendental over $K(\gamma)$. Therefore $z_0 - \gamma$ is transcendental over $K(\gamma)$. Since γ is transcendental over K and $e_0 \in K^\times$, the two numbers γ and $\Delta = e_0(z_0 - \gamma)$ are algebraically independent over K . $\square$

We apply the theorem over $K = \mathcal{E}_{\text{exp}}$. For $N, n \geq 1$, define

$$Y_N = (N+1)\text{Ein}(2^N) - N\text{Ein}(2^{N+1}), \quad \xi_n = \text{Ein}(-n), \quad (70)$$

$$R_N = e((N+1)E_1(2^N) - NE_1(2^{N+1})), \quad S_n = e(\text{Ei}(n) - \log n). \quad (71)$$

The connection formulas give

$$R_N = e(Y_N - \gamma), \quad -S_n = e(\xi_n - \gamma), \quad \delta = e(\text{Ein}(1) - \gamma). \quad (72)$$

Lemma 11.2 (Independent residual coordinates). *The countable family*

$$\{\text{Ein}(1)\} \cup \{Y_N : N \geq 1\} \cup \{\xi_n : n \geq 1\}$$

is algebraically independent over $\mathcal{E}_{\text{exp}}$, in the finite-subset sense.

Proof. The pure-exponential base-change theorem makes the distinct coordinates

$$\text{Ein}(1), \quad \text{Ein}(2^k) \ (k \geq 1), \quad \text{Ein}(-n) \ (n \geq 1)$$

algebraically independent over $\mathcal{E}_{\text{exp}}$. The negative coordinates are disjoint from the positive ones. It remains only to check the row rank of the Y_N . In a nonzero finite linear combination of their defining rows, choose the smallest occurring index N . The coordinate $\text{Ein}(2^N)$ occurs only in that smallest row and has coefficient $N+1$, so its row coefficient must vanish. Iteration proves that every row coefficient vanishes. Lemma 5.2 now gives the assertion. $\square$

Corollary 11.3 (Countable residual dichotomy). *Exactly one of the following alternatives holds.*

(i) *The family*

$$\{R_N : N \geq 1\} \cup \{S_n : n \geq 1\}$$

is algebraically independent over $\mathcal{E}_{\text{exp}}$.

(ii) *Some finite subfamily is algebraically dependent over $\mathcal{E}_{\text{exp}}$, and γ, δ are then algebraically independent over $\mathcal{E}_{\text{exp}}$. In particular e, γ, δ are algebraically independent over $\overline{\mathbb{Q}}$.*

Unconditionally, the anchored family

$$\boxed{\{R_N - R_1 : N \geq 2\} \cup \{S_n + R_1 : n \geq 1\}} \quad (73)$$

is algebraically independent over $\mathcal{E}_{\text{exp}}$.

Proof. Apply Theorem 11.1 to every finite selection from (72), with $z_0 = \text{Ein}(1)$, common scale $c = e$, and $\Delta = \delta$. Lemma 11.2 verifies the hypothesis. A countable family is algebraically dependent exactly when one of its finite subfamilies is dependent, which proves the dichotomy.

For the final assertion, anchor the common-shift variables at R_1 . Their differences are

$$R_N - R_1 = e(Y_N - Y_1), \quad (-S_n) - R_1 = -(S_n + R_1),$$

and the anchored-difference assertion of Theorem 11.1 proves their joint algebraic independence. $\square$

Remark 11.4 (Logical scope). The corollary is an exact alternative, not a construction of the relation in its second branch. If no finite residual relation is supplied, the result is the unconditional algebraic independence of the anchored family (73); it does not decide γ or δ individually.

12 A six-constant consequence

Put

$$A = \text{Ein}(1), \quad B = \text{Ein}(-1), \quad C = \text{Cin}(1) = \frac{\text{Ein}(i) + \text{Ein}(-i)}{2}.$$

Applying theorem 4.1 to $\{1, -1, i, -i\}$ shows that

$$e, A, B, C \text{ are algebraically independent over } \overline{\mathbb{Q}}. \quad (74)$$

The identities needed below are

$$\begin{aligned} E_1(1) &= A - \gamma, & \text{Ei}(1) &= \gamma - B, \\ \text{Ci}(1) &= \gamma - C, & \text{Chi}(1) &= \gamma - \frac{A+B}{2}, & \delta &= e(A - \gamma). \end{aligned} \quad (75)$$

Proposition 12.1 (The remaining dimension). *For*

$$\mathcal{S} = \{\gamma, \delta, E_1(1), \text{Ei}(1), \text{Ci}(1), \text{Chi}(1)\}$$

one has

$$\overline{\mathbb{Q}}(\mathcal{S}) = \overline{\mathbb{Q}}(\gamma, e, A, B, C)$$

and therefore

$$4 \leq \text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(\mathcal{S}) \leq 5.$$

In particular,

$$\text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(\gamma, \delta, E_1(1), \text{Ei}(1)) \geq 3.$$

Proof. The forward inclusion follows from $e = \delta/E_1(1)$, $A = \gamma + E_1(1)$, $B = \gamma - \text{Ei}(1)$, and $C = \gamma - \text{Ci}(1)$; here $E_1(1) > 0$. The reverse inclusion follows from (75). Algebraic independence of e, A, B, C gives the lower bound, and adjoining γ gives the upper bound. For the final claim, the four-number field contains $e = \delta/E_1(1)$, $A = \gamma + E_1(1)$, and $B = \gamma - \text{Ei}(1)$, which are algebraically independent. $\square$

Theorem 12.2. *No two members of*

$$\mathcal{S} = \{\gamma, \delta, E_1(1), \text{Ei}(1), \text{Ci}(1), \text{Chi}(1)\}$$

are algebraic. Hence at least five members of $\mathcal{S}$ are transcendental.

Proof. First omit δ . If any two of the remaining five numbers were algebraic, adding or subtracting their expressions in (75) would put one of the following nonzero linear forms in A, B, C in $\overline{\mathbb{Q}}$:

$$A, B, C, A + B, A - B, A - C, C - B, \frac{A + B}{2} - C.$$

This contradicts (74).

Now pair δ with each of the other constants. If δ and γ were algebraic, the identity $\delta = e(A - \gamma)$ would be a nonzero algebraic relation between e and A . If δ and $E_1(1)$ were algebraic, positivity of $E_1(1)$ would give $e = \delta/E_1(1) \in \overline{\mathbb{Q}}$. Finally, algebraicity of δ together with, respectively, $\text{Ei}(1), \text{Ci}(1), \text{Chi}(1)$, would give one of the relations

$$\delta = e(A - B - \text{Ei}(1)), \quad \delta = e(A - C - \text{Ci}(1)),$$

or

$$\delta = e\left(\frac{A - B}{2} - \text{Chi}(1)\right),$$

contradicting algebraic independence of the indicated subset of e, A, B, C . All fifteen pairs are therefore excluded. $\square$

Corollary 12.3. *If γ is algebraic, then $E_1(1), \text{Ei}(1), \delta$ are algebraically independent. More generally, if one of $E_1(1), \text{Ei}(1), \delta$ is algebraic, then γ and the other two are algebraically independent.*

Proof. Use $A = \gamma + E_1(1)$, $B = \gamma - \text{Ei}(1)$, and $e = \delta/E_1(1)$ when the relevant constant is assumed algebraic and nonzero. A polynomial relation among the claimed triple would then induce a polynomial relation among an appropriate subset of e, A, B . The changes of coordinates are birational, so nonzero relations remain nonzero. $\square$

Theorem 12.4 (Conditional relation ideal after one algebraic specialization). *Write*

$$(\Gamma, D, X, Y, Z, W) = (\gamma, \delta, E_1(1), \text{Ei}(1), \text{Ci}(1), \text{Chi}(1))$$

and let $I_{\mathcal{S}}$ be the kernel of the corresponding evaluation homomorphism from $\overline{\mathbb{Q}}[\Gamma, D, X, Y, Z, W]$ to $\mathbb{C}$. If one displayed constant κ is algebraic, say $\kappa = c \in \overline{\mathbb{Q}}$, then

$$\text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(\mathcal{S}) = 4$$

and

$$\boxed{I_{\mathcal{S}} = (Y - X - 2W, T_{\kappa} - c),} \quad (76)$$

where T_{κ} denotes the coordinate occupied by κ . In particular the Euler-level point w_* of Theorem 26.7 is transcendental.

Proof. Recall that e, A, B, C are algebraically independent and that $\overline{\mathbb{Q}}(\mathcal{S}) = \overline{\mathbb{Q}}(\gamma, e, A, B, C)$. In the six possible cases, algebraicity of $\kappa = c$ expresses γ inside $\overline{\mathbb{Q}}(e, A, B, C)$ as follows:

κ	γ	κ	γ	κ	γ
γ	c	δ	$A - c/e$	$E_1(1)$	$A - c$
$\text{Ei}(1)$	$B + c$	$\text{Ci}(1)$	$C + c$	$\text{Chi}(1)$	$(A + B)/2 + c$

Hence $\overline{\mathbb{Q}}(\mathcal{S}) = \overline{\mathbb{Q}}(e, A, B, C)$, of transcendence degree four. The evaluation kernel therefore has height two. It contains the two independent linear polynomials in (76); these generate a height-two prime ideal and hence the complete kernel. If w_* were algebraic, Corollary 26.8 would instead give transcendence degree five, a contradiction. $\square$

13 Conditional relation ideals and gamma detectors

The structural theorems above do not decide the arithmetic nature of γ . They do, however, turn several concrete relations into exact certificates against algebraicity. The point of this section is to state those certificates with their full quantifiers and relation ideals.

Put

$$D = \delta, \quad X = E_1(1), \quad Y = \text{Ein}(1), \quad Z = \text{Ci}(1), \quad W = \text{Chi}(1),$$

and retain the pure exponential field $\mathcal{E}_{\text{exp}} = \overline{\mathbb{Q}}(e^\alpha : \alpha \in \overline{\mathbb{Q}})$.

Proposition 13.1 (The conditional ideal over the pure exponential field). *If $\gamma \in \overline{\mathbb{Q}}$, then X, Y, Z are algebraically independent over $\mathcal{E}_{\text{exp}}$, and the complete relation ideal of D, X, Y, Z, W over $\mathcal{E}_{\text{exp}}$ is*

$$(\text{D} - eX, 2W - Y + X). \quad (77)$$

Consequently

$$\boxed{\gamma \text{ is transcendental, or } \delta \text{ is transcendental over } \mathcal{E}_{\text{exp}}.} \quad (78)$$

Proof. Under the hypothesis, the connection identities give

$$X = \text{Ein}(1) - \gamma, \quad Y = \gamma - \text{Ein}(-1), \quad Z = \gamma - \frac{\text{Ein}(i) + \text{Ein}(-i)}{2}.$$

They are three linearly independent combinations of the four Ein-coordinates at $1, -1, i, -i$. Theorem 5.1 and Lemma 5.2 therefore make them algebraically independent over $\mathcal{E}_{\text{exp}}$. The remaining coordinates satisfy, and are determined by,

$$D = eX, \quad 2W = Y - X.$$

This proves the ideal statement. In particular $D = eX$ is transcendental over $\mathcal{E}_{\text{exp}}$ whenever γ is algebraic, proving (78). $\square$

For $m \geq 1$, set

$$T_m = e^{\gamma^m}.$$

Theorem 13.2 (Exact T_m -ideals under algebraic γ). *Assume $\gamma \in \overline{\mathbb{Q}}$, fix $m \geq 1$, and put $\eta = \gamma^m$. If $\eta \notin \mathbb{Q}$, then*

$$T_m, D, X, Y, Z$$

are algebraically independent over $\overline{\mathbb{Q}}$. If $\eta = a/b \in \mathbb{Q}_{>0}$, where $(a, b) = 1$, then the complete relation ideal of these five coordinates is the principal prime ideal

$$(\text{D}^a - \text{T}^b X^a). \quad (79)$$

After adjoining W , in the first case one adds only $2W - Y + X$; in the second case the full ideal is generated by that linear form and (79).

In either case T_m and D are algebraically independent. Hence, unconditionally,

$$\boxed{P(e^{\gamma^m}, \delta) = 0 \text{ for some } 0 \neq P \in \overline{\mathbb{Q}}[U, V] \implies \gamma \text{ is transcendental.}} \quad (80)$$

Proof. Since $0 < \gamma < 1$, the number η is nonzero and differs from $1, -1, i, -i$. Apply Theorem 4.1 to $\{1, -1, i, -i, \eta\}$. With

$$A = \text{Ein}(1), \quad B = \text{Ein}(-1), \quad C = \frac{\text{Ein}(i) + \text{Ein}(-i)}{2},$$

the relevant coordinates are birationally related by

$$e = \frac{D}{X}, \quad A = X + \gamma, \quad B = \gamma - Y, \quad C = \gamma - Z.$$

If 1 and η are $\mathbb{Q}$ -independent, the exponential coordinates e, T_m have no toric relation, and the five displayed coordinates are independent. If $\eta = a/b$, their only toric relation is $T_m^b = e^a$, which becomes $D^a = T_m^b X^a$. The exponent vector is primitive, so the binomial is prime; the transcendence-degree formula gives height one and excludes any additional relation. The identity $2W = Y - X$ handles W .

In both cases contraction to $\overline{\mathbb{Q}}[\mathbb{T}, \mathbb{D}]$ is zero. Thus a relation as in (80) contradicts the temporary hypothesis $\gamma \in \overline{\mathbb{Q}}$. $\square$

An exact zero is not the only possible certificate. A quantitative E -function theorem also rules out fixed-degree super-Liouville approximation under the algebraicity hypothesis.

Corollary 13.3 (Fixed-degree approximation detector). *Fix $m, d \geq 1$. Suppose that nonzero polynomials $P_n \in \mathbb{Z}[U, V]$, of degree at most d , satisfy*

$$H(P_n) \longrightarrow \infty, \quad P_n(T_m, D) \neq 0,$$

and

$$\frac{-\log |P_n(T_m, D)|}{\log H(P_n)} \longrightarrow \infty. \quad (81)$$

Then γ is transcendental.

Proof. Assume $\gamma \in \overline{\mathbb{Q}}$. Then

$$f_1(z) = e^{\gamma^m z}, \quad f_2(z) = e^z (\text{Ein}(z) - \gamma)$$

are E -functions with algebraic coefficients and $(f_1(1), f_2(1)) = (T_m, D)$. Theorem 13.2 makes the two values algebraically independent. The fixed-degree case of the Adamczewski–Favre–Jon algebraic-independence measure [1] supplies constants $c_d, \kappa_d > 0$ with

$$|P(T_m, D)| \geq c_d H(P)^{-\kappa_d}$$

for every nonzero integral polynomial of degree at most d . This contradicts (81). $\square$

Remark 13.4 (Logical boundary). Theorems 13.2 and Corollary 13.3 are one-way detectors. They do not construct a polynomial relation or a super-small sequence. Moreover e^{γ^m} is an E -value with algebraic coefficients only inside the contradiction hypothesis. Treating it as an unconditional E -value would reverse the logic.

14 Finite Ein witnesses and one-variable Euler traces

The pure-exponential base-change theorem also controls finite representations of constants by algebraic-point Ein-values. We apply that observation first to $\text{Ein}(z) - \gamma$, and then to the rank-one witness family $n \text{Ein}(z) - \text{Ein}(z^n) - (n-1)\gamma$. No appeal to Picard’s theorem or to a numerical zero count is needed.

14.1 Rigidity of finite algebraic-point witnesses

Put

$$\mathcal{V} = \text{span}_{\overline{\mathbb{Q}}} \{ \text{Ein}(\alpha) : \alpha \in \overline{\mathbb{Q}}^\times \} \subset \mathbb{C},$$

and let $\mathcal{E}_{\text{exp}}^{\text{alg}}$ denote the relative algebraic closure of $\mathcal{E}_{\text{exp}}$ in $\mathbb{C}$.

Proposition 14.1 (Finite-witness rigidity). *The linear map*

$$\bigoplus_{\alpha \in \overline{\mathbb{Q}}^\times} \overline{\mathbb{Q}} e_\alpha \longrightarrow \mathbb{C}, \quad \sum_{\alpha} c_\alpha e_\alpha \longmapsto \sum_{\alpha} c_\alpha \text{Ein}(\alpha),$$

where the direct sum means finite support, is injective. Moreover

$$\mathcal{V} \cap \mathcal{E}_{\text{exp}}^{\text{alg}} = \{0\}. \quad (82)$$

Consequently, if $\gamma \in \mathcal{V}$, then its finite Ein-representation is unique and γ is transcendental over $\mathcal{E}_{\text{exp}}$.

Proof. Every finite family of distinct nonzero algebraic-point values $\text{Ein}(\alpha)$ is algebraically independent over $\mathcal{E}_{\text{exp}}$ by Theorem 5.1; it is therefore linearly independent. This proves injectivity. A nonzero linear form in algebraically independent indeterminates is itself transcendental over the base field: extend that linear form to an invertible linear coordinate system. Hence every nonzero element of $\mathcal{V}$ is transcendental over $\mathcal{E}_{\text{exp}}$, proving (82) and the final assertion. $\square$

Let $q_* \in (0, 1)$ be the unique real solution of

$$\text{Ein}(q_*) = \gamma. \quad (83)$$

Indeed, $\text{Ein}'(x) = (1 - e^{-x})/x > 0$ on $\mathbb{R}$, after filling the removable value at 0, while $\text{Ein}(0) - \gamma = -\gamma < 0$ and $\text{Ein}(1) - \gamma = \delta/e > 0$.

Theorem 14.2 (The one-variable Euler-level dichotomy). *Exactly one of the following alternatives holds.*

- (a) *Every zero of $\text{Ein}(z) - \gamma$ is transcendental.*
- (b) *The number q_* is algebraic and is the unique algebraic zero of $\text{Ein}(z) - \gamma$. In this case γ and δ are algebraically independent over $\mathcal{E}_{\text{exp}}$.*

Under alternative (b), the six-constant field

$$\overline{\mathbb{Q}}(\gamma, \delta, E_1(1), \text{Ei}(1), \text{Ci}(1), \text{Chi}(1))$$

has transcendence degree 5, and its complete relation ideal is generated by

$$\text{Ei}(1) - E_1(1) - 2 \text{Chi}(1). \quad (84)$$

Proof. Two distinct algebraic zeros a, b would give $\text{Ein}(a) = \text{Ein}(b)$, contradicting Theorem 5.1. A nonreal algebraic zero is also impossible, since its distinct conjugate would be a second algebraic zero. Thus the only possible algebraic zero is the unique real zero q_* .

Assume that q_* is algebraic and put $Y_a = \text{Ein}(a)$. The values Y_{q_*} and Y_1 are independent over $\mathcal{E}_{\text{exp}}$, and

$$\gamma = Y_{q_*}, \quad \frac{\delta}{e} = Y_1 - Y_{q_*}.$$

This invertible linear change proves the asserted independence of γ, δ over $\mathcal{E}_{\text{exp}}$.

For the last assertion, put

$$A = \text{Ein}(1), \quad B = \text{Ein}(-1), \quad C = \frac{\text{Ein}(i) + \text{Ein}(-i)}{2}.$$

The five arguments $q_*, 1, -1, i, -i$ are distinct. Pure-exponential base change shows that $\gamma = \text{Ein}(q_*)$, A, B, C are independent over $\mathcal{E}_{\text{exp}}$; together with the transcendence of e , the five numbers e, γ, A, B, C are algebraically independent over $\overline{\mathbb{Q}}$. The six-constant field is $\overline{\mathbb{Q}}(e, \gamma, A, B, C)$. Identity (84) is therefore a height-one prime relation and generates the complete kernel. $\square$

14.2 The rank-one witness family

For $n \geq 2$, define

$$G_n(z) = n \text{Ein}(z) - \text{Ein}(z^n) - (n-1)\gamma. \quad (85)$$

Proposition 14.3 (Real-zero classification for G_n). *For every $n \geq 2$, the function G_n has exactly one positive real zero $\beta_n \in (0, 1)$. If n is even, it has no negative real zero. If n is odd, it has exactly one negative real zero $\lambda_n \in (-\infty, -1)$. All these real zeros are simple.*

Proof. For $x \neq 0$, direct differentiation gives

$$G'_n(x) = \frac{n}{x}(e^{-x^n} - e^{-x}). \quad (86)$$

Thus G_n is strictly increasing on $(0, 1)$. Since

$$G_n(0) = -(n-1)\gamma < 0, \quad G_n(1) = \frac{(n-1)\delta}{e} > 0,$$

there is exactly one zero β_n in that interval. For $x > 1$,

$$G_n(x) = nE_1(x) - E_1(x^n) > (n-1)E_1(x) > 0,$$

so there are no further positive zeros.

If n is even and $x < 0$, then $x^n > 0 > x$; hence the numerator in (86) is negative and $G'_n(x) > 0$. Moreover $G_n(x) \rightarrow -\infty$ as $x \rightarrow -\infty$, while $G_n(0) = -(n-1)\gamma < 0$. Strict monotonicity therefore excludes a negative zero. The asserted limit also follows directly, on writing $x = -t$, from

$$G_n(-t) = -n \text{Ei}(t) - E_1(t^n) \quad (t > 0).$$

If n is odd, (86) shows that G_n is strictly decreasing on $(-\infty, -1)$ and strictly increasing on $(-1, 0)$. One has

$$\begin{aligned} G_n(x) &\longrightarrow +\infty \quad (x \rightarrow -\infty), \\ G_n(-1) &= -(n-1) \text{Ei}(1) < 0, \quad G_n(0) = -(n-1)\gamma < 0. \end{aligned}$$

Here the first limit follows from

$$G_n(-t) = \text{Ei}(t^n) - n \text{Ei}(t) \quad (t > 0)$$

and the standard positive-axis asymptotic for Ei . This gives exactly one zero in $(-\infty, -1)$ and none in $(-1, 0)$. Formula (86) is nonzero at each listed root, so they are simple. $\square$

Theorem 14.4 (Algebraic-zero rigidity for G_n). *For every $n \geq 2$, the function G_n has at most one algebraic zero, and every algebraic zero is real. Consequently:*

- (i) *if n is even, either every zero is transcendental or β_n is the unique algebraic zero;*
- (ii) *if n is odd, every nonreal zero is transcendental and at most one of β_n, λ_n is algebraic.*

If α is an algebraic zero and $D_\alpha = \text{Ein}(\alpha) - \text{Ein}(\alpha^n)$, then

$$\gamma, \quad \delta, \quad D_\alpha \quad \text{are algebraically independent over } \mathcal{E}_{\text{exp}}. \quad (87)$$

The six-constant field then has transcendence degree 5, with complete relation ideal generated by (84).

Proof. Suppose that distinct algebraic numbers α, β were zeros. They are nonzero, and subtraction gives

$$n \text{Ein}(\alpha) - \text{Ein}(\alpha^n) = n \text{Ein}(\beta) - \text{Ein}(\beta^n).$$

After collecting equal arguments, this is a linear relation among distinct algebraic-point Ein-values. Its coefficient at $\text{Ein}(\alpha)$ is

$$n - \mathbf{1}_{\alpha^n=\alpha} + \mathbf{1}_{\beta^n=\alpha} \geq n - 1 > 0,$$

because $\beta \neq \alpha$. This contradicts Proposition 14.1. Hence there is at most one algebraic zero. Since G_n has real coefficients, a nonreal algebraic zero and its conjugate would give two, so every algebraic zero is real. Proposition 14.3 now gives the two alternatives.

For either possible algebraic real zero, the three arguments $\alpha, \alpha^n, 1$ are distinct. The corresponding three Ein-values are independent over $\mathcal{E}_{\text{exp}}$, while

$$\gamma = \frac{n \text{Ein}(\alpha) - \text{Ein}(\alpha^n)}{n-1}, \quad \frac{\delta}{e} = \text{Ein}(1) - \gamma, \quad D_\alpha = \text{Ein}(\alpha) - \text{Ein}(\alpha^n).$$

This is an invertible linear change of three coordinates, proving (87). Moreover $\{\alpha, \alpha^n\}$ is disjoint from $\{1, -1, i, -i\}$, so the final six-constant assertion follows exactly as in Theorem 14.2. $\square$

14.3 Two exact traces of the one-variable Euler divisor

Let Z_γ be the multiset of zeros of $F_\gamma(z) = \text{Ein}(z) - \gamma$, and put

$$S_m = \sum_{\rho \in Z_\gamma} \rho^{-m} \quad (m \geq 2). \quad (88)$$

The sums converge absolutely because F_γ is entire of order one.

Theorem 14.5 (Trace recovery for $\text{Ein} - \gamma$). *One has*

$$S_2 = \frac{1}{\gamma^2} - \frac{1}{2\gamma}, \quad (89)$$

$$S_3 = \frac{1}{\gamma^3} - \frac{3}{4\gamma^2} + \frac{1}{6\gamma}. \quad (90)$$

Conversely,

$$\boxed{\gamma = \frac{24S_2 + 1}{24S_3 + 6S_2}}. \quad (91)$$

In particular

$$\overline{\mathbb{Q}}(S_2, S_3) = \overline{\mathbb{Q}}(\gamma),$$

and the two traces satisfy

$$144S_2^3 + 21S_2^2 + S_2 - 144S_3^2 + 3S_3 = 0. \quad (92)$$

Moreover $S_2 \in \overline{\mathbb{Q}}$ if and only if $\gamma \in \overline{\mathbb{Q}}$, and the same equivalence holds with S_3 in place of S_2 .

Proof. A genus-one Hadamard factorization of F_γ gives, for $m \geq 2$, the inverse power sums from the Taylor coefficients of $\log(F_\gamma(z)/F_\gamma(0))$. The origin jet is

$$F_\gamma(0) = -\gamma, \quad F'_\gamma(0) = 1, \quad F''_\gamma(0) = -\frac{1}{2}, \quad F'''_\gamma(0) = \frac{1}{3}.$$

Hence

$$S_2 = -(\log F_\gamma)''(0), \quad S_3 = -\frac{1}{2}(\log F_\gamma)'''(0),$$

which gives (89) and (90). With $x = 1/\gamma$, these read

$$S_2 = x^2 - \frac{x}{2}, \quad S_3 = x^3 - \frac{3}{4}x^2 + \frac{x}{6}.$$

Eliminating x^2 gives

$$x \left(S_2 + \frac{1}{24} \right) = S_3 + \frac{S_2}{4},$$

which proves (91); its denominator is nonzero because $x = 1/\gamma > 1$. A resultant calculation, or direct substitution, gives (92). Finally, each S_j is a nonconstant polynomial in x over $\overline{\mathbb{Q}}$, so algebraicity of either trace forces x , hence γ , to be algebraic. The converse is immediate. $\square$

Remark 14.6 (All higher traces). The same calculation gives

$$S_m = -m[z^m] \log \left(1 + \frac{1}{\gamma} \sum_{r \geq 1} \frac{(-1)^r z^r}{r r!} \right) \quad (m \geq 2).$$

Thus every higher inverse trace is a rational polynomial in $1/\gamma$. Unlike the phase recovery from the zero set of the γ -free reciprocal function, this is a spectral reparametrization of a γ -dependent level.

15 Euler zero lifting and denominator explosion

Fix $0 \neq \xi \in \mathbb{Z}$ and set

$$\mathcal{E}_\xi(z) = \sum_{n \geq 0} (-1)^n n! \xi^n z^n, \quad S_n(\xi) = \sum_{k=0}^n (-1)^k k! \xi^k.$$

Coefficient comparison gives the exact identity

$$\frac{\mathcal{E}_\xi(z) - \alpha}{z - 1} = \sum_{n \geq 0} n! b_n z^n, \quad b_n = \frac{\alpha - S_n(\xi)}{n!}. \quad (93)$$

Theorem 15.1. *Let K contain α , and let Δ_N be the least positive integer with $\Delta_N b_0, \dots, \Delta_N b_N \in \mathcal{O}_K$. Then*

$$\lim_{N \rightarrow \infty} \frac{\log \Delta_N}{N} = +\infty.$$

The series in (93) is therefore not arithmetic Gevrey of order one.

Proof. Suppose first that $\alpha \notin \mathbb{Q}$. There is a $\mathbb{Q}$ -linear functional $\ell : K \rightarrow \mathbb{Q}$, a positive integer M , and a nonzero $r \in \mathbb{Q}$ such that

$$\ell(1) = 0, \quad \ell(\alpha) = r, \quad M\ell(\mathcal{O}_K) \subset \mathbb{Z}.$$

Applying $M\ell$ to $\Delta_N b_N \in \mathcal{O}_K$ gives $M\Delta_N r/N! \in \mathbb{Z}$. After clearing the fixed numerator and denominator of Mr , this implies

$$\Delta_N \geq \frac{N!}{C}$$

for a fixed $C > 0$. Hence $\log \Delta_N/N \rightarrow \infty$.

Now let $\alpha = a/b \in \mathbb{Q}$ in lowest terms. For every prime $p \nmid b$, the series

$$W_{p,\xi} = \sum_{k \geq 0} (-1)^k k! \xi^k$$

converges in $\mathbb{Q}_p$. Call p *bad* if $W_{p,\xi} \neq a/b$. At a bad prime, $v_p(a/b - S_N(\xi))$ is eventually constant, and therefore

$$v_p(\Delta_N) \geq v_p(N!) - O_p(1). \quad (94)$$

We claim that

$$\sum_{p \text{ bad}} \frac{\log p}{p-1} = \infty. \quad (95)$$

Indeed, otherwise let $\mathcal{P}$ be the set of good primes, after deleting the finitely many primes dividing b . The product formula and $v_p(N!) \leq N/(p-1)$ give, for a fixed $C_\xi > 0$,

$$C_\xi^N N! \prod_{p \in \mathcal{P}} |N!|_p^2 \leq \frac{C^N}{N!} \rightarrow 0.$$

Theorem 1 of Matala-aho–Zudilin [36], applied at their variable ξ , says that either one $W_{p,\xi}$, $p \in \mathcal{P}$, is irrational or two such values are distinct. Both alternatives contradict $W_{p,\xi} = a/b$ for every $p \in \mathcal{P}$. Thus (95) holds.

For any finite set B of bad primes, (94) yields

$$\liminf_{N \rightarrow \infty} \frac{\log \Delta_N}{N} \geq \sum_{p \in B} \frac{\log p}{p-1}.$$

Letting B increase proves the theorem. Finally, $\mathcal{E}_\xi$ is holonomic and division by $z-1$ preserves holonomy; the conjugates of the normalized coefficients b_n have exponential growth. The superexponential denominator is therefore the precise failed arithmetic-Gevrey condition. $\square$

Remark 15.2. The theorem is a global statement. At a good prime the factorial tail makes the local denominator disappear, while every fixed bad prime contributes the asymptotic rate $\log p/(p-1)$. The global-relation theorem forces the sum of these bad rates to diverge; compare the stronger residue-class results in [20].

16 A sharp p -adic Padé radius transition

Matala-aho and Zudilin use [36]

$$Q_n(z) = \sum_{k=0}^n k! \binom{n}{k}^2 z^k,$$

$$Q_n(z)\mathcal{E}_1(z) - P_n(z) = n!^2 z^{2n} \sum_{k \geq 0} (-1)^k k! \binom{n+k}{k}^2 z^k.$$

Set $q_n = Q_n(1)$, $s_n = P_n(1)$, with $q_0 = 1, s_0 = 0$, and $r_n(\alpha) = q_n \alpha - s_n$. Both Padé coordinates satisfy

$$u_{n+1} = 2(n+1)u_n - n^2 u_{n-1} \quad (n \geq 1),$$

and $r_1(\alpha) = 2\alpha - 1$. Hence

$$F_\alpha(z) = \sum_{n \geq 0} r_n(\alpha) \frac{z^n}{n!}$$

satisfies

$$(1-z)^2 F'_\alpha(z) + (z-2)F_\alpha(z) + 1 = 0. \quad (96)$$

Theorem 16.1. *For a prime p , put $\delta_p = \sum_{m \geq 0} (-1)^m m! \in \mathbb{Z}_p$. For every $\alpha \in \mathbb{Q}_p$,*

$$R_p(F_\alpha) = \begin{cases} p^{1/(p-1)}, & \alpha = \delta_p, \\ p^{-1/(p-1)}, & \alpha \neq \delta_p. \end{cases}$$

Furthermore $F_\alpha \in \mathbb{Z}_p[[z]]$ exactly in the matched case $\alpha = \delta_p$.

Proof. In the matched case the Padé remainder identity gives

$$r_n(\delta_p) = n!^2 \sum_{k \geq 0} (-1)^k k! \binom{n+k}{k}^2,$$

so $r_n(\delta_p)/n! \in n!\mathbb{Z}_p$. Thus the radius is at least $\rho_p = p^{1/(p-1)}$, and the coefficients are integral.

To see exactness, expand a hypothetical analytic solution of (96) at $z = 1$. Writing $t = z - 1$, the unique formal solution regular there is

$$\sum_{m \geq 0} m! t^m,$$

whose p -adic radius is exactly ρ_p . Since nonarchimedean discs of radius greater than one centered at 0 and 1 coincide, a radius larger than ρ_p for F_{δ_p} would contradict this local expansion.

The normalized homogeneous solution of (96) is

$$H(z) = \frac{1}{1-z} \exp\left(\frac{z}{1-z}\right) = \sum_{n \geq 0} q_n \frac{z^n}{n!}. \quad (97)$$

Therefore

$$F_\alpha - F_{\delta_p} = (\alpha - \delta_p)H.$$

For $n \equiv 0 \pmod{p}$, the formula for q_n gives $q_n \equiv 1 \pmod{p}$: terms $1 \leq k < p$ have $p \mid \binom{n}{k}$, while terms $k \geq p$ contain $p \mid k!$. It follows that H has exact radius $p^{-1/(p-1)}$. If $\alpha \neq \delta_p$, this smaller-radius term dominates F_{δ_p} , proving the second case. Along the same subsequence the coefficient valuations are unbounded below; in particular no fixed scalar makes the full series integral. Conversely the matched coefficients were already shown integral. $\square$

The preceding theorem is the $\xi = 1$ instance of the all-prime, all-parameter result below. We retain it as a transparent model calculation; the next section supplies the general Laguerre recurrence proof and removes every coprimality restriction.

17 The all-prime Padé transition and its arithmetic obstructions

Fix $0 \neq \xi \in \mathbb{Z}$ and write

$$\mathcal{E}(u) = \sum_{m \geq 0} (-1)^m m! u^m, \quad \delta_p(\xi) = \sum_{m \geq 0} (-1)^m m! \xi^m \in \mathbb{Z}_p.$$

We use the classical Padé pair [36]

$$Q_n(u) = \sum_{k=0}^n k! \binom{n}{k}^2 u^k, \quad (98)$$

$$Q_n(u)\mathcal{E}(u) - P_n(u) = n!^2 u^{2n} \sum_{k \geq 0} (-1)^k k! \binom{n+k}{k}^2 u^k. \quad (99)$$

Here $Q_0 = 1$ and $P_0 = 0$. Put

$$q_n(\xi) = Q_n(\xi), \quad s_n(\xi) = P_n(\xi), \quad F_\alpha^{(\xi)}(z) = \sum_{n \geq 0} (q_n(\xi)\alpha - s_n(\xi)) \frac{z^n}{n!}.$$

Lemma 17.1 (Padé recurrence and affine equation). *For $n \geq 1$, both $q_n(\xi)$ and $s_n(\xi)$ satisfy*

$$u_{n+1} = (1 + (2n+1)\xi)u_n - n^2 \xi^2 u_{n-1}. \quad (100)$$

Consequently $F = F_\alpha^{(\xi)}$ satisfies

$$(1 - \xi z)^2 F'(z) - (1 + \xi - \xi^2 z) F(z) + 1 = 0. \quad (101)$$

Its normalized homogeneous solution is

$$H_\xi(z) = \frac{1}{1 - \xi z} \exp\left(\frac{z}{1 - \xi z}\right) = \sum_{n \geq 0} q_n(\xi) \frac{z^n}{n!}. \quad (102)$$

Proof. The polynomial identity

$$Q_n(u) = n! u^n L_n(-1/u)$$

and the Laguerre recurrence give

$$Q_{n+1}(u) = (1 + (2n+1)u)Q_n(u) - n^2 u^2 Q_{n-1}(u).$$

The Padé determinant identity

$$D_n(u) := Q_n(u)P_{n+1}(u) - Q_{n+1}(u)P_n(u) = n!^2 u^{2n} \quad (103)$$

then yields, as a polynomial identity,

$$\begin{aligned} Q_n(u)(P_{n+1}(u) - (1 + (2n+1)u)P_n(u) + n^2 u^2 P_{n-1}(u)) \\ = D_n(u) - n^2 u^2 D_{n-1}(u) = 0. \end{aligned}$$

Thus P_n obeys the same recurrence. Evaluation at $u = \xi$ proves (100).

For $r_n = q_n(\xi)\alpha - s_n(\xi)$ one has $r_0 = \alpha$ and $r_1 = (1 + \xi)\alpha - 1$. Multiplying (100) by $z^n/n!$ and summing gives (101). Solving its homogeneous part with value 1 at $z = 0$ gives (102). $\square$

Theorem 17.2 (All-prime radius transition). *Let p be any prime, set $s = v_p(\xi)$, and let $\alpha \in \mathbb{Q}_p$. Then*

$$R_p(F_\alpha^{(\xi)}) = \begin{cases} p^{\frac{1}{p-1}+2s} = p^{1/(p-1)}|\xi|_p^{-2}, & \alpha = \delta_p(\xi), \\ p^{-1/(p-1)}, & \alpha \neq \delta_p(\xi). \end{cases} \quad (104)$$

Moreover

$$F_\alpha^{(\xi)} \in \mathbb{Z}_p[[z]] \iff \alpha = \delta_p(\xi). \quad (105)$$

If the right-hand condition fails, the valuations of the coefficients of $F_\alpha^{(\xi)}$ are unbounded below; in particular no fixed nonzero scalar makes the whole series integral.

Proof. Specializing (99) at $u = \xi$ gives

$$\frac{q_n(\xi)\delta_p(\xi) - s_n(\xi)}{n!} = n!\xi^{2n} \sum_{k \geq 0} (-1)^k k! \binom{n+k}{k}^2 \xi^k \in n!\xi^{2n}\mathbb{Z}_p. \quad (106)$$

Hence the matched radius is at least $p^{1/(p-1)+2s}$.

Let $c = 1/\xi$ and $t = z - c$. Equation (101) becomes

$$\xi^2 t^2 F'(t) + (\xi^2 t - 1)F(t) + 1 = 0. \quad (107)$$

Its unique formal solution regular at $t = 0$ is

$$F_{\text{reg}}(t) = \sum_{m \geq 0} m! \xi^{2m} t^m, \quad (108)$$

whose radius is exactly $p^{1/(p-1)+2s}$. This number is strictly larger than $|c|_p = p^s$. The lower bound from (106) therefore places c inside the disc of the matched solution, and its re-expansion at c must be (108). If its radius at 0 were larger than $p^{1/(p-1)+2s}$, the ultrametric discs with centers 0 and c and that larger radius would coincide, contradicting the exact radius of (108). This proves the first case of (104).

The coefficients of H_ξ have radius at least $p^{-1/(p-1)}$, since $q_n(\xi) \in \mathbb{Z}$. If $p \mid \xi$, then $q_n(\xi) \equiv 1 \pmod{p}$ for every n . If $p \nmid \xi$ and $n \equiv 0 \pmod{p}$, then the terms in (98) with $1 \leq k < p$ contain $p \mid \binom{n}{k}$, while those with $k \geq p$ contain $p \mid k!$; hence again $q_n(\xi) \equiv 1 \pmod{p}$. Along the indicated subsequence, Legendre's formula gives

$$\frac{1}{n} v_p \left(\frac{q_n(\xi)}{n!} \right) \longrightarrow -\frac{1}{p-1}.$$

Thus $R_p(H_\xi) = p^{-1/(p-1)}$. Finally,

$$F_\alpha^{(\xi)} - F_{\delta_p(\xi)}^{(\xi)} = (\alpha - \delta_p(\xi))H_\xi.$$

When the scalar is nonzero, the term on the right has strictly smaller radius than the matched series, proving the second case of (104). Formula (106) proves integrality in the matched case. In the unmatched case, on the same unit subsequence,

$$v_p \left((\alpha - \delta_p(\xi)) \frac{q_n(\xi)}{n!} \right) = v_p(\alpha - \delta_p(\xi)) - v_p(n!) \longrightarrow -\infty,$$

whereas the matched coefficient is integral. This proves (105) and the final assertion. $\square$

Proposition 17.3 (Local denominator–radius equivalence). *Put*

$$S_n(\xi) = \sum_{k=0}^n (-1)^k k! \xi^k, \quad b_n = \frac{\alpha - S_n(\xi)}{n!},$$

and, for $\alpha \in \mathbb{Q}_p$, define

$$d_{p,N} = \max_{0 \leq n \leq N} \max\{0, -v_p(b_n)\}.$$

Then the following are equivalent:

$$\alpha = \delta_p(\xi), \quad b_n \in \mathbb{Z}_p \ (n \geq 0), \quad d_{p,N} = 0 \ (N \geq 0), \quad R_p(F_\alpha^{(\xi)}) > 1. \quad (109)$$

If $\alpha \neq \delta_p(\xi)$ and $c_p = v_p(\alpha - \delta_p(\xi))$, then, for all sufficiently large N ,

$$d_{p,N} = v_p(N!) - c_p, \quad \lim_{N \rightarrow \infty} \frac{d_{p,N}}{N} = \frac{1}{p-1}. \quad (110)$$

Proof. In the matched case,

$$b_n = \sum_{k \geq n+1} (-1)^k \frac{k!}{n!} \xi^k \in \mathbb{Z}_p.$$

In the unmatched case, $S_n(\xi) \rightarrow \delta_p(\xi)$ and ultrametric stability gives $v_p(\alpha - S_n(\xi)) = c_p$ for all sufficiently large n . Hence $-v_p(b_n) = v_p(n!) - c_p$ eventually. This is nondecreasing apart from plateaux and ultimately exceeds the maximum contributed by the finitely many earlier terms, proving (110). The radius equivalence is Theorem 17.2. $\square$

Corollary 17.4 (Relative adelic radius defect). *Let $\alpha = a/b \in \mathbb{Q}$ in lowest terms and define*

$$\mathcal{B}(\alpha, \xi) = \{p : p \nmid b\xi, \ \alpha \neq \delta_p(\xi)\}.$$

Then

$$\sum_{p \in \mathcal{B}(\alpha, \xi)} \frac{\log p}{p-1} = +\infty. \quad (111)$$

Consequently, with $\rho_{p,\xi} = p^{1/(p-1)} |\xi|_p^{-2}$,

$$\prod_{p \nmid b\xi} \frac{R_p(F_\alpha^{(\xi)})}{\rho_{p,\xi}} = 0, \quad (112)$$

where the product is the limit over finite sets of primes.

Proof. Suppose that the sum in (111) were finite, and let $\mathcal{P}$ be the primes $p \nmid b\xi$ for which $\delta_p(\xi) = \alpha$. The complement of $\mathcal{P}$ then has finite $\sum_p \log p/(p-1)$. Since

$$\prod_{p \in \mathcal{P}} |n!|_p^2 = \frac{1}{(n!)^2} \prod_{p \notin \mathcal{P}} p^{2v_p(n!)} \leq \frac{C^n}{(n!)^2}$$

for a fixed $C > 0$, the hypothesis of the global-relation theorem of Matala-aho and Zudilin [36] is satisfied. That theorem forbids all values $\delta_p(\xi)$, $p \in \mathcal{P}$, from being the same rational number, a contradiction. Thus (111) holds. For $p \nmid b\xi$, Theorem 17.2 gives a local ratio 1 at a matched prime and $p^{-2/(p-1)}$ at an unmatched prime. Taking logarithms proves (112). $\square$

Remark 17.5 (What the adelic product does and does not say). The product in (112) is normalized by the matched local benchmark. It is not itself an application of the Borel–Dwork–Pólya–Bertrandias rationality criterion. Rather, it records why the direct application fails here: for rational α there are infinitely many bad integral lattices, with divergent total radius defect. It does not exclude a different auxiliary series or different adelic domains; compare the original Dwork criterion [19] and its capacity-theoretic extension [6].

17.1 Frobenius and Borel obstructions

Proposition 17.6 (Exact p -curvature). *Let $u = 1 - \xi z$. The homogeneous equation in Lemma 17.1 is the rank-one connection*

$$\nabla_{\partial} = \partial - a_{\xi}(z), \quad a_{\xi}(z) = \frac{\xi}{u} + \frac{1}{u^2}.$$

At every prime p , its reduction has p -curvature

$$\psi_p(\partial) = -\frac{1}{(1 - \bar{\xi}z)^{2p}}, \quad (113)$$

up to the conventional overall sign. In particular it is nonzero, hence not nilpotent, at every prime.

Proof. In characteristic p , the rank-one Jacobson formula gives

$$(\partial - a_{\xi})^p = -(\partial^{p-1}a_{\xi} + a_{\xi}^p),$$

because $\partial^{[p]} = 0$ on $\mathbb{F}_p(z)$. Wilson's congruence gives

$$\partial^{p-1}(\xi u^{-1}) = -\xi^p u^{-p}, \quad \partial^{p-1}(u^{-2}) = p! \xi^{p-1} u^{-p-1} = 0,$$

whereas Frobenius gives

$$a_{\xi}^p = \xi^p u^{-p} + u^{-2p}.$$

Substitution proves (113). □

Corollary 17.7 (Global-nilpotence no-go). *The minimal rank-one operator annihilating H_{ξ} is not globally nilpotent. Consequently it is not a subquotient of an ordinary smooth proper Gauss–Manin connection and is not a G -operator in the usual globally nilpotent category.*

Proof. For a rank-one connection, nilpotence of p -curvature is equivalent to its vanishing. Proposition 17.6 therefore rules out global nilpotence. The stated consequences are the standard Gauss–Manin/global-nilpotence theorems of Katz [30] and the André–Chudnovsky–Katz theory of G -operators [2]. □

Remark 17.8. This is deliberately a statement about the ordinary globally nilpotent category. It does not rule out every notion of irregular exponential geometric origin. Moreover the connection is independent of α : p -curvature cannot distinguish the matched initial value. Matching is a property of an overconvergent horizontal lift, not of the differential module alone.

Proposition 17.9 (Dieudonné–Dwork defect). *Put $w(z) = z/(1 - \xi z)$. For every prime p ,*

$$[z^p](w(z^p) - pw(z)) = 1 - p\xi^{p-1} \notin p\mathbb{Z}_p. \quad (114)$$

Consequently neither $\exp(w(z))$ nor $H_{\xi}(z)$ belongs to $\mathbb{Z}_p[[z]]$.

Proof. The Dieudonné–Dwork criterion [16] says that, for $S \in z\mathbb{Q}_p[[z]]$,

$$\exp(S) \in 1 + z\mathbb{Z}_p[[z]] \iff S(z^p) - pS(z) \in pz\mathbb{Z}_p[[z]].$$

Since $w(z) = \sum_{m \geq 1} \xi^{m-1} z^m$, equation (114) follows immediately and violates the criterion. Finally $(1 - \xi z)^{-1}$ and its inverse both lie in $\mathbb{Z}_p[[z]]$, so multiplication by this factor cannot repair integrality. □

Proposition 17.10 (Fixed Borel logarithm). *Let*

$$\frac{E_\xi(z) - \alpha}{z - 1} = \sum_{n \geq 0} n! b_n z^n, \quad E_\xi(z) = \sum_{n \geq 0} (-1)^n n! \xi^n z^n,$$

and let $\mathcal{B}_1$ denote the order-one formal Borel transform. Then

$$B_{\alpha, \xi}(t) := \mathcal{B}_1 \left(\frac{E_\xi - \alpha}{z - 1} \right) (t) = \sum_{n \geq 0} b_n t^n$$

satisfies

$$B'_{\alpha, \xi} - B_{\alpha, \xi} = \frac{\xi}{(1 + \xi t)^2}, \quad B_{\alpha, \xi}(0) = \alpha - 1. \quad (115)$$

If $t_0 = -1/\xi$ and $x = t - t_0$, then on every local complex branch

$$B_{\alpha, \xi}(t) = -\frac{e^x}{\xi x} - \frac{e^x}{\xi} \log x + e^x h_{\alpha, \xi}(x), \quad h_{\alpha, \xi} \in \mathbb{C}\{x\}. \quad (116)$$

In particular no choice of α makes this Borel transform meromorphic at t_0 , and none makes it rational.

Proof. Coefficient comparison gives $b_n = (\alpha - S_n(\xi))/n!$ and hence

$$(n+1)b_{n+1} = b_n + (n+1)(-1)^n \xi^{n+1}.$$

Summation proves (115), equivalently

$$B_{\alpha, \xi}(t) = e^t \left(\alpha - 1 + \int_0^t \frac{\xi e^{-v}}{(1 + \xi v)^2} dv \right).$$

At $t = t_0 + x$, equation (115) reads $B' - B = 1/(\xi x^2)$. Equivalently,

$$(e^{-x} B)' = \frac{e^{-x}}{\xi x^2}.$$

Since $e^{-x} x^{-2} = x^{-2} - x^{-1} +$ a holomorphic germ, integration and multiplication by e^x give (116). The homogeneous correction Ce^t , which is the only dependence on α , is holomorphic and cannot cancel either singular term. $\square$

Corollary 17.11 (Polynomial reduction in the matched case). *If $p \nmid \xi$, then*

$$\overline{F_{\delta_p(\xi)}^{(\xi)}}(z) = \sum_{m=0}^{p-1} m! \bar{\xi}^{2m} (z - \bar{\xi}^{-1})^m \in \mathbb{F}_p[z]. \quad (117)$$

If $p \mid \xi$, then $\overline{F_{\delta_p(\xi)}^{(\xi)}}(z) = 1$. Thus the matched reduction has a finite-support, hence p -automatic, coefficient sequence. In the unmatched case no fixed scalar gives an integral series to reduce.

Proof. For $p \nmid \xi$, reduce the local expansion (108); all terms with $m \geq p$ vanish modulo p , giving (117). If $p \mid \xi$, equation (106) shows that every positive-degree coefficient is divisible by p , while $F_{\delta_p(\xi)}^{(\xi)}(0) = \delta_p(\xi) \equiv 1 \pmod{p}$. The last two claims follow from Theorem 17.2; the automaticity statement is also the elementary polynomial case of Christol's theorem [10]. $\square$

Remark 17.12 (Kurepa specialization). For $\xi = -1$,

$$\delta_p(-1) \equiv \sum_{m=0}^{p-1} m! \equiv -1 \pmod{p}, \quad \overline{F_{\delta_p(-1)}^{(-1)}}(z) = \sum_{m=0}^{p-1} m! (z+1)^m.$$

Thus Kurepa's conjecture is equivalent to the assertion that the constant term of this matched polynomial is nonzero for every odd prime. This is an exact reformulation, not a proof of the conjecture.

18 Bell–Gould prime-power filters and mixed integrals

The purpose of this section is twofold. First we isolate an exact integer-valued-polynomial mechanism which annihilates the Bell and Gould functionals simultaneously modulo an arbitrary integer. The boundary term in the Gould functional is retained throughout; omitting it makes the prime-power lift invalid. Second, we match the resulting reciprocal integrals with logarithmic moments. The matching places the answer in the pure E -value direction $e^\lambda \text{Ein}(\lambda)$, where Theorem 5.1 turns the construction into an unconditional algebraic-independence theorem.

18.1 The Bell–Gould adjoint calculus

Let $(B_n)_{n \geq 0}$ be the Bell sequence and let $(G_n)_{n \geq 0}$ be the Gould sequence, normalized by

$$B_0 = 1, \quad B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k,$$

and

$$G_0 = 0, \quad G_1 = 1, \quad G_{n+1} = \sum_{k=0}^n \binom{n}{k} G_k \quad (n \geq 1).$$

Define the two integral linear functionals on $\mathbb{Z}[x]$ by

$$\mathfrak{B}(x^n) = B_n, \quad \mathfrak{G}(x^n) = G_n.$$

For $r \geq 1$, write $(x)_r = x(x-1) \cdots (x-r+1)$, and put

$$\Theta_r(f) = \sum_{j=0}^{r-1} (-1)^{r-1-j} (r-1-j)! f(j). \quad (118)$$

Lemma 18.1 (Bell–Gould adjoints). *For every $r \geq 1$ and $f \in \mathbb{Z}[x]$,*

$$\mathfrak{B}((x)_r f(x)) = \mathfrak{B}(f(x+r)), \quad (119)$$

$$\mathfrak{G}((x)_r f(x)) = \mathfrak{G}(f(x+r)) + \Theta_r(f). \quad (120)$$

Proof. For $r = 1$, the two identities are precisely the defining binomial recurrences, with the exceptional initial value $G_1 = 1$ producing the term $f(0)$ in the second identity. Iteration proves (119). For the Gould identity, suppose it holds for r . Apply the case $r = 1$ to $h(x) = (x-1)_r f(x)$, and then apply the induction hypothesis to $f(x+1)$. The new boundary contribution is

$$(-1)^r r! f(0) + \sum_{j=0}^{r-1} (-1)^{r-1-j} (r-1-j)! f(j+1),$$

which is $\Theta_{r+1}(f)$. This completes the induction. $\square$

Let

$$\tau_n(x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} (-x)^k \quad (121)$$

be the signed Touchard polynomial. The elementary reciprocal integral is

$$\mathcal{R}_n := \int_0^\infty \frac{e^{-x} \tau_n(x)}{1+x} dx = B_n \delta - G_n. \quad (122)$$

Indeed, the exponential generating function of the left side is

$$\int_0^\infty \frac{e^{-e^z x}}{1+x} dx = e^{e^z} E_1(e^z).$$

The function $\delta \exp(e^z - 1)$ minus this generating function has derivative equal to itself multiplied by e^z , plus 1, and hence has coefficients (G_n) with the displayed normalization.

18.2 A boundary-aware prime-power lift

Fix a prime p . Put

$$D_p(x) = x^p - x.$$

There is a $G_p \in \mathbb{Z}[x]$ such that

$$D_p(x) = (x)_p + pG_p(x). \quad (123)$$

To see this, subtract the two monic degree- p polynomials. The difference has degree at most $p-1$ and vanishes at all p elements of $\mathbf{F}_p$, so every coefficient is divisible by p .

For $r \geq 1$, set

$$m_{p,r} = \min\{m \geq 0 : v_p(m!) \geq r\}.$$

Choose $a \in \mathbb{Z}$, put $W_{p,r,a}(x) = (x-a)_{m_{p,r}}$, and define

$$\mathcal{L}_p[P](x) = D_p(x)P(x) - P(x+p). \quad (124)$$

For a polynomial $P \in \mathbb{Z}[x]$, use the abbreviations

$$\begin{aligned} \text{FD}_r(P) &: \iff P(k) \in p^r \mathbb{Z} \quad (k \in \mathbb{Z}), \\ \text{AN}_s(P) &: \iff \mathfrak{B}(Pf), \mathfrak{G}(Pf) \in p^s \mathbb{Z} \quad (f \in \mathbb{Z}[x]). \end{aligned}$$

Lemma 18.2 (One-step lift). *Suppose that $1 \leq s < r$ and that P satisfies $\text{FD}_r(P)$ and $\text{AN}_s(P)$. Then $Q = \mathcal{L}_p[P]$ satisfies $\text{FD}_r(Q)$ and $\text{AN}_{s+1}(Q)$.*

Proof. First note the shifted-annihilator consequence of Lemma 18.1. If $r \geq s$, then for every $g \in \mathbb{Z}[x]$,

$$\begin{aligned} \mathfrak{B}(P(x+p)g(x)) &= \mathfrak{B}(P(x)(x)_p g(x-p)), \\ \mathfrak{G}(P(x+p)g(x)) &= \mathfrak{G}(P(x)(x)_p g(x-p)) - \Theta_p(P(x)g(x-p)). \end{aligned}$$

The first terms on the right are divisible by p^s , and every summand in the boundary term is divisible by p^r . Thus both shifted functionals are divisible by p^s .

The fixed-divisor assertion for Q follows immediately from (124). Write

$$f(x+p) - f(x) = p\Delta_p f(x), \quad \Delta_p f \in \mathbb{Z}[x].$$

Using (123) and Lemma 18.1 gives

$$\begin{aligned} \mathfrak{B}(Qf) &= p\mathfrak{B}(P(x+p)\Delta_p f) + p\mathfrak{B}(PG_p f), \\ \mathfrak{G}(Qf) &= p\mathfrak{G}(P(x+p)\Delta_p f) + p\mathfrak{G}(PG_p f) + \Theta_p(Pf). \end{aligned}$$

The first two terms in each line are divisible by p^{s+1} . The last term in the second line is divisible by p^r , hence by p^{s+1} . $\square$

Theorem 18.3 (Common Bell–Gould prime-power annihilator). *For every prime p , $r \geq 1$, and $a \in \mathbb{Z}$, the polynomial*

$$S_{p,r}^{(a)}(x) = \mathcal{L}_p^{r-1} [W_{p,r,a}(x)(D_p(x) - 1)] \quad (125)$$

is monic and primitive, has degree $m_{p,r} + rp$, and satisfies

$$p^r \mid \mathfrak{B}(S_{p,r}^{(a)}f), \quad p^r \mid \mathfrak{G}(S_{p,r}^{(a)}f) \quad (f \in \mathbb{Z}[x]). \quad (126)$$

Proof. Put $P_0 = W_{p,r,a}(D_p - 1)$. Since

$$W_{p,r,a}(k) = m_{p,r}! \binom{k-a}{m_{p,r}} \quad (k \in \mathbb{Z}),$$

the polynomial P_0 satisfies $\text{FD}_r(P_0)$. For $h = W_{p,r,a}f$, equations (119)–(123) give

$$\begin{aligned} \mathfrak{B}(P_0f) &= \mathfrak{B}(h(x+p) - h(x)) + p\mathfrak{B}(G_ph), \\ \mathfrak{G}(P_0f) &= \mathfrak{G}(h(x+p) - h(x)) + p\mathfrak{G}(G_ph) + \Theta_p(h). \end{aligned}$$

The difference $h(x+p) - h(x)$ belongs to $p\mathbb{Z}[x]$, while every value $h(j)$ occurring in $\Theta_p(h)$ is divisible by p^r . Hence P_0 satisfies $\text{AN}_1(P_0)$. Iterating Lemma 18.2 $r-1$ times proves (126).

The starting polynomial is monic of degree $m_{p,r} + p$. The operator $\mathcal{L}_p$ preserves monicity and raises the degree by p , since its first summand has degree $p + \deg P$ and its second has degree $\deg P$. This proves the degree formula. A monic integral polynomial is primitive. $\square$

The carrier $W_{p,r,a}$ has the smallest possible degree among primitive integer polynomials whose fixed divisor is divisible by p^r : the fixed divisor of a primitive degree- d polynomial divides $d!$, while $W_{p,r,a}$ attains $d = m_{p,r}$; see [47]. This is an optimality statement for the fixed-divisor carrier, not for all possible simultaneous Bell–Gould annihilators.

18.3 Coefficientwise CRT for an arbitrary modulus

Theorem 18.4 (Arbitrary-modulus Bell–Gould filter). *Let*

$$M = \prod_{p \mid M} p^{r_p} \geq 2, \quad d_p = m_{p,r_p} + r_p p, \quad D_M = \max_{p \mid M} d_p.$$

There is a monic primitive $C_M \in \mathbb{Z}[x]$, of degree D_M , such that

$$M \mid \mathfrak{B}(C_M f), \quad M \mid \mathfrak{G}(C_M f) \quad (f \in \mathbb{Z}[x]). \quad (127)$$

Every nonleading coefficient of C_M may be chosen to have absolute value at most $M/2$.

Proof. For every $p \mid M$, choose an integer a_p and put

$$T_p(x) = x^{D_M - d_p} S_{p,r_p}^{(a_p)}(x).$$

All T_p are monic of degree D_M . Apply the Chinese remainder theorem independently to their nonleading coefficients and set the leading coefficient equal to 1. This gives

$$C_M \equiv T_p \pmod{p^{r_p}} \quad (p \mid M),$$

with centered representatives bounded by $M/2$. Multiplication by a power of x preserves (126); moreover $C_M - T_p \in p^{r_p}\mathbb{Z}[x]$. Since the two functionals take integral values on $\mathbb{Z}[x]$, both values in (127) are divisible by every p^{r_p} , and hence by M . $\square$

Let (E) denote the forward shift in (n) ,

$$(Ea)_n = a_{n+1},$$

acting coefficientwise on polynomial-valued sequences.

Corollary 18.5 (Factorial specialization). *For $M = N!$, $N \geq 2$, one may take a monic common filter of degree D_N satisfying*

$$3N - 2\lfloor \log_2 N \rfloor - 3 \leq D_N \leq 3N. \quad (128)$$

Consequently, for every $n \geq 0$, there are integers $P_{N,n}, Q_{N,n}$ such that

$$\frac{1}{N!} \int_0^\infty \frac{e^{-x} [C_{N!}(E)\tau_n](x)}{1+x} dx = Q_{N,n}\delta - P_{N,n}. \quad (129)$$

Proof. If $r_p = v_p(N!)$, then

$$m_{p,r_p} = p\lfloor N/p \rfloor, \quad d_p = p\lfloor N/p \rfloor + pv_p(N!).$$

The quotient of $N!$ by $(p\lfloor N/p \rfloor)!$ is a p -adic unit, and deleting the last multiple of p lowers the valuation, proving the first equality. Also

$$d_p \leq N + p \sum_{j \geq 1} \frac{N}{p^j} \leq 3N.$$

For $p = 2$, Legendre's formula $v_2(N!) = N - s_2(N)$, together with $s_2(N) \leq \lfloor \log_2 N \rfloor + 1$, gives the lower bound in (128). Finally apply (122) and Theorem 18.4 with $f = x^n$. $\square$

The congruence tradition begins with Touchard's classical congruence for Bell numbers and its polynomial variants [53, 52]; the adjoint language belongs to the classical umbral calculus [48], and the companion Gould recurrence was studied systematically by Gould and Quaintance [45]. The preceding proof records a particular boundary-aware simultaneous lift and its arbitrary-modulus CRT packaging. No claim of historical priority for that exact packaging is made without a separate search through the older congruence and umbral literature.

18.4 Matched mixed moments

Retain the pure exponential field

$$\mathcal{E}_{\text{exp}} = \overline{\mathbb{Q}}(e^\alpha : \alpha \in \overline{\mathbb{Q}}).$$

For $U(t) = \sum_k u_k t^k \in \overline{\mathbb{Q}}[t]$, define

$$\beta(U) = \sum_k u_k k!, \quad R(U) = \sum_k u_k k! H_k. \quad (130)$$

For $\lambda \in \overline{\mathbb{Q}}_{>0}$ and $V(t) = \sum_k v_k t^k \in \overline{\mathbb{Q}}[t]$, put

$$S_\lambda(V) = \sum_{k \geq 1} v_k \sum_{j=0}^{k-1} (-\lambda)^j (k-1-j)!. \quad (131)$$

Theorem 18.6 (Matched mixed-integral independence). *Let $\lambda_1, \dots, \lambda_s$ be distinct positive algebraic numbers. For each i , choose $U_i, V_i \in \overline{\mathbb{Q}}[t]$ such that*

$$V_i(-\lambda_i) = -\beta(U_i) \neq 0, \quad (132)$$

and define

$$\mathcal{J}_i = e^{\lambda_i} \int_0^\infty e^{-t} U_i(t) \log\left(\frac{t}{\lambda_i}\right) dt + \int_0^\infty \frac{e^{-t} V_i(t)}{t + \lambda_i} dt. \quad (133)$$

Then $\mathcal{J}_1, \dots, \mathcal{J}_s$ are algebraically independent over $\mathcal{E}_{\text{exp}}$.

Proof. The logarithmic moments are

$$\int_0^\infty e^{-t} t^k \log(t/\lambda) dt = k!(H_k - \gamma - \log \lambda). \quad (134)$$

If $J_k(\lambda) = \int_0^\infty e^{-t} t^k / (t + \lambda) dt$, then

$$J_0(\lambda) = e^\lambda E_1(\lambda), \quad J_k(\lambda) = (k-1)! - \lambda J_{k-1}(\lambda).$$

Therefore

$$\int_0^\infty \frac{e^{-t} V(t)}{t + \lambda} dt = S_\lambda(V) + V(-\lambda) e^\lambda E_1(\lambda). \quad (135)$$

Equations (132)–(135) and $\text{Ein}(\lambda) = \gamma + \log \lambda + E_1(\lambda)$ give the exact identity

$$\boxed{\mathcal{J}_i = e^{\lambda_i} R(U_i) + S_{\lambda_i}(V_i) - e^{\lambda_i} \beta(U_i) \text{Ein}(\lambda_i).} \quad (136)$$

The constant term and the nonzero coefficient of $\text{Ein}(\lambda_i)$ lie in $\mathcal{E}_{\text{exp}}$. Thus the displayed values are obtained from the distinct $\text{Ein}(\lambda_i)$ by an invertible diagonal affine transformation over $\mathcal{E}_{\text{exp}}$. Theorem 5.1 proves the claim. $\square$

There is also an exact relation-ideal version. For a fixed λ_i , let (U_{ij}, V_{ij}) , $1 \leq j \leq r_i$, satisfy (132), and put

$$a_{ij} = e^{\lambda_i} \beta(U_{ij}), \quad c_{ij} = e^{\lambda_i} R(U_{ij}) + S_{\lambda_i}(V_{ij}).$$

If T_{ij} is the coordinate corresponding to $\mathcal{J}_{ij}$, then the complete relation ideal over $\mathcal{E}_{\text{exp}}$ is generated by

$$a_{i1}(T_{ij} - c_{ij}) - a_{ij}(T_{i1} - c_{i1}), \quad 1 \leq i \leq s, \quad 2 \leq j \leq r_i. \quad (137)$$

Indeed, $\mathcal{J}_{ij} - c_{ij} = -a_{ij} \text{Ein}(\lambda_i)$, and the s remaining Ein -coordinates are algebraically independent.

The smallest specialization already gives a countable independent family. Taking $U = 1$ and $V = -1$ yields

$$\boxed{\mathcal{J}(\lambda) = e^\lambda \int_0^\infty e^{-t} \log(t/\lambda) dt - \int_0^\infty \frac{e^{-t}}{t + \lambda} dt = -e^\lambda \text{Ein}(\lambda).} \quad (138)$$

Hence $\{\mathcal{J}(\lambda) : \lambda \in \overline{\mathbb{Q}}_{>0}\}$ is algebraically independent over $\mathcal{E}_{\text{exp}}$, in the finite-subset sense.

Corollary 18.7 (Filtered Touchard integrals). *For every $n \geq 0$,*

$$\mathcal{R}_n - eB_n \int_0^\infty e^{-t} \log t \, dt = eB_n \operatorname{Ein}(1) - G_n \quad (139)$$

is transcendental over $\mathcal{E}_{\exp}$.

More generally, let $P \in \mathbb{Z}[X]$, and put

$$q_n = P(E)B_n, \quad p_n = P(E)G_n.$$

If $q_n \neq 0$, then

$$\int_0^\infty \frac{e^{-x}[P(E)\tau_n](x)}{1+x} dx - eq_n \int_0^\infty e^{-t} \log t \, dt = q_n e \operatorname{Ein}(1) - p_n \quad (140)$$

is transcendental over $\mathcal{E}_{\exp}$. If P is monic, then $q_n > 0$ and $q_n \rightarrow \infty$ for all sufficiently large n .

Proof. The identities follow from (122) and $\int_0^\infty e^{-t} \log t \, dt = -\gamma$. Their right sides are nonconstant affine expressions in $\operatorname{Ein}(1)$ over $\mathcal{E}_{\exp}$, so they are transcendental.

For the final assertion, Dobiński's formula gives

$$q_n = e^{-1} \sum_{k \geq 0} \frac{k^n P(k)}{k!}.$$

A monic real polynomial is positive at all sufficiently large nonnegative integers. Its negative terms in the displayed sum therefore have finite support. Choose a positive integer L beyond that support with $P(L) > 0$. The single positive term at L eventually dominates the finite negative sum, whose bases are all smaller than L . It follows that $q_n > 0$ eventually; choosing $L \geq 2$ also gives $q_n \rightarrow \infty$. $\square$

In particular, for the polynomial C_M of Theorem 18.4, the two integers q_n, p_n are divisible by M . Dividing (140) by M therefore gives, for all sufficiently large n , an explicit normalized mixed integral which is transcendental over $\mathcal{E}_{\exp}$ and whose $\operatorname{Ein}(1)$ -coefficient tends to infinity. This closes nonvanishing and coefficient growth in the matched $e\gamma + \delta = e \operatorname{Ein}(1)$ direction. It does not prove nonvanishing or smallness for the unpaired δ -linear form and therefore gives no irrationality or transcendence theorem for δ itself.

19 Hankel determinants and critical capacity

Let

$$D_n(x) = \det(H_{i+j} - x)_{0 \leq i, j \leq n}.$$

Theorem 19.1. *For every $n \geq 0$ and $x \in \mathbb{C}$,*

$$D_n(x) = (-1)^n \frac{2H_n - x}{n! \prod_{j=1}^n \binom{2j}{j} \binom{2j-1}{j}}.$$

For fixed x , $|D_n(x)|^{1/n^2} \rightarrow 1/4$.

Proof. Starting from the bottom, replace row i by row i minus row $i - 1$ for $1 \leq i \leq n$. The bottom n rows become

$$\left(\frac{1}{i+j} \right)_{1 \leq i \leq n, 0 \leq j \leq n}.$$

Its null vector is the coefficient vector $(c_0, \dots, c_n)$ of the shifted Legendre polynomial

$$P_n(2t-1) = \sum_{j=0}^n c_j t^j,$$

because $\int_0^1 t^{i-1} P_n(2t-1) dt = 0$ for $1 \leq i \leq n$. The normalizations give

$$\sum_{j=0}^n c_j = 1, \quad \sum_{j=0}^n c_j H_j = 2H_n.$$

Laplace expansion along the first row is consequently a constant multiple of $2H_n - x$. The cofactor of its last entry is the n -by- n Hilbert determinant, while $c_n = \binom{2n}{n}$. The Cauchy determinant formula then gives the displayed constant

$$\frac{(-1)^n}{n! \prod_{j=1}^n \binom{2j}{j} \binom{2j-1}{j}}.$$

Finally Stirling's formula yields

$$\log \left(n! \prod_{j=1}^n \binom{2j}{j} \binom{2j-1}{j} \right) = (\log 4)n^2 + O(n \log n).$$

Since $2H_n - x$ has subexponential size and is nonzero for all sufficiently large n , the limit follows. $\square$

The generating function is

$$B_x(z) = \sum_{n \geq 0} (H_n - x) z^n = \frac{-x - \log(1-z)}{1-z}. \quad (141)$$

After the moment transformation implicit in the row differences, the archimedean singular compact is $[0, 1]$, whose logarithmic capacity is $1/4$. Thus the determinant reaches, rather than beats, the extremal capacity scale. The parameter x enters the Hankel matrix through the rank-one perturbation $-x \mathbf{1} \mathbf{1}^T$ and changes only the $O(n \log n)$ part. Consequently this particular Pólya–Bertrandias/Hankel normalization has no strict inequality left with which to distinguish $x = \gamma$. This is a statement about this normalization, not a no-go theorem for every possible capacity construction.

20 Positive mixed Hankel determinants and finite arithmetic reconstruction

Throughout this section put $K = \mathcal{E}_{\text{exp}}$.

Put $H_0 = 0$, $H_m = \sum_{j=1}^m 1/j$, and define

$$G_m = \int_0^\infty x^m e^{-x} (x - \log x) dx = m!(\gamma + m + 1 - H_m), \quad (142)$$

$$I_{k+1} = \int_0^1 t^k \frac{e^{1-1/t}}{t} dt = \int_0^\infty \frac{e^{-x}}{(1+x)^{k+1}} dx. \quad (143)$$

Integration by parts gives $I_{m+1} = (1 - I_m)/m$. Hence

$$k!I_{k+1} = A_k + (-1)^k\delta, \quad A_0 = 0, \quad A_k = (k-1)! - A_{k-1} \in \mathbb{Z}. \quad (144)$$

For $d \geq 1$ set

$$M_k^{(d)} = G_{k+d-1}I_{k+1}, \quad \mathcal{P}_d(\gamma, \delta) = \det(M_{i+j}^{(d)})_{0 \leq i, j < d}. \quad (145)$$

Theorem 20.1 (Positive full-degree mixed determinant). *For every $d \geq 1$,*

$$\mathcal{P}_d(\gamma, \delta) \in \mathbb{Q}[\gamma, \delta], \quad \mathcal{P}_d(\gamma, \delta) > 0.$$

Its total degree is exactly $2d$, and its top mixed coefficient is

$$[\gamma^d \delta^d] \mathcal{P}_d = (-1)^{d(d-1)/2} ((d-1)!)^d \neq 0. \quad (146)$$

More precisely, it has the Heine representation

$$\begin{aligned} \mathcal{P}_d &= \frac{1}{d!} \int_{((0, \infty) \times (0, 1))^d} \prod_{1 \leq i < j \leq d} (x_i t_i - x_j t_j)^2 \\ &\quad \times \prod_{\ell=1}^d x_\ell^{d-1} e^{-x_\ell} (x_\ell - \log x_\ell) \frac{e^{1-1/t_\ell}}{t_\ell} dx_\ell dt_\ell. \end{aligned} \quad (147)$$

Proof. The product variable $y = xt$ under the positive measure

$$x^{d-1} e^{-x} (x - \log x) \frac{e^{1-1/t}}{t} dx dt$$

on $(0, \infty) \times (0, 1)$ has k th moment $M_k^{(d)}$. Andreief's identity gives (147). Since $x - \log x \geq 1$ and the product measure is non-atomic with infinite support, the Vandermonde is nonzero almost everywhere off a null set; the integral is therefore strictly positive.

Equations (142) and (144) prove polynomial membership and the upper bound $2d$ for the total degree. The coefficient of $\gamma\delta$ in $M_k^{(d)}$ is

$$(-1)^k \frac{(k+d-1)!}{k!} = (-1)^k (k+1)_{d-1},$$

where $(x)_n$ is the rising factorial. Consequently the coefficient of $\gamma^d \delta^d$ in the determinant equals

$$\det((-1)^{i+j} (i+j+1)_{d-1})_{0 \leq i, j < d}.$$

Factoring the row and column signs has no net effect. For $p(k) = (k+1)_{d-1}$, apply the unitriangular forward-difference transform to both rows and columns. The resulting matrix has entries $D^{i+j}p(0)$; it vanishes past the anti-diagonal, while every anti-diagonal entry is $D^{d-1}p(0) = (d-1)!$. Its determinant is therefore the right side of (146). This coefficient is nonzero, so the total degree is exactly $2d$. $\square$

Proposition 20.2 (Finite reconstruction from the first two mixed determinants). *Put $U = \mathcal{P}_1$ and $V = \mathcal{P}_2$ for the determinants in Theorem 20.1. Then the extension*

$$K(\gamma, \delta)/K(U, V)$$

is algebraic of degree at most four. More explicitly,

$$U = (\gamma + 1)\delta,$$

and δ is a root of

$$2T^4 + (U - 4)T^3 + (2U^2 - 16U + 2V + 2)T^2 + (-16U^2 + 8U)T + 8U^2 = 0, \quad (148)$$

after which $\gamma = U/\delta - 1$. Consequently

$$\text{trdeg}_K K(U, V) = \text{trdeg}_K K(\gamma, \delta).$$

In particular, at least one of $\mathcal{P}_1, \mathcal{P}_2$ is transcendental over K , and the explicit number

$$\mathcal{P}_1 + i\mathcal{P}_2$$

is transcendental over K . Moreover $\mathcal{P}_1, \mathcal{P}_2$ are algebraically independent over K if and only if γ, δ are.

Proof. The formulas for G_m and I_m give $\mathcal{P}_1 = (\gamma + 1)\delta$. The explicit second determinant is

$$2\mathcal{P}_2 = -18 + 36\delta - 5\delta^2 - 24\gamma + 48\gamma\delta - 5\gamma\delta^2 - 8\gamma^2 + 16\gamma^2\delta - 2\gamma^2\delta^2.$$

Substituting $\gamma = U/\delta - 1$ and clearing δ^2 gives (148). Since $\delta > 0$, the displayed recovery of γ is valid. This proves finiteness and equality of transcendence degrees over K .

The pure exponential-base theorem makes $\eta = \text{Ein}(1) = \gamma + \delta/e$ transcendental over K . Since $e \in K$, the two numbers γ, δ cannot both be algebraic over K . Therefore U, V cannot both be algebraic over K . The field K is stable under complex conjugation. If $U + iV$ were algebraic over K , then its conjugate would be algebraic over K , and hence so would its real and imaginary parts U, V , a contradiction. The final equivalence is the case of transcendence degree two. $\square$

21 Geronimus–Markov reduction and the Laguerre–Kummer tower

This section replaces finite symbolic verification by an all-degree argument. It also separates three different statements that should not be conflated: the exact determinant identity, the classical Painlevé specialization, and the arithmetic usefulness of the resulting linear forms.

Let $d\nu$ be a positive measure of infinite support on $[0, \infty)$, with finite moments

$$m_k = \int_0^\infty x^k d\nu(x) \quad (k \geq 0).$$

Write p_j for its monic orthogonal polynomials and

$$\Delta_j = \det(m_{r+c})_{0 \leq r, c < j}, \quad \Delta_0 = 1.$$

For $s > 0$, set

$$\mu_k(s) = \int_0^\infty \frac{x^k}{x+s} d\nu(x), \quad \mathcal{D}_n(s) = \det(\mu_{r+c}(s))_{0 \leq r, c < n}.$$

Theorem 21.1 (Geronimus–Markov determinant reduction). *For every $n \geq 1$,*

$$\boxed{\mathcal{D}_n(s) = (-1)^{n-1} \Delta_{n-1} \int_0^\infty \frac{p_{n-1}(x)}{x+s} d\nu(x).} \quad (149)$$

If $F(s) = \mu_0(s)$, then equivalently

$$\begin{aligned} \mathcal{D}_n(s) &= \Delta_{n-1}(Q_{n-1}(s)F(s) - P_{n-2}(s)), \\ Q_{n-1}(s) &= (-1)^{n-1}p_{n-1}(-s) > 0, \end{aligned} \tag{150}$$

where P_{n-2} is a polynomial of degree at most $n-2$, with the convention $P_{-1} = 0$. Thus the Cauchy-transform Hankel determinant is precisely the classical Markov–Padé remainder multiplied by the preceding moment determinant.

Proof. In the defining matrix for $\mathcal{D}_n$, perform the column operations

$$C_j \leftarrow C_j + sC_{j-1}, \quad j = n-1, n-2, \dots, 1,$$

in this order. Since $\mu_k + s\mu_{k-1} = m_{k-1}$ for $k \geq 1$, the resulting columns are

$$(\mu_i)_{i=0}^{n-1}, (m_i)_{i=0}^{n-1}, \dots, (m_{i+n-2})_{i=0}^{n-1}.$$

Expansion in the first column is the functional $f \mapsto \int f(x)(x+s)^{-1}d\nu(x)$ applied to a polynomial orthogonal, with respect to $d\nu$, to $1, x, \dots, x^{n-2}$. Its leading coefficient is the cofactor $(-1)^{n-1}\Delta_{n-1}$. The polynomial is therefore $(-1)^{n-1}\Delta_{n-1}p_{n-1}$, proving (149).

Now write

$$\frac{p_{n-1}(x)}{x+s} = \frac{p_{n-1}(-s)}{x+s} + \frac{p_{n-1}(x) - p_{n-1}(-s)}{x+s}.$$

The second quotient is a polynomial of degree at most $n-2$, which gives (150). Finally, all zeros of p_{n-1} lie in $(0, \infty)$, so $(-1)^{n-1}p_{n-1}(-s) > 0$. $\square$

Remark 21.2 (Necessary scope). For an arbitrary signed or finitely supported weight, the rank-one argument only proves degree at most one in F . Exact degree one follows under the positive infinite-support assumptions above. Moreover, (150) is a structural Padé identity; it does not by itself compare arithmetic heights. Any assertion that the determinant is arithmetically weaker than Padé requires a separate estimate for Δ_{n-1} .

We now specialize to the Laguerre measure. For $a > -1$, put

$$\begin{aligned} d\nu_a(x) &= x^a e^{-x} dx, \\ D_n^{(a)}(s) &= \det \left(\int_0^\infty \frac{x^{i+j+a} e^{-x}}{x+s} dx \right)_{0 \leq i, j < n}, \\ C_n(a) &= \prod_{k=0}^{n-1} k! \Gamma(k+a+1) = \frac{G(n+1)G(n+a+1)}{G(a+1)}. \end{aligned}$$

Here and below G denotes the Barnes G -function.

Theorem 21.3 (Exact Laguerre–Kummer collapse). *For every $n \geq 1$, $a > -1$, and $s > 0$,*

$$\boxed{D_n^{(a)}(s) = C_n(a)U(n, 1-a, s).} \tag{151}$$

Consequently $y = D_n^{(a)}$ satisfies

$$sy'' + (1-a-s)y' - ny = 0. \tag{152}$$

If $\Sigma_n = s(\log D_n^{(a)})'$, then

$$s\Sigma'_n = (s+a)\Sigma_n - \Sigma_n^2 + ns. \tag{153}$$

In addition, if $u_n = U(n, 1-a, s)$ and $u_0 = 1$, then

$$n(n+a)u_{n+1} = (s+2n+a-1)u_n - u_{n-1} \quad (n \geq 1). \tag{154}$$

Proof. The monic Laguerre polynomial is $p_{n-1}(x) = (-1)^{n-1}(n-1)!L_{n-1}^{(a)}(x)$, and $\Delta_{n-1} = C_{n-1}(a)$. The Laplace representation of $(x+s)^{-1}$ and the Laguerre transform give

$$\int_0^\infty x^a e^{-(1+t)x} L_{n-1}^{(a)}(x) dx = \frac{\Gamma(n+a)}{(n-1)!} \frac{t^{n-1}}{(1+t)^{n+a}}. \quad (155)$$

Substitution in (149), followed by

$$U(n, 1-a, s) = \frac{1}{\Gamma(n)} \int_0^\infty e^{-st} \frac{t^{n-1}}{(1+t)^{n+a}} dt, \quad (156)$$

yields (151), because

$$C_{n-1}(a)\Gamma(n)\Gamma(n+a) = C_n(a).$$

Equations (152) and (153) follow from Kummer's equation. Equation (154) is the monic Laguerre three-term recurrence for its function of the second kind. It can also be checked directly from (156): integrate the derivative of

$$e^{-st}t^{n-1}(1+t)^{-n-a}.$$

For $n \geq 2$ both endpoint terms vanish and elementary rearrangement gives (154); for $n = 1$, the endpoint at zero is -1 , which is precisely the term $-u_0$. $\square$

Remark 21.4 (Literature status). The mathematical content of (151) is classical: the Stieltjes transform of a monic Laguerre polynomial is its function of the second kind and is a Kummer U -function. See, for example, Deaño–Huertas–Román [13], especially their equations (4)–(5), and DLMF [17], §13.4. The point here is to provide a complete determinant proof and to fix the normalization, not to claim a new special-function identity.

21.1 Rigorous growth and the classical Padé remainder

Theorem 21.5 (Superfactorial divergence). *For fixed $a > -1$ and $s > 0$,*

$$\frac{e^{-2s}}{3^{a+1}2^{n-1}\Gamma(n)} \leq U(n, 1-a, s) \leq \frac{1}{s\Gamma(n)}. \quad (157)$$

Consequently

$$\boxed{\log D_n^{(a)}(s) = n^2 \log n - \frac{3}{2}n^2 + O(n \log n)}, \quad D_n^{(a)}(s) \longrightarrow +\infty. \quad (158)$$

Proof. For the lower bound in (157), restrict (156) to $1 \leq t \leq 2$ and write its integrand as

$$e^{-st} \left(\frac{t}{1+t} \right)^{n-1} (1+t)^{-a-1}.$$

On this interval the three factors are bounded below by e^{-2s} , $2^{-(n-1)}$, and 3^{-a-1} , respectively. For the upper bound, the last two factors have product at most one, and $\int_0^\infty e^{-st} dt = s^{-1}$. Using

$$C_n(a) = \frac{G(n+1)G(n+a+1)}{G(a+1)}$$

and the standard Barnes- G asymptotic now gives (158); see DLMF [17, §5.17]. $\square$

For $a = 0$, let $B(s) = e^s E_1(s) = U(1, 1, s)$, so that $B(1) = \delta$, and put

$$Q_{n-1}(s) = (n-1)!L_{n-1}(-s).$$

Theorem 21.1 gives a polynomial $P_{n-2}(s) \in \mathbb{Z}[s]$ such that

$$\boxed{Q_{n-1}(s)B(s) - P_{n-2}(s) = (n-1)!^2 U(n, 1, s).} \quad (159)$$

Indeed, $(n-1)!L_{n-1}(x) \in \mathbb{Z}[x]$, polynomial division by $x+s$ stays in $\mathbb{Z}[x, s]$, and the moments of $e^{-x}dx$ are factorials. At $s = 1$, (159) is the classical Laguerre–Stieltjes Padé remainder for the Euler–Gompertz constant. In particular,

$$Q_{n-1}(1)\delta - P_{n-2}(1) \geq \frac{e^{-2}}{3} \frac{(n-1)!}{2^{n-1}} \longrightarrow +\infty. \quad (160)$$

Thus both the determinant and its underlying classical Padé linear form diverge; this is a theorem, not numerical evidence. The continued-fraction identification is classical; see Lagarias [34, §2.5] and Ernvall-Hytönen–Matala-aho–Seppälä [21].

21.2 Hausdorff moments, total positivity, and random matrices

Proposition 21.6 (Complete monotonicity and Turán inequality). *Define*

$$h_m(a, s) = m!U(m+1, 1-a, s) \quad (m \geq 0).$$

Then

$$\boxed{h_m(a, s) = \int_0^1 r^m (1-r)^{a-1} \exp\left(-\frac{sr}{1-r}\right) dr.} \quad (161)$$

Consequently $(h_m)_{m \geq 0}$ is a strict Hausdorff moment sequence. With $\Delta f_m = f_{m+1} - f_m$,

$$(-1)^k \Delta^k h_m > 0 \quad (k, m \geq 0). \quad (162)$$

It is strictly log-convex, and therefore

$$0 < n \frac{U(n+1, 1-a, s)}{U(n, 1-a, s)} < 1, \quad n \frac{U(n+1, 1-a, s)}{U(n, 1-a, s)} \nearrow 1, \quad (n \geq 1). \quad (163)$$

and

$$\frac{n}{n+1} U(n+1, 1-a, s)^2 < U(n, 1-a, s) U(n+2, 1-a, s) \quad (n \geq 1). \quad (164)$$

Proof. In (156), set $r = t/(1+t)$ to obtain (161). Formula (162) follows by inserting the factor $(1-r)^k$ under the integral. Strict log-convexity is Cauchy–Schwarz, since the representing measure is not a point mass. The successive moment ratios are strictly increasing and tend to the essential supremum of the support, which is one; this gives (163). Rewriting strict log-convexity for $h_n = n!U(n+1, 1-a, s)$ gives (164). $\square$

Proposition 21.7 (Total positivity and determinacy). *For $a > -1$ and $s > 0$, the infinite Hankel matrix*

$$\left(\int_0^\infty \frac{x^{i+j+a} e^{-x}}{x+s} dx \right)_{i,j \geq 0}$$

is strictly totally positive. Its Stieltjes moment problem is determinate.

Proof. Andreief's identity expresses a minor with increasing row exponents $r_1 < \dots < r_k$ and column exponents $c_1 < \dots < c_k$ as the integral of

$$\det(x_j^{r_i})_{i,j=1}^k \det(x_j^{c_i})_{i,j=1}^k \prod_{j=1}^k \frac{x_j^a e^{-x_j}}{x_j + s} dx_j.$$

On $0 < x_1 < \dots < x_k$, each alternant is a Vandermonde determinant times a Schur polynomial and is strictly positive. Hence every minor is positive. For determinacy, if μ_k denotes the moment sequence, then

$$\mu_{2k} \leq s^{-1} \Gamma(2k + a + 1),$$

so $\sum_k \mu_{2k}^{-1/(2k)} = \infty$; Carleman's criterion applies. $\square$

Corollary 21.8 (Laguerre-unitary-ensemble interpretation). *Let X have the n -dimensional Laguerre unitary ensemble with parameter $a > -1$, normalized so that its eigenvalue density is proportional to*

$$\prod_{i < j} (x_i - x_j)^2 \prod_{i=1}^n x_i^a e^{-x_i}.$$

Then

$$\boxed{\mathbb{E} \det(X + sI)^{-1} = U(n, 1 - a, s).} \quad (165)$$

Proof. Andreief's identity identifies the ratio of the pole-deformed and undeformed partition functions with $D_n^{(a)}(s)/C_n(a)$. Apply (151). $\square$

Equation (165) is the standard average inverse characteristic polynomial, or Laguerre function of the second kind. It is a useful interpretation, not a new random-matrix problem solved here.

21.3 The exact Painlevé V specialization and its boundary condition

The Riccati equation (153) is not itself the standard Jimbo–Miwa–Okamoto sigma form. Put

$$H_n(s) = \Sigma_n(s) + n.$$

We use the JMO convention [28, Appendix C, Eq. (C.45)]

$$(sH'')^2 = [H - sH' + 2(H')^2 + (\nu_0 + \nu_1 + \nu_2 + \nu_3)H']^2 - 4 \prod_{j=0}^3 (H' + \nu_j).$$

Direct substitution gives

$$\begin{aligned} (sH_n'')^2 &= [H_n - sH_n' + 2(H_n')^2 - (2n + a + 1)H_n']^2 \\ &\quad - 4H_n'(H_n' - 1)(H_n' - n)(H_n' - n - a). \end{aligned} \quad (166)$$

This is the JMO sigma form of P_V with

$$(\nu_0, \nu_1, \nu_2, \nu_3) = (0, -1, -n, -n - a).$$

For $a > 0$, it is the $\lambda = -1$ specialization of the deformed Laguerre theorem of Mu–Lyu [40]; for $-1 < a \leq 0$, the identification follows directly from (153). Equation (153) also shows that this specialization lies on a classical Riccati locus.

For $a = 0$, the identity $B' = B - s^{-1}$ is invariant under $B \mapsto B + ce^s$. At the level of moments this is

$$\mu_k(s) \mapsto \mu_k(s) + ce^s(-s)^k, \quad (167)$$

namely the Uvarov mass modification

$$\frac{e^{-x}}{x+s} dx \mapsto \frac{e^{-x}}{x+s} dx + ce^s \delta_{-s}.$$

The transformed determinant is another linear combination of the two Kummer solutions and remains on the same Riccati/Painlevé locus. However, it does not preserve the distinguished Stieltjes boundary when $c \neq 0$. For every $a > -1$, that boundary is

$$D_n^{(a)}(s) \sim C_n(a)s^{-n} \quad (s \rightarrow +\infty), \quad (168)$$

or equivalently

$$H_n(s) = \frac{n(n+a)}{s} + O(s^{-2}).$$

If $c < 0$, the added atom has negative mass; if $c \notin \mathbb{R}$, the functional is not a positive real measure; and if $c > 0$, positive mass is placed at $-s \notin [0, \infty)$. Thus no $c \neq 0$ preserves the Stieltjes class under discussion. Thus the differential equation without boundary data has the translation symmetry, whereas the actual Stieltjes and random-matrix solution is canonically selected. No arithmetic “no-go” statement follows from the formal shift alone. The direct B -shift applies to $a = 0$, and after reduction to B to nonnegative integral a ; it is not an action on the arbitrary-real- a family.

22 A confluent reciprocal-Gamma matching determinant

Put $H_0 = 0$, $H_m = \sum_{r=1}^m r^{-1}$, and

$$K_N(x) = \det \left(\frac{H_{i+j} - x}{(i+j)!} \right)_{0 \leq i, j \leq N}, \quad M_N = \binom{N+1}{2}.$$

Let $\mathfrak{M}_N$ be the set of partial matchings of $\{0, 1, \dots, N\}$, and write $V(\mathfrak{m})$ for the matched vertices. Define

$$\begin{aligned} h_{N,j} &= H_{N+j} - H_j + H_{N-j}, \\ u_{N,j}(x) &= x - h_{N,j}, \\ B_N &= \frac{\prod_{k=1}^N k!}{\prod_{m=N}^{2N} m!}. \end{aligned}$$

Theorem 22.1 (Reciprocal-Gamma matching formula). *For every $N \geq 1$,*

$$\boxed{K_N(x) = (-1)^{N+1+M_N} B_N Z_N(x)}, \quad (169)$$

where

$$Z_N(x) = \sum_{\mathfrak{m} \in \mathfrak{M}_N} \left(\prod_{\{j,k\} \in \mathfrak{m}} \frac{1}{(j-k)^2} \right) \left(\prod_{j \notin V(\mathfrak{m})} u_{N,j}(x) \right). \quad (170)$$

In particular $\deg K_N = N + 1$, with leading coefficient $(-1)^{N+1+M_N} B_N$.

Proof. Introduce one auxiliary variable per column and set

$$F_N(\boldsymbol{\alpha}) = \det \left(\frac{1}{\Gamma(i+j+1+\alpha_j)} \right)_{0 \leq i, j \leq N}.$$

A reciprocal-Gamma Vandermonde evaluation gives

$$F_N(\boldsymbol{\alpha}) = (-1)^{M_N} \frac{\prod_{0 \leq j < k \leq N} (k-j+\alpha_k-\alpha_j)}{\prod_{j=0}^N \Gamma(N+j+1+\alpha_j)}. \quad (171)$$

Since

$$-(\partial_{\alpha_j} + x - \gamma) \frac{1}{\Gamma(i+j+1+\alpha_j)} \Big|_{\alpha_j=0} = \frac{H_{i+j} - x}{(i+j)!},$$

we apply one such operator in every column. At $\boldsymbol{\alpha} = \mathbf{0}$, one has $F_N(\mathbf{0}) = (-1)^{M_N} B_N \neq 0$, so a local logarithm can be chosen. Equivalently, put

$$\tilde{F}_N(\boldsymbol{\alpha}) = e^{(x-\gamma) \sum_j \alpha_j} F_N(\boldsymbol{\alpha}).$$

Then

$$\prod_{j=0}^N (\partial_{\alpha_j} + x - \gamma) F_N \Big|_{\mathbf{0}} = \partial_{\alpha_0} \cdots \partial_{\alpha_N} \tilde{F}_N \Big|_{\mathbf{0}}.$$

The distinct-variable logarithmic cumulants of $\tilde{F}_N$ are

$$\begin{aligned} \kappa_{\{j\}} &= \partial_{\alpha_j} \log F_N \Big|_{\mathbf{0}} + x - \gamma = u_{N,j}(x), \\ \kappa_{\{j,k\}} &= \partial_{\alpha_j} \partial_{\alpha_k} \log F_N \Big|_{\mathbf{0}} = (j-k)^{-2}, \\ \kappa_S &= \prod_{j \in S} \partial_{\alpha_j} \log F_N \Big|_{\mathbf{0}} = 0 \quad (|S| \geq 3). \end{aligned}$$

The multivariate moment-cumulant formula therefore sums precisely over set partitions into singletons and pairs, which are the partial matchings in (170). Evaluation of $F_N(\mathbf{0})$, together with the $N+1$ minus signs, gives (169). $\square$

The matching sum has two exact cross-field interpretations. First, it is the monomer-dimer partition function of the complete graph on $\{0, \dots, N\}$, with monomer activities $u_{N,j}(x)$ and positive dimer activities $(j-k)^{-2}$. Second, let A_N be a random real skew-symmetric matrix whose upper-triangular entries are independent and centered, with

$$\mathbb{E}(A_N)_{jk}^2 = (j-k)^{-2} \quad (j < k),$$

and let $\mathbf{H}_N = \text{diag}(h_{N,0}, \dots, h_{N,N})$. Expansion by permutations gives

$$Z_N(x) = \mathbb{E} \det(xI - \mathbf{H}_N + A_N). \quad (172)$$

Indeed, every cycle of length at least three contains independent centered entries to the first power and has zero expectation, while the transpositions give the dimer factors. Formula (172) is an expected characteristic-polynomial representation, not a random-matrix limit theorem.

Theorem 22.2 (Heilmann–Lieb complex-zero strip). *Every complex zero ζ of K_N satisfies*

$$H_{2N} - H_N \leq \Re \zeta \leq 2H_N. \quad (173)$$

Proof. A direct difference calculation shows that $j \mapsto h_{N,j}$ is strictly decreasing, with

$$h_{N,0} = 2H_N, \quad h_{N,N} = H_{2N} - H_N.$$

If $\Re x > 2H_N$, every monomer activity $u_{N,j}(x) = x - h_{N,j}$ lies in the open right half-plane. If $\Re x < H_{2N} - H_N$, they all lie in the open left half-plane. The Heilmann–Lieb left–right half-plane theorem for a monomer–dimer partition function with nonnegative dimer activities [25, Theorem 4.6 and Lemma 4.7] says that $Z_N(x) \neq 0$ in either case; see also the precise multivariate restatement in [9, Theorem 10.1]. Equation (169) proves the claim. $\square$

Remark 22.3. Theorem 22.2 does not assert real-rootedness. In fact

$$K_1(x) = -\frac{2x^2 - 5x + 4}{4}$$

already has nonreal roots. Heilmann–Lieb supplies a zero-free complement of a vertical strip, not a real-rooted matching polynomial.

Proposition 22.4 (Cleared determinants below the strip). *Fix a real $x < \log 2$. If q_N is any positive integer that clears all coefficient denominators of K_N , then, for all sufficiently large N ,*

$$q_N |K_N(x)| \geq \prod_{j=0}^N (h_{N,j} - x) \longrightarrow +\infty. \quad (174)$$

Proof. For large N , $x < h_{N,N} = H_{2N} - H_N$. Hence all singleton factors $u_{N,j}(x)$ are negative. A matching with r pairs leaves $N + 1 - 2r$ singletons, so every term in (170) has the same sign $(-1)^{N+1}$. There is no cancellation, and the empty matching gives

$$|Z_N(x)| \geq \prod_{j=0}^N (h_{N,j} - x).$$

The leading coefficient of K_N is $\pm B_N$. Since $q_N K_N$ has integer coefficients, $q_N B_N$ is a positive integer and is at least one. This proves the inequality. Finally, for $j \leq N - 1$,

$$h_{N,j} \geq h_{N,N-1} = H_{2N-1} - H_{N-1} + 1 \longrightarrow 1 + \log 2,$$

whereas $h_{N,N} - x \rightarrow \log 2 - x > 0$. At least N factors are therefore eventually bounded below by a fixed number greater than one, proving divergence. $\square$

In particular, (174) applies at $x = \gamma$: an elementary estimate gives $\gamma < \log 2$. For completeness, using

$$\gamma = \sum_{k \geq 1} \left(\frac{1}{k} - \log \left(1 + \frac{1}{k} \right) \right), \quad u - \log(1 + u) < \frac{u^2}{2},$$

one obtains $\gamma < 11/8 - \log 2 < \log 2$; the last inequality follows from $\log 2 > 2(1/3 + 1/81) = 56/81 > 11/16$. Hence

$$q_N |K_N(\gamma)| \longrightarrow +\infty.$$

This is an obstruction for this cleared-determinant normalization, not an irrationality or transcendence statement about γ .

Remark 22.5 (Priority and scope). The Geronimus reduction and the Laguerre function of the second kind are classical. So are the continued fraction, the Uvarov modification, and the Painlevé V embedding, or they are direct specializations of cited results. The zero strip is a direct application of the classical Heilmann–Lieb theorem to the exact matching formula. These statements give rigorous structural closures. They also rule out the specified determinant normalizations as sources of small linear forms. They do not solve a named open problem in orthogonal polynomials, moment theory, continued fractions, isomonodromy, or random matrices. The exact priority of the differentiated reciprocal-Gamma matching formula itself requires a specialist literature search before any novelty claim.

Proposition 22.6 (Exact adelic denominator wall for the one-pole lift). *Let $M \geq N + 1$, put $m = M - N - 1$, and let*

$$K_{N,M}(z) = \frac{z^N}{N!} {}_1F_1(M; N + 1; -z), \quad \mathcal{H}[K_{N,M}](y) = \frac{y^N}{(1 + y)^M}.$$

Then

$$e^z K_{N,M}(z) = \sum_{j=0}^m (-1)^j \binom{m}{j} \frac{z^{N+j}}{(N+j)!}. \quad (175)$$

The least positive common denominator of this polynomial is exactly $(M - 1)!$. Equivalently, at every prime p its local denominator exponent is exactly

$$\max_{0 \leq j \leq m} \max \left\{ 0, -v_p \left(\frac{\binom{m}{j}}{(N+j)!} \right) \right\} = v_p((M - 1)!). \quad (176)$$

Consequently

$$\widetilde{K}_{N,M}(z) := (M - 1)! e^z K_{N,M}(z) \in \mathbb{Z}[z] \quad (177)$$

is primitive. Its coefficient height satisfies

$$\frac{(M - 1)!}{N!} \leq H(\widetilde{K}_{N,M}) \leq 2^m \frac{(M - 1)!}{N!}. \quad (178)$$

If $M = \alpha N$ with fixed $\alpha > 1$ along an integral subsequence, then

$$\log H(\widetilde{K}_{N,M}) = (\alpha - 1)N \log N + O_\alpha(N), \quad (179)$$

$$\log \left((M - 1)! \max_{y \geq 0} \frac{y^N}{(1 + y)^M} \right) = \alpha N \log N + O_\alpha(N). \quad (180)$$

In fact the restriction to a fixed ratio is unnecessary. For fixed N , the quantity

$$\mathfrak{F}_{N,M} := (M - 1)! \max_{y \geq 0} \frac{y^N}{(1 + y)^M}$$

is strictly increasing for integers $M \geq N + 1$. Hence

$$\mathfrak{F}_{N,M} \geq \mathfrak{F}_{N,N+1} = N! \frac{N^N}{(N + 1)^{N+1}} = \exp(N \log N - N + O(\log N)). \quad (181)$$

Thus the archimedean exponential decay of the one-pole bump is defeated by an exact all-prime factorial denominator for every possible choice of the pole order $M = M(N) \geq N + 1$. This closes this scalar rational-pole family, but it does not cover coupled determinants or nonseparable Borel–Rodrigues kernels.

Proof. Kummer's transformation gives

$$K_{N,M}(z) = \frac{z^N e^{-z}}{N!} {}_1F_1(-m; N+1; z).$$

Expanding the terminating hypergeometric series yields (175). Every denominator $(N+j)!$ divides $(M-1)!$, while the top coefficient, corresponding to $j = m$, is $(-1)^m/(M-1)!$. Hence the global least common denominator is exactly $(M-1)!$ and its p -part is exactly $p^{v_p((M-1)!)}$, proving (176). After clearing, the leading coefficient is $(-1)^m$, so the integer polynomial is primitive.

The coefficient with $j = 0$ is $(M-1)!/N!$, which proves the lower height bound. For every j ,

$$\binom{m}{j} \frac{(M-1)!}{(N+j)!} \leq \binom{m}{j} \frac{(M-1)!}{N!} \leq 2^m \frac{(M-1)!}{N!},$$

giving the upper bound. Stirling's formula proves (179). Finally,

$$\max_{y \geq 0} \frac{y^N}{(1+y)^M} = \frac{N^N (M-N)^{M-N}}{M^M};$$

its logarithm is $O_\alpha(N)$ when $M = \alpha N$, whereas $\log((M-1)!) = \alpha N \log N + O_\alpha(N)$. This proves (180).

For the uniform assertion, write $m = M - N$. Direct division gives

$$\frac{\mathfrak{F}_{N,M+1}}{\mathfrak{F}_{N,M}} = M \frac{(m+1)^{m+1}}{m^m} \frac{M^M}{(M+1)^{M+1}}.$$

Its logarithm is

$$\log M - (\phi(M) - \phi(m)), \quad \phi(x) = (x+1) \log(x+1) - x \log x.$$

Since

$$\phi(M) - \phi(m) = \int_m^M \log(1+x^{-1}) dx < \int_m^M \frac{dx}{x} = \log(M/m) \leq \log M,$$

the ratio is strictly greater than one. Its minimum is therefore at $M = N+1$, and Stirling's formula gives (181). $\square$

23 The explicit mixed family

Let

$$\mathcal{E}(w) = \sum_{n \geq 0} (-1)^n n! w^n, \quad \Phi_\beta(z) = e^z (\text{Ein}(z) - \beta) - \mathcal{E}(1/z), \quad \beta \in \overline{\mathbb{Q}}.$$

The direction-zero Borel sum of $\mathcal{E}(1/z)$ on $|\arg z| < \pi$ is

$$\mathcal{S}_0 \mathcal{E}(1/z) = z e^z E_1(z).$$

Using $\text{Ein}(z) = \gamma + \log z + E_1(z)$ gives

$$\mathcal{S}_0 \Phi_\beta(z) = e^z [\gamma - \beta + \log z + (1-z)E_1(z)], \quad \mathcal{S}_0 \Phi_\beta(1) = e(\gamma - \beta). \quad (182)$$

Let $\mathcal{M}$ denote the mixed space consisting of formal objects $A(z) + a(1/z)$, where A is an E -function, a is arithmetic Gevrey of order one, and $a(0) = 0$. The harmless normalization $a(0) = 0$ makes the splitting unique up to an algebraic constant.

Proposition 23.1 (A mixed divisibility obstruction). *For every $\beta \in \overline{\mathbb{Q}}$,*

$$\Phi_\beta \notin (z-1)\mathcal{M}.$$

Proof. Suppose $\Phi_\beta = (z-1)(A(z) + a(1/z))$ with $A(z) + a(1/z) \in \mathcal{M}$. Rearranging gives

$$e^z(\text{Ein}(z) - \beta) - (z-1)A(z) = \mathcal{E}(1/z) + (z-1)a(1/z).$$

The left side has only nonnegative powers of z . Because $a(0) = 0$, the right side has only nonpositive powers. Their common value must therefore be a constant $c \in \overline{\mathbb{Q}}$. Evaluation at $z = 1$ gives

$$e(\text{Ein}(1) - \beta) = c,$$

contradicting the algebraic independence of e and $\text{Ein}(1)$ supplied by [theorem 4.1](#). $\square$

There is an independent coefficient-level form of the obstruction. In the canonical split expansion of the formal quotient, its inverse tail is $w\mathcal{E}(w)/(w-1)$. [Theorem 15.1](#), with $\xi = 1$, shows that even after allowing an arbitrary algebraic overlap constant c , the series $(\mathcal{E}(w) - c)/(w-1)$ has superexponential normalized denominators.

Consider the restricted zero-removal assertion

$$\text{ZR}(\beta) : \quad \mathcal{S}_0\Phi_\beta(1) = 0 \implies \Phi_\beta \in (z-1)\mathcal{M}. \quad (183)$$

Theorem 23.2 (Exact logical status of the proposed bridge). *For a fixed $\beta \in \overline{\mathbb{Q}}$, $\text{ZR}(\beta)$ holds if and only if $\gamma \neq \beta$. Consequently*

$$[\text{ZR}(\beta) \text{ for every } \beta \in \overline{\mathbb{Q}}] \iff \gamma \text{ is transcendental.}$$

Proof. By [\(182\)](#), the antecedent of $\text{ZR}(\beta)$ holds exactly when $\gamma = \beta$. By [proposition 23.1](#), its consequent is always false. Thus the implication holds exactly when its antecedent is false. Taking the conjunction over algebraic β gives the result. $\square$

This theorem explains why the apparently modest one-family lemma is not a shortcut: it is the original transcendence problem in vacuous-implication form. There is also an analytic reflection of the obstruction. Under the hypothetical equality $\beta = \gamma$, the summed quotient is locally holomorphic at 1 and equals

$$q(z) = e^z \left(\frac{\log z}{z-1} - E_1(z) \right). \quad (184)$$

A positive circuit about 0 sends $\log z$ to $\log z + 2\pi i$ and $E_1(z)$ to $E_1(z) - 2\pi i$. Therefore

$$\Delta q(z) = \frac{2\pi i z e^z}{z-1},$$

which has a genuine pole at $z = 1$. In the Fourier–Laplace/ E -operator formulation of mixed arithmetic–Gevrey functions [\[3, 22\]](#), nonzero finite points are ordinary; every continued solution germ must remain holomorphic there. Thus the local monodromy records the same obstruction as the denominator theorem.

24 A compact Euler integral and its unavoidable logarithmic contamination

The arithmetic framework of this section—Beukers–Sondow double integrals and Nesterenko-type rational functions producing linear forms in $1, \gamma$, and logarithms—goes back at least to Sondow [\[50, 51\]](#). We make no priority claim for that framework. We instead isolate the exact asymmetric contiguous family needed here, correct its asymptotic and height normalization, and prove that its logarithmic contamination is strictly positive for every index.

24.1 The floor-weighted rational transform

For a rational function g which is integrable against the indicated weight, define

$$\mathcal{T}(g) = \int_0^\infty (\lfloor u \rfloor + 1)g(u) du. \quad (185)$$

Write $H_m^{(r)} = \sum_{j=1}^m j^{-r}$, with $H_0^{(r)} = 0$, and $H_m = H_m^{(1)}$.

Proposition 24.1 (Complete pole-order transducer). *Suppose that $g \in \overline{\mathbb{Q}}(u)$ is $O(u^{-3})$ at infinity, all its poles belong to $\{-1, -2, \dots\}$, and*

$$g(u) = \sum_{p \geq 1} \sum_{r=1}^{m_p} \frac{c_{p,r}}{(u+p)^r}$$

is its finite partial-fraction expansion. Then

$$\begin{aligned} \mathcal{T}(g) = & \sum_p c_{p,1} \left(\log((p-1)!) + p + 1 - \frac{1}{2} \log(2\pi) \right) \\ & + \sum_p c_{p,2} (\gamma - 1 - H_{p-1}) \\ & + \sum_p \sum_{r=3}^{m_p} \frac{c_{p,r}}{r-1} \left(\zeta(r-1) - H_{p-1}^{(r-1)} \right). \end{aligned} \quad (186)$$

In particular, simple poles carry factorial logarithms, double poles carry γ , and poles of order $r \geq 3$ carry $\zeta(r-1)$.

Proof. For a simple pole, summation over the intervals $[j, j+1]$ gives

$$\sum_{j=0}^J (j+1) \log \frac{j+p+1}{j+p} = (J+1) \log(J+p+1) - \log \Gamma(J+p+1) + \log \Gamma(p).$$

Stirling's formula isolates the finite part $\log \Gamma(p) + p + 1 - \frac{1}{2} \log(2\pi)$. For a double pole the corresponding finite part is

$$\frac{1}{p} - 1 - \psi(p+1) = \gamma - 1 - H_{p-1}.$$

For $r \geq 3$, absolute convergence and summation by parts give

$$\begin{aligned} \sum_{j \geq 0} (j+1) \int_j^{j+1} \frac{du}{(u+p)^r} \\ = \frac{1}{r-1} \sum_{j \geq 0} (j+p)^{1-r} = \frac{\zeta(r-1, p)}{r-1}. \end{aligned}$$

Since p is a positive integer, $\zeta(r-1, p) = \zeta(r-1) - H_{p-1}^{(r-1)}$.

The hypothesis $g(u) = O(u^{-3})$ gives

$$\sum_p c_{p,1} = 0, \quad \sum_p c_{p,2} = \sum_p p c_{p,1}.$$

These identities cancel the linear and logarithmic divergences of the simple- and double-pole partial sums, so the sum of their finite parts is exactly (186). $\square$

Remark 24.2 (Necessary correction of scope). A product of Beta factors can have poles of multiplicity greater than two. In that situation a partial-fraction expansion containing only simple and double poles is incomplete; the last line of (186) is indispensable. Restricting to products with pole multiplicity at most two recovers the $(1, \gamma, \log)$ -valued special case.

There is also a one-variable endpoint obstruction which explains why the compact construction below is genuinely two-dimensional.

Lemma 24.3 (Endpoint removal loses γ). *Let $P(x) = \sum_{k=0}^d a_k x^k \in \overline{\mathbb{Q}}[x]$ satisfy $P(1) = 0$. Then*

$$\int_0^1 P(x) \left(\frac{1}{1-x} + \frac{1}{\log x} \right) dx = \int_0^1 \frac{P(x)}{1-x} dx + \sum_{k=0}^d a_k \log(k+1). \quad (187)$$

The first term on the right is algebraic; in particular no γ -term survives.

Proof. The first integrand is a polynomial because $P(1) = 0$. The same condition allows one to write $P(x) = \sum_k a_k (x^k - 1)$, and Frullani's identity

$$\int_0^1 \frac{x^k - 1}{\log x} dx = \log(k+1)$$

proves (187). □

24.2 The asymmetric compact family

For $n \geq 1$, set

$$J_n = \int_0^1 \int_0^1 \frac{[x(1-x)y(1-y)]^n (1-x)}{(1-xy)(-\log(xy))} dx dy. \quad (188)$$

Using $1/(-\log(xy)) = \int_0^\infty (xy)^t dt$ and $(1-xy)^{-1} = \sum_{j \geq 0} (xy)^j$, Tonelli's theorem gives

$$J_n = \mathcal{T}(g_n), \quad g_n(u) = B(u+n+1, n+2)B(u+n+1, n+1). \quad (189)$$

Put $L = n+1$, $M = 2n+2$, and $C = n!(n+1)!$. Then

$$g_n(u) = \frac{C}{(u+M) \prod_{p=L}^{M-1} (u+p)^2}. \quad (190)$$

Theorem 24.4 (Exact coefficients and strict logarithmic contamination). *The partial-fraction expansion of g_n is*

$$g_n(u) = \sum_{p=L}^M \frac{\alpha_p}{u+p} + \sum_{p=L}^{M-1} \frac{\beta_p}{(u+p)^2},$$

where, for $0 \leq k \leq n$,

$$\beta_{L+k} = \binom{n}{k} \binom{n+1}{k}, \quad (191)$$

$$\alpha_{L+k} = \beta_{L+k} \left(2H_k - 2H_{n-k} - \frac{1}{n+1-k} \right), \quad (192)$$

$$\alpha_M = \frac{1}{n+1}. \quad (193)$$

Consequently

$$J_n = \binom{2n+1}{n} \gamma + r_n + \mathcal{L}_n, \quad (194)$$

where

$$r_n = \sum_{p=L}^M \alpha_p(p+1) + \sum_{p=L}^{M-1} \beta_p \left(\frac{1}{p} - 1 - H_p \right) \in \mathbb{Q}, \quad (195)$$

$$\mathcal{L}_n = \sum_{p=L}^M \alpha_p \log((p-1)!) = \sum_{j=L}^{M-1} T_j \log j, \quad T_j = \sum_{p>j} \alpha_p. \quad (196)$$

Every proper tail is strictly positive:

$$T_j > 0 \quad (L \leq j \leq M-1). \quad (197)$$

In particular

$$\boxed{\mathcal{L}_n > 0 \text{ for every } n \geq 1.} \quad (198)$$

Thus this entire compact family is never logarithm-free.

Proof. At the double pole $p = L + k$, evaluation of $(u+p)^2 g_n(u)$ at $u = -p$ gives

$$\beta_{L+k} = \frac{n!(n+1)!}{(n+1-k)(k!)^2((n-k)!)^2} = \binom{n}{k} \binom{n+1}{k}.$$

Logarithmic differentiation of the same expression gives

$$\frac{\alpha_{L+k}}{\beta_{L+k}} = -\frac{1}{n+1-k} - 2 \sum_{\substack{0 \leq j \leq n \\ j \neq k}} \frac{1}{j-k} = 2H_k - 2H_{n-k} - \frac{1}{n+1-k},$$

which proves (192). The simple pole at M gives (193). Since $g_n(u) = O(u^{-2n-3})$, comparison at infinity gives

$$\sum_{p=L}^M \alpha_p = 0. \quad (199)$$

Proposition 24.1 now yields (194)–(196); the $\log(2\pi)$ -term disappears by (199). Vandermonde's identity gives the γ -coefficient:

$$\sum_{k=0}^n \beta_{L+k} = \sum_{k=0}^n \binom{n}{k} \binom{n+1}{k} = \binom{2n+1}{n}.$$

It remains to prove the strict obstruction. Write

$$b_k = 2H_k - 2H_{n-k} - \frac{1}{n+1-k}.$$

Then

$$b_{k+1} - b_k = \frac{2}{k+1} + \frac{1}{n-k} + \frac{1}{n+1-k} > 0,$$

while $b_0 < 0$ and $b_n = 2H_n - 1 > 0$. Since every β_{L+k} is positive, the sequence $\alpha_L, \dots, \alpha_{M-1}$ changes sign at most once, from negative to positive, and $\alpha_M > 0$. Together with the zero total sum (199), this implies that every proper tail T_j is positive. Formula (196) then gives (198). $\square$

24.3 Corrected asymptotic and integer height

Theorem 24.5 (Corrected asymptotic and integerization). *As $n \rightarrow \infty$,*

$$\boxed{J_n \sim \frac{\pi}{12 \log 2} \frac{16^{-n}}{n}}. \quad (200)$$

Let $d_n = \text{lcm}(1, \dots, 2n+2)$. There are integers $A_n, B_n, C_{n,q}$, with q running over the primes at most $2n+1$, such that

$$d_n J_n = A_n + B_n \gamma + \sum_{\substack{q \leq 2n+1 \\ q \text{ prime}}} C_{n,q} \log q, \quad B_n = d_n \binom{2n+1}{n}. \quad (201)$$

For every prime $q \in [n+1, 2n+1]$,

$$C_{n,q} = d_n T_q > 0. \quad (202)$$

Thus the number of nonzero logarithmic coordinates tends to infinity and is at least $\pi_{\text{pr}}(2n+1) - \pi_{\text{pr}}(n) \sim n/\log n$, where π_{pr} is the prime-counting function.

If

$$\mathcal{H}_n = \max \left(1, |A_n|, |B_n|, \max_{q \leq 2n+1} |C_{n,q}| \right),$$

then

$$\log \mathcal{H}_n = (2 + \log 4)n + o(n), \quad (203)$$

$$-\log |d_n J_n| = (\log 16 - 2)n + o(n). \quad (204)$$

Hence the exponent of this explicit integer form is

$$\frac{-\log |d_n J_n|}{\log \mathcal{H}_n} \longrightarrow \frac{\log 16 - 2}{\log 4 + 2} = 0.2281516725 \dots \quad (205)$$

The exponent is positive, but the ambient dimension grows; this is not a fixed-dimensional Nesterenko criterion and proves neither irrationality nor transcendence of γ .

Proof. The function

$$\phi(x, y) = \log(x(1-x)y(1-y))$$

has its unique maximum $-\log 16$ at $(1/2, 1/2)$, with Hessian $-8I_2$. The remaining factor in (188) has value

$$\frac{1 - 1/2}{(1 - 1/4)(-\log(1/4))} = \frac{1}{3 \log 2}$$

there. The two-dimensional Laplace method therefore gives

$$J_n \sim 16^{-n} \frac{2\pi/n}{8} \frac{1}{3 \log 2},$$

which is (200).

Equations (191)–(193) show that the denominators of all α_p divide $\text{lcm}(1, \dots, n+1)$, while the denominators in (195) divide d_n . Expanding every $\log((p-1)!)$ into prime logarithms proves (201). If $q \in [n+1, 2n+1]$ is prime, then $2q > 2n+1$, so

$$v_q((p-1)!) = \mathbf{1}_{p>q} \quad (L \leq p \leq M).$$

Equation (202) follows from (197); the prime-number theorem gives the count.

Finally

$$\log d_n = 2n + o(n), \quad \log \binom{2n+1}{n} = n \log 4 + o(n).$$

The coefficient formulas show that every coefficient in (201) is at most $d_n 4^n \exp(o(n))$, while B_n has exactly that exponential scale. This proves (203). Combining (200) with $\log d_n = 2n + o(n)$ proves (204), and division gives (205). $\square$

Remark 24.6 (Comparison with the reciprocal height barrier). The reciprocal truncations studied earlier saturate their full defining height so completely that primitive integerization destroys the small value. The compact forms (201) behave differently: after integerization they remain exponentially small. Their obstruction is instead the rigorously growing set of positive logarithmic coordinates in (202). These are two distinct boundary mechanisms and should not be conflated.

25 A recent logarithmic consequence

The preceding results concern algebraic relations. A recent theorem of Fischler and Rivoal on logarithms of E -values [23] gives a complementary one-number statement.

Proposition 25.1. *The number*

$$\log \text{Ein}(1) = \log \left(\gamma + \frac{\delta}{e} \right)$$

is transcendental. It is also not an ultra-Liouville number.

Proof. The specialization of Fischler–Rivoal’s logarithm theorem [23] to Ein leaves only the exceptional alternatives $\text{Ein}(1) = 1$ and $\text{Ein}(1) = e^{-1}$. They are excluded by the elementary integral bounds

$$e^{-1} < \int_0^1 \frac{1 - e^{-t}}{t} dt < 1.$$

Indeed $(1 - e^{-t})/t$ is the average of e^{-u} on $[0, t]$, so it is strictly above e^{-1} on a set of positive measure and strictly below 1. Their quantitative theorem supplies the non-ultra-Liouville conclusion. $\square$

This proposition is an application of the cited general theorem, not a new proof of an older open problem. It does not imply the transcendence of either summand in $\text{Ein}(1) = \gamma + \delta/e$ separately.

26 Joukowski descent and the arithmetic of reciprocal levels

The reciprocal equation admits a useful quotient which separates its complex-analytic geometry from its arithmetic level. Put

$$J(z) = z + z^{-1} \quad (z \in \mathbb{C}^\times).$$

The involution $z \mapsto z^{-1}$ is the deck involution of J . In this section we descend the reciprocal Ein -sum through J , determine the critical points of the descended entire function, and then apply Theorem 4.1 to its values at algebraic points.

26.1 The entire quotient and its critical points

Let $C_0(w) = 2$, $C_1(w) = w$, and

$$C_{n+1}(w) = wC_n(w) - C_{n-1}(w) \quad (n \geq 1).$$

Thus $C_n(w) = 2T_n(w/2)$ and

$$C_n(J(z)) = z^n + z^{-n}. \quad (206)$$

Proposition 26.1 (Entire Joukowski descent). *There is a unique entire function H such that*

$$H(J(z)) = \text{Ein}(z) + \text{Ein}(z^{-1}) \quad (z \in \mathbb{C}^\times). \quad (207)$$

It is given by the locally uniformly convergent series

$$H(w) = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n n!} C_n(w). \quad (208)$$

Moreover

$$H'(w) = 2e^{-w/2} \frac{\sinh\left(\frac{1}{2}\sqrt{w^2 - 4}\right)}{\sqrt{w^2 - 4}} \quad (209)$$

$$= e^{-w/2} \prod_{n \geq 1} \left(1 + \frac{w^2 - 4}{4\pi^2 n^2}\right). \quad (210)$$

The right side of (209) is understood by its removable continuation at $w = \pm 2$; in particular $H'(2) = e^{-1}$ and $H'(-2) = e$.

Proof. If $K \subset \mathbb{C}$ is compact, there is an $R_K > 1$ such that both roots of $X^2 - wX + 1$ have modulus at most R_K for every $w \in K$. Hence (206) gives $|C_n(w)| \leq 2R_K^n$ on K . The series (208) therefore converges locally uniformly and defines an entire function. Substitution of $w = J(z)$ and the power series of Ein proves (207). Uniqueness follows because J is surjective: the equation $z^2 - wz + 1 = 0$ has a nonzero root for every $w \in \mathbb{C}$.

For $z \neq \pm 1$, differentiation of (207) gives

$$H'(J(z)) = \frac{e^{-1/z} - e^{-z}}{z - z^{-1}}.$$

Write $w = z + z^{-1}$ and $v = z - z^{-1}$, so $v^2 = w^2 - 4$. Then

$$e^{-1/z} - e^{-z} = e^{-w/2} (e^{v/2} - e^{-v/2}),$$

which proves (209) away from $w = \pm 2$. The quotient $2 \sinh(v/2)/v$ is an entire even function of v , so the apparent singularities are removable. Finally Euler's product

$$\frac{\sinh u}{u} = \prod_{n \geq 1} \left(1 + \frac{u^2}{\pi^2 n^2}\right)$$

with $u = \frac{1}{2}\sqrt{w^2 - 4}$ gives (210). □

Corollary 26.2 (Critical lattice and the real diffeomorphism). *The critical points of H are exactly the simple points*

$$\xi_n^\pm = \pm 2i\sqrt{\pi^2 n^2 - 1} \quad (n \geq 1). \quad (211)$$

In particular every critical point is transcendental. On the real axis $H'(x) > 0$, and H maps $\mathbb{R}$ increasingly and bijectively onto $\mathbb{R}$.

Proof. The product (210) gives (211); each zero is simple because it occurs in exactly one factor and is nonzero. If one of these points were algebraic, then $\pi^2 = ((\xi_n^\pm)^2/(-4) + 1)/n^2$ would be algebraic, a contradiction. For real x , every factor in (210) is positive, since

$$1 + \frac{x^2 - 4}{4\pi^2 n^2} \geq 1 - \frac{1}{\pi^2 n^2} > 0.$$

Thus $H'(x) > 0$.

For $w \rightarrow +\infty$, choose the root $z > 1$ of $J(z) = w$. Then $z = w + O(w^{-1})$ and

$$H(w) = \text{Ein}(z) + \text{Ein}(z^{-1}) = \gamma + \log w + O(w^{-1}) \rightarrow +\infty.$$

For $w \rightarrow -\infty$, write the root of modulus greater than one as $z = -a$, $a \rightarrow +\infty$. Since $\text{Ein}(-a) = \gamma + \log a - \text{Ei}(a)$ and $\text{Ei}(a) \sim e^a/a$, one has $H(w) \rightarrow -\infty$. Strict monotonicity now proves the last assertion. $\square$

Proposition 26.3 (A rational differential equation). *The function H satisfies*

$$w(w^2 - 4)H''' + (w^3 + 2w^2 - 4w + 4)H'' + (w^2 - w + 2)H' = 0. \quad (212)$$

Consequently every additive translate $H - 2c$ satisfies the same equation.

Proof. Put $P(w) = e^{w/2}H'(w)$. By (209),

$$P(w) = 2 \frac{\sinh(\frac{1}{2}\sqrt{w^2 - 4})}{\sqrt{w^2 - 4}}.$$

Direct differentiation gives

$$w(w^2 - 4)P'' + 2(w^2 + 2)P' - \frac{w^3}{4}P = 0.$$

Substitution of $P = e^{w/2}H'$ and collection of terms gives (212). The equation contains no undifferentiated H , so it is invariant under additive translation. $\square$

26.2 Algebraic values after descent

For $w \in \overline{\mathbb{Q}}$, let

$$\Lambda(w) = \{\lambda \in \overline{\mathbb{Q}}^\times : \lambda + \lambda^{-1} = w\}$$

as a set. At $w = 2$ or $w = -2$ this set has one element, but the identity (207) counts that element twice.

Theorem 26.4 (Multipoint arithmetic descent). *Let $w_1, \dots, w_s \in \overline{\mathbb{Q}}$ be distinct, let $\Lambda = \bigcup_{j=1}^s \Lambda(w_j)$, and put*

$$d = \dim_{\mathbb{Q}} \text{span}_{\mathbb{Q}} \Lambda.$$

Then

$$\text{trdeg}_{\overline{\mathbb{Q}}}(\{e^\lambda\}_{\lambda \in \Lambda}, H(w_1), \dots, H(w_s)) = d + s. \quad (213)$$

In particular $H(w_1), \dots, H(w_s)$ are algebraically independent over the field generated by the exponential coordinates. Hence $H(w)$ is transcendental for every algebraic w .

Proof. By Theorem 4.1, the coordinates $Y_\lambda = \text{Ein}(\lambda)$, $\lambda \in \Lambda$, are algebraically independent over $E = \overline{\mathbb{Q}}(\{e^\lambda\}_{\lambda \in \Lambda})$, and $\text{trdeg}_{\overline{\mathbb{Q}}} E = d$. Each $H(w_j)$ is a nonzero linear form in the Y_λ supported on $\Lambda(w_j)$, with coefficient 2 at a double root. The supports for distinct j are disjoint. These s linear forms can therefore be extended to a linear coordinate system in the Y_λ , and are algebraically independent over E . This proves (213). The last assertion is the case $s = 1$. $\square$

Corollary 26.5 (Algebraic levels have transcendental preimages). *For every $c \in \overline{\mathbb{Q}}$, every solution of*

$$H(w) = 2c \quad (214)$$

is transcendental. Equivalently, every solution of

$$\text{Ein}(z) + \text{Ein}(z^{-1}) = 2c \quad (215)$$

is transcendental.

Proof. An algebraic solution of (214) would contradict the last assertion of Theorem 26.4. If z in (215) were algebraic, then $w = J(z)$ would be algebraic and would solve (214). $\square$

Corollary 26.6 (The unique real orbit of a real level). *For every $c \in \mathbb{R}$ there is a unique $w_c \in \mathbb{R}$ with $H(w_c) = 2c$. It satisfies*

condition on c	location of w_c	preimage under J
$\text{Ein}(-1) < c < \text{Ein}(1)$	$-2 < w_c < 2$	one conjugate pair on $ z = 1$,
$c > \text{Ein}(1)$	$w_c > 2$	two positive reciprocal points,
$c < \text{Ein}(-1)$	$w_c < -2$	two negative reciprocal points.

At $c = \text{Ein}(1)$ or $c = \text{Ein}(-1)$, respectively, $z = 1$ or $z = -1$ is a double zero of (215). If c is algebraic, the points in the table are all transcendental.

More generally, for arbitrary real c , every nonreal algebraic solution of $H(w) = 2c$ is impossible. Thus at most the unique real preimage w_c can be algebraic.

Proof. Existence and uniqueness follow from Corollary 26.2. The thresholds follow from $H(2) = 2 \text{Ein}(1)$ and $H(-2) = 2 \text{Ein}(-1)$, and the three descriptions are the elementary inverse images of the intervals $(-2, 2)$, $(2, \infty)$ and $(-\infty, -2)$ under J . Since

$$J(z) - 2 = \frac{(z-1)^2}{z}, \quad J(z) + 2 = \frac{(z+1)^2}{z},$$

and $H'(\pm 2) \neq 0$, the endpoint zeros have order two. The assertion for algebraic c is Corollary 26.5.

Finally, if a nonreal algebraic w solved $H(w) = 2c$ with c real, then $\bar{w}$ would be a distinct algebraic point with $H(\bar{w}) = \overline{H(w)} = 2c$. This contradicts the algebraic independence of $H(w)$ and $H(\bar{w})$ in Theorem 26.4. $\square$

26.3 The Euler level and witness amplification

Put

$$G_\gamma(w) = H(w) - 2\gamma.$$

Theorem 26.7 (Euler level in the quotient). *The function G_γ has exactly one real zero w_* , and*

$$-2 < w_* < 2, \quad w_* = 2 \cos \theta_*, \quad (216)$$

A high-precision Newton iteration gives the non-rigorous decimal values

$$w_* = 1.0073046182687490652\dots, \quad \theta_* = 1.0429750684400255695\dots$$

The two preimages of w_ are the unit-circle zeros $\alpha_* = e^{i\theta_*}$ and $\bar{\alpha}_*$ of the reciprocal Ein equation. Every algebraic zero of G_γ must equal w_* ; equivalently every algebraic zero of the reciprocal equation belongs to $\{\alpha_*, \bar{\alpha}_*\}$.*

If w_* , equivalently α_* , is algebraic, then γ and δ are algebraically independent. More precisely, put $a = \alpha_*$, $b = a^{-1}$,

$$D_* = \text{Ein}(a) - \text{Ein}(b), \quad d_* = \dim_{\mathbb{Q}} \text{span}_{\mathbb{Q}}\{1, a, b\}.$$

Then

$$\text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(e, e^a, e^b, \gamma, \delta, D_*) = d_* + 3, \quad (217)$$

and γ, δ, D_* are algebraically independent over $\overline{\mathbb{Q}}(e, e^a, e^b)$.

Proof. Since

$$H(-2) = 2\gamma - 2\text{Ei}(1) < 2\gamma < 2\gamma + 2E_1(1) = H(2),$$

Corollary 26.2 gives the unique zero w_* and (216). If an algebraic zero w were nonreal, the last assertion of Corollary 26.6 would exclude it; if it were real, uniqueness would force $w = w_*$. Passing between w and the roots of $z^2 - wz + 1$ proves the equivalent statement in the z -plane.

Now assume that w_* is algebraic. Then a and b are distinct algebraic numbers. Apply Theorem 4.1 to $\Lambda = \{1, a, b\}$. Over $E = \overline{\mathbb{Q}}(e, e^a, e^b)$ the three numbers

$$Y_1 = \text{Ein}(1), \quad Y_a = \text{Ein}(a), \quad Y_b = \text{Ein}(b)$$

are algebraically independent. The reciprocal equation and $\delta = eE_1(1)$ give

$$\gamma = \frac{Y_a + Y_b}{2}, \quad \frac{\delta}{e} = Y_1 - \frac{Y_a + Y_b}{2}, \quad D_* = Y_a - Y_b.$$

This is an invertible linear change of the three Y -coordinates; replacing δ/e by δ is invertible over E . This proves (217). $\square$

Corollary 26.8 (Conditional completion of the six-constant field). *If α_* is algebraic, then*

$$\text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(\gamma, \delta, E_1(1), \text{Ei}(1), \text{Ci}(1), \text{Chi}(1)) = 5.$$

The complete ideal of algebraic relations among these six displayed coordinates is generated by

$$\text{Ei}(1) - E_1(1) - 2\text{Chi}(1) = 0. \quad (218)$$

Proof. Use the notation

$$A = \text{Ein}(1), \quad B = \text{Ein}(-1), \quad C = \frac{\text{Ein}(i) + \text{Ein}(-i)}{2}.$$

The point α_* is distinct from $\pm 1, \pm i$. Indeed $w_* \in (-2, 2)$ excludes ± 1 , while

$$H(0) = 2\gamma - 2\text{Ci}(1) < 2\gamma$$

excludes $\pm i$; here $\text{Ci}(1) = \gamma + \int_0^1 (\cos t - 1) dt/t > \gamma - 1/4 > 0$. Theorem 4.1, applied to $\{1, -1, i, -i, \alpha_*, \alpha_*^{-1}\}$, now shows that

$$\gamma, \quad e, \quad A, \quad B, \quad C$$

are algebraically independent: they are one exponential coordinate and four independent linear forms with disjoint supports in the relevant Ein -coordinates. The six-constant field equals $\overline{\mathbb{Q}}(\gamma, e, A, B, C)$, so its transcendence degree is five.

Identity (218) follows from $E_1(1) = A - \gamma$, $\text{Ei}(1) = \gamma - B$, and $\text{Chi}(1) = \gamma - (A + B)/2$. The kernel of the evaluation map from a polynomial ring in the six displayed coordinates is a height-one prime. It contains the irreducible linear polynomial in (218); hence it is precisely the principal ideal generated by that polynomial. $\square$

26.4 Every level has asymptotic transcendental fibres

The following strengthens the existence part of Picard's theorem for this specific function: there is no exceptional finite level, and an explicit sequence of preimages is available at every level.

Theorem 26.9 (Asymptotic level fibres). *Fix $c \in \mathbb{C}$. For every sufficiently large integer k there is a simple solution $w_{k,c}$ of $H(w) = 2c$ such that*

$$w_{k,c} = -\log(2\pi k) - \log \log(2\pi k) + i \left(2\pi k + \frac{\pi}{2} \right) + O_c \left(\frac{1}{\log k} \right). \quad (219)$$

Consequently every finite value of H is assumed infinitely often. If c is algebraic, every member of these fibres is transcendental. If $c \in \mathbb{R}$, the conjugates $\overline{w_{k,c}}$ give a second family.

Proof. It is convenient first to work in the z -plane. Put

$$\mathcal{F}_c(z) = \text{Ein}(z) + \text{Ein}(z^{-1}) - 2c = H(J(z)) - 2c.$$

On $\mathbb{C} \setminus (-\infty, 0]$, with principal logarithm, set

$$L_c(z) = \text{Log } z + \gamma - 2c.$$

The connection formula for Ein gives

$$\mathcal{F}_c(z) = L_c(z) + E_1(z) + \text{Ein}(z^{-1}). \quad (220)$$

Uniformly in a fixed closed sector about the positive imaginary axis, the standard expansion of E_1 [17] and the Taylor expansion of Ein give

$$E_1(z) = \frac{e^{-z}}{z} (1 + O(z^{-1})), \quad \text{Ein}(z^{-1}) = z^{-1} + O(z^{-2}). \quad (221)$$

Let

$$R_k = 2\pi k, \quad a_k = \log R_k + \log \log R_k, \quad \zeta_k = -a_k + i \left(R_k + \frac{\pi}{2} \right).$$

On $D_k = \{z : |z - \zeta_k| \leq 1\}$, for all sufficiently large k , the principal branches of $\text{Log } z$ and $\text{Log } L_c(z)$ are analytic. Uniformly on D_k ,

$$\text{Log } z = \log R_k + \frac{\pi i}{2} + O(a_k/R_k), \quad (222)$$

$$\text{Log } L_c(z) = \log \log R_k + O_c((\log R_k)^{-1}). \quad (223)$$

Define

$$\Phi_{k,c}(z) = (2k+1)\pi i - \text{Log } z - \text{Log } L_c(z).$$

Equations (222)–(223) give

$$\Phi_{k,c}(z) = \zeta_k + O_c((\log R_k)^{-1}), \quad \Phi'_{k,c}(z) = -\frac{1}{z} - \frac{1}{zL_c(z)} = O(R_k^{-1}).$$

Thus $\Phi_{k,c}$ maps D_k into itself and is a contraction for large k . Its fixed point $u_{k,c}$ satisfies

$$u_{k,c} = \zeta_k + O_c((\log R_k)^{-1}) \quad (224)$$

and

$$u_{k,c} e^{u_{k,c}} L_c(u_{k,c}) = -1.$$

Hence $u_{k,c}$ is a zero of the model function

$$h_c(z) = 1 + z e^z L_c(z).$$

Write $q_c(z) = z + \text{Log } z + \text{Log } L_c(z)$, so $h_c = 1 + e^{q_c}$ on D_k . At $u_{k,c}$,

$$q'_c(u_{k,c}) = 1 + \frac{1}{u_{k,c}} + \frac{1}{u_{k,c} L_c(u_{k,c})} = 1 + O(R_k^{-1}). \quad (225)$$

In particular the model zero is simple.

Now multiply the exact function by its nonvanishing normalizing factor:

$$A_c(z) = z e^z \mathcal{F}_c(z).$$

From (220)–(221),

$$A_c(z) - h_c(z) = z e^z \left(E_1(z) - \frac{e^{-z}}{z} \right) + z e^z \text{Ein}(z^{-1}). \quad (226)$$

The first term is $O(R_k^{-1})$ near $u_{k,c}$. The model equation gives

$$|e^{u_{k,c}}| = \frac{1}{|u_{k,c}| |L_c(u_{k,c})|} = O_c((R_k \log R_k)^{-1}),$$

so the second term in (226) is $O_c((R_k \log R_k)^{-1})$. Consequently there is a constant C_c such that

$$|A_c(z) - h_c(z)| \leq \frac{C_c}{R_k} \quad (227)$$

uniformly on $|z - u_{k,c}| \leq 1$ for all sufficiently large k .

Fix $M > 2C_c$. Equation (225) and Taylor's theorem give, on $|z - u_{k,c}| = M/R_k$,

$$|h_c(z)| \geq \frac{M}{2R_k}$$

for large k . A second application of the same Taylor estimate, now comparing h_c with $-q'_c(u_{k,c})(z - u_{k,c})$, shows that h_c has exactly one zero, counted with multiplicity, in this circle. By (227) and Rouché's theorem, A_c also has exactly one zero there. Since $z e^z$ does not vanish, this is a simple zero $z_{k,c}$ of $\mathcal{F}_c$, and

$$z_{k,c} = u_{k,c} + O_c(R_k^{-1}).$$

Finally put $w_{k,c} = J(z_{k,c})$. Since $z_{k,c}^{-1} = O(R_k^{-1})$ and $J'(z_{k,c}) = 1 + O(R_k^{-2})$, this is a simple zero of $H(w) - 2c$ and has the same asymptotic (219). The arithmetic assertion follows from Corollary 26.5, and conjugation gives the lower family for real c . $\square$

Corollary 26.10 (Order and a zero-count lower bound). *For every $c \in \mathbb{C}$, the entire function $H - 2c$ has order one and exponential type one. If $n_c(r)$ counts its zeros in $|w| \leq r$, with multiplicity, then*

$$\frac{r}{\pi} - O_c(\log r) \leq n_c(r) = O_c(r).$$

Proof. Formula (209) gives exponential type at most one, and the negative-real asymptotic used in Corollary 26.2 gives type at least one. Jensen's formula then gives $n_c(r) = O_c(r)$. For real c , Theorem 26.9 and conjugation give the lower bound directly. For general c , apply the same contraction argument with negative indices as well as positive ones; equivalently repeat the proof in a fixed sector about the negative imaginary axis. The two families have modulus $2\pi k + O_c(\log k)$, which gives the stated bound. $\square$

26.5 The derivative-level de Branges structure

Define

$$K(t) = e^{it/2} H'(it). \quad (228)$$

Proposition 26.11 (Rotated derivative). *One has*

$$K(t) = \frac{2 \sin\left(\frac{1}{2}\sqrt{t^2 + 4}\right)}{\sqrt{t^2 + 4}} \quad (229)$$

$$= \sin(1) \prod_{n \geq 1} \left(1 - \frac{t^2}{4(\pi^2 n^2 - 1)}\right). \quad (230)$$

Thus K belongs to the Laguerre–Pólya class and has only the simple real zeros

$$\pm 2\sqrt{\pi^2 n^2 - 1} \quad (n \geq 1).$$

Furthermore

$$E(t) = K(t) + iK'(t) \quad (231)$$

is a Hermite–Biehler function and therefore defines a de Branges space.

Proof. Equations (229) and (230) follow from (209) and Euler’s product for $\sin u/u$; the constant is $\prod_{n \geq 1} (1 - 1/(\pi^2 n^2)) = \sin(1)$. The partial products in (230) are real-rooted polynomials and converge locally uniformly, which is exactly the required Laguerre–Pólya approximation.

For $\Im t > 0$, the logarithmic derivative of the product satisfies

$$\Im \frac{K'(t)}{K(t)} = \sum_{n \geq 1} \Im \left(\frac{1}{t - a_n} + \frac{1}{t + a_n} \right) < 0, \quad a_n = 2\sqrt{\pi^2 n^2 - 1}.$$

The paired series converges locally uniformly off the real zero set. With $E^\#(t) = \overline{E(\bar{t})} = K(t) - iK'(t)$, it follows that

$$|E(t)|^2 - |E^\#(t)|^2 = -4|K(t)|^2 \Im \frac{K'(t)}{K(t)} > 0 \quad (\Im t > 0).$$

This is the Hermite–Biehler inequality. □

Remark 26.12 (Scope of the real-zero analogy). The terminology above is classical in the sense of Pólya–Schur and de Branges.¹ It applies one derivative below the reciprocal-level problem. Every translate $H - 2c$ has the same derivative, hence the same K , the same Hermite–Biehler function (231), and the same canonical de Branges space. This construction therefore forgets the level c and in particular cannot distinguish $c = \gamma$ from another real level.

Conversely, Theorem 26.9 gives infinitely many nonreal zeros of $H - 2c$. For real c , the function $H - 2c$ is therefore not in the Laguerre–Pólya class and cannot be the real part of a Hermite–Biehler function, whose real and imaginary parts have only real zeros. Likewise, although the symmetry $z \leftrightarrow z^{-1}$ resembles a self-inversive or Lee–Yang symmetry, the off-circle fibres show that no full circle theorem holds here. The statement that algebraic zeros at the Euler level must lie on the unit circle is an arithmetic restriction, not a Lee–Yang theorem.

¹G. Pólya and J. Schur, *Über zwei Arten von Faktorenfolgen in der Theorie der algebraischen Gleichungen*, J. Reine Angew. Math. 144 (1914), 89–113; L. de Branges, *Hilbert Spaces of Entire Functions*, Prentice–Hall, 1968.

Remark 26.13 (Novelty and logical status). The analytic descent and product formulas above are elementary consequences of the Joukowski symmetry and Euler’s sine product. The arithmetic statements are corollaries of Theorem 4.1; their novelty is the packaging, not a new transcendence mechanism. The conditional hypothesis that α_* is algebraic is not known, so Theorem 26.7 does not prove the transcendence of γ or δ . No claim of priority should be made for the Joukowski packaging before a specialist literature search.

27 Spectral stability, arithmetic Clark data, and a second recovery of γ

The Joukowski quotient has a positive Laplace representation which places its zero divisor strictly in the left half-plane. After rotation and a centering exponential, this stability becomes a Hermite–Biehler inequality. The resulting meromorphic inner function has two complementary features: its values at algebraic real points are algebraically independent, whereas their large-point phase recovers Euler’s constant. We develop these statements in a form which keeps the unconditional arithmetic assertions separate from the still open arithmetic of γ .

27.1 A positive Bessel–Laplace representation

Proposition 27.1 (Bessel–Laplace representation). *For every $w \in \mathbb{C}$,*

$$H'(w) = \int_0^1 e^{-sw} J_0\left(2\sqrt{s(1-s)}\right) ds. \quad (232)$$

The kernel in (232) is strictly positive on $[0, 1]$.

Proof. Set $t = 2s - 1$. The classical Sonine integral, obtained for example by expanding the exponential and the Bessel function and integrating termwise, is

$$\int_{-1}^1 e^{at} J_0\left(b\sqrt{1-t^2}\right) dt = 2 \frac{\sinh \sqrt{a^2 - b^2}}{\sqrt{a^2 - b^2}},$$

with removable continuation when $a^2 = b^2$. Taking $a = -w/2$ and $b = 1$ gives

$$\int_0^1 e^{-sw} J_0\left(2\sqrt{s(1-s)}\right) ds = 2e^{-w/2} \frac{\sinh(\frac{1}{2}\sqrt{w^2 - 4})}{\sqrt{w^2 - 4}},$$

which is (209). Finally $0 \leq 2\sqrt{s(1-s)} \leq 1$, and the alternating series for J_0 shows that $J_0(x) > 0$ for $0 \leq x \leq 1$. $\square$

Theorem 27.2 (Rightmost point of every real fibre). *Let $a \in \mathbb{R}$. If*

$$H(w) = H(a), \quad w \neq a,$$

then

$$\Re w < a. \quad (233)$$

Thus a is the unique rightmost point of its full complex fibre.

Proof. Integrating (232) from a to w gives

$$H(w) - H(a) = \int_0^1 e^{-as} \frac{1 - e^{-s(w-a)}}{s} J_0\left(2\sqrt{s(1-s)}\right) ds, \quad (234)$$

where the integrand is continued at $s = 0$. Put $d = w - a$. If $\Re d \geq 0$, then

$$\Re(1 - e^{-sd}) = 1 - e^{-s\Re d} \cos(s\Im d) \geq 1 - e^{-s\Re d} \geq 0.$$

All remaining factors in (234) are positive. If the integral vanishes, its real part therefore vanishes identically. This first forces $\Re d = 0$, and then $1 - \cos(s\Im d) = 0$ for every $s \in [0, 1]$, which forces $\Im d = 0$. Hence $d = 0$, contrary to the hypothesis. Thus $\Re d < 0$. $\square$

Corollary 27.3 (Hurwitz stability of the level-zero divisors). *Let ω_0 denote the unique real zero of H . Then $\omega_0 < 0$, and every other zero ρ of H satisfies*

$$\Re \rho < \omega_0 < 0.$$

All zeros of H are simple and transcendental. Moreover every zero z of

$$\Psi(z) = H(z + z^{-1})$$

is simple, transcendental, and lies in the open left half-plane.

Proof. The real diffeomorphism in Corollary 26.2, together with $H(0) = 2\text{Cin}(1) > 0$, gives the unique real zero $\omega_0 < 0$. Theorem 27.2 gives the asserted strict inequality for every other zero. The critical points of H are the purely imaginary points (211), so none is a zero; the real zero is simple because $H'(\omega_0) > 0$. Transcendence follows from Corollary 26.5 at the algebraic level zero.

If $\Psi(z) = 0$, put $\rho = J(z)$. Since

$$\Re J(z) = \Re z (1 + |z|^{-2}),$$

the inequality $\Re \rho < 0$ implies $\Re z < 0$. The points $z = \pm 1$, at which J' vanishes, are not zeros because $H(2) > 0$ and $H(-2) < 0$. Hence simplicity follows from the chain rule. Finally an algebraic zero z would make $J(z)$ an algebraic zero of H , which has just been excluded. $\square$

27.2 The Euler point as an explicit E - D collision

Recall that $w_* \in (-2, 2)$ is the unique real solution of $H(w) = 2\gamma$, and let α_* and α_*^{-1} be its two preimages under J . Define

$$\mathcal{G}(u) := eH'(2+u) = e^{-u/2} \sum_{m \geq 0} \frac{(u(u+4)/4)^m}{(2m+1)!}, \quad (235)$$

$$\mathcal{K}(u) := -\frac{1}{2} \int_0^u \mathcal{G}(t) dt. \quad (236)$$

Both functions are entire, D -finite, and have rational Taylor coefficients. Direct integration gives the useful identity

$$\mathcal{K}(u) = \frac{e}{2} (H(2) - H(2+u)). \quad (237)$$

There is a tempting but false shortcut here: rational Taylor coefficients and holonomicity do not make $\mathcal{K}$ a Siegel E -function.

Proposition 27.4 (The auxiliary primitive is not an E -function). *Neither $\mathcal{G}$ nor $\mathcal{K}$ is a Siegel E -function. Nevertheless, if $u \in \overline{\mathbb{Q}}$, then $\mathcal{K}(u)$ belongs to the ring $\mathbf{E}$ of finite E -values.*

Proof. Writing $y = \mathcal{G}$, the differential equation (212) shifted by $w = 2 + u$ gives

$$u(u+2)(u+4)y'' + (u^3 + 8u^2 + 16u + 12)y' + (u^2 + 3u + 4)y = 0. \quad (238)$$

The function $y = eH'(2+u)$ has infinitely many zeros, by (211). It therefore cannot satisfy a first-order homogeneous equation over $\overline{\mathbb{Q}}(u)$: a nonzero entire solution of such an equation can have zeros only among the finitely many poles of its rational logarithmic derivative. Thus the order-two operator in (238) is minimal. At $u = -4$ its indicial polynomial is

$$8r(r-1) + 12r = 4r(2r+1),$$

so its local exponents are 0 and $-1/2$. Hence -4 is a non-apparent nonzero finite singularity. The finite-singularity theorem for minimal annihilators of E -functions says that every nonzero finite singularity is apparent [3]. Consequently $\mathcal{G}$ is not an E -function. Since E -functions are closed under differentiation and $\mathcal{K}' = -\mathcal{G}/2$, neither is $\mathcal{K}$.

For the last assertion, choose algebraic $a \neq 0$ satisfying $a + a^{-1} = 2 + u$. Equations (207) and (237) give

$$\mathcal{K}(u) = \frac{e}{2}(2\text{Ein}(1) - \text{Ein}(a) - \text{Ein}(a^{-1})). \quad (239)$$

The right side belongs to $\mathbf{E}$, since e and the three displayed finite algebraic-point values are E -values and $\mathbf{E}$ is a ring. $\square$

Proposition 27.5 (Equivalent algebraic witnesses). *The following assertions are equivalent:*

- (i) $w_* \in \overline{\mathbb{Q}}$;
- (ii) $\alpha_* \in \overline{\mathbb{Q}}$;
- (iii) *there exists $a \in \overline{\mathbb{Q}}^\times \setminus (-\infty, 0]$ such that, on the principal branch,*

$$E_1(a) + E_1(a^{-1}) = 0;$$

- (iv) *there exists $u \in \overline{\mathbb{Q}} \cap (-4, 0)$ such that*

$$\mathcal{K}(u) = \delta.$$

If these conditions hold, then γ and δ are algebraically independent.

Proof. The equivalence of (i) and (ii) follows from $\alpha_*^2 - w_*\alpha_* + 1 = 0$ and $w_* = \alpha_* + \alpha_*^{-1}$. For a point in the domain specified in (iii), the inverse-compatible principal branches satisfy

$$E_1(z) + E_1(z^{-1}) = H(J(z)) - 2\gamma$$

and this identity shows that (ii) implies (iii). Conversely an algebraic solution in (iii) gives an algebraic zero $J(a)$ of $H - 2\gamma$. By Theorem 26.7, every algebraic zero of this Euler-level translate must equal w_* , proving (i); the same theorem then forces $a \in \{\alpha_*, \alpha_*^{-1}\}$.

Put $u_* = w_* - 2 \in (-4, 0)$. Since

$$H(2) = 2\text{Ein}(1) = 2\gamma + \frac{2\delta}{e},$$

identity (237) gives $\mathcal{K}(u_*) = \delta$, so (i) implies (iv). Conversely, if $u \in (-4, 0)$ satisfies $\mathcal{K}(u) = \delta$, the same identity gives $H(2+u) = 2\gamma$. Uniqueness on the real axis forces $2+u = w_*$, proving (i). The final assertion is the witness amplification statement in Theorem 26.7. $\square$

Corollary 27.6 (Exact E – D consequence and the Euler dichotomy). *Let $\mathbf{E}$ and $\mathbf{D}$ denote the rings of E -values and arithmetic–Gevrey values of order one, respectively, in the sense of Fischler–Rivoal [22]. If w_* is algebraic, then*

$$\delta \in (\mathbf{E} \cap \mathbf{D}) \setminus \overline{\mathbb{Q}}.$$

Consequently the conjectural equality $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ implies that w_ is transcendental. Unconditionally, exactly one of the following alternatives holds:*

- (a) *every zero of $H - 2\gamma$, and every zero of the reciprocal Euler equation, is transcendental;*
- (b) *the Euler level has the unique algebraic reciprocal pair $\{\alpha_*, \alpha_*^{-1}\}$, and γ, δ are algebraically independent.*

Proof. Under the algebraicity hypothesis, Proposition 27.5 writes δ as $\mathcal{K}(u_*)$ at an algebraic point. Formula (239), rather than the false assertion that $\mathcal{K}$ itself is an E -function, gives $\delta \in \mathbf{E}$. The inclusion $\delta \in \mathbf{D}$ is the standard Euler-series realization of the Gompertz constant, and the algebraic independence conclusion in Proposition 27.5 shows that δ is transcendental. This proves the first assertion.

Theorem 26.7 says that w_* is the only possible algebraic zero at the Euler level. If it is transcendental, an algebraic reciprocal zero is impossible because its Joukowski image would be algebraic. If it is algebraic, its two Joukowski preimages are algebraic, they are the only algebraic reciprocal zeros, and witness amplification gives the stated algebraic independence. $\square$

27.3 A level-zero de Branges–Clark model

For an entire function E , write $E^\#(z) = \overline{E(\bar{z})}$. Define

$$E_0(z) = e^{-iz/2} H(-iz), \quad \Theta(z) = \frac{E_0^\#(z)}{E_0(z)} = e^{iz} \frac{H(iz)}{H(-iz)}. \quad (240)$$

Theorem 27.7 (Hermite–Biehler and Clark structure). *The function E_0 is a Cartwright-class Hermite–Biehler function of exponential type $1/2$. Consequently Θ is a meromorphic inner function in the upper half-plane and has zero mean type. The zeros of Θ are the points $-i\rho$, where ρ ranges over the zeros of H . Thus E_0 defines a de Branges space, and for each $\alpha \in \mathbb{T}$ the real solutions of*

$$\Theta(x) = \alpha \quad (241)$$

form the corresponding discrete Clark, equivalently self-adjoint, spectrum.

Proof. Corollary 27.3 places every zero ρ of H in the open left half-plane. Hence the zeros $i\rho$ of E_0 lie in the lower half-plane, and E_0 has no real zeros. The connection formula and the standard sectorial expansion of E_1 give, as $x \rightarrow +\infty$,

$$H(ix) = \gamma + \log x + \frac{\pi i}{2} + O(x^{-1}), \quad H(-ix) = \overline{H(ix)}. \quad (242)$$

The two-exponential formula (209), followed by one integration, gives the indicator

$$h_{E_0}(\vartheta) = \frac{1}{2} |\sin \vartheta|.$$

In particular E_0 has exponential type $1/2$. Equation (242) shows that $|E_0(x)| = O(\log(2 + |x|))$ on the real axis, so E_0 belongs to the Cartwright class.

It follows that $\Theta = E_0^\# / E_0$ is analytic and of bounded type in the upper half-plane, with unimodular boundary values. For $y \rightarrow +\infty$, the real-axis asymptotics obtained from the Joukowski representation are

$$H(y) = \gamma + \log y + O(y^{-1}), \quad H(-y) = -\frac{e^y}{y}(1 + O(y^{-1})).$$

Therefore

$$\Theta(iy) = e^{-y} \frac{H(-y)}{H(y)} = -\frac{1 + o(1)}{y \log y}. \quad (243)$$

Thus Θ has zero mean type. Bounded-type factorization, together with its unimodular boundary values, now shows that Θ is inner: meromorphic continuation across the real axis removes the singular-inner measure, while zero mean type removes the exponential factor in the canonical factorization [14, 11]. It is nonconstant, so the Hermite–Biehler inequality is strict. Meromorphic continuation follows from (240). The description of its zeros is immediate from $H(iz) = 0$, and the final assertion is the standard meromorphic-inner Clark/de Branges correspondence [11, 14]. $\square$

Theorem 27.8 (Arithmetic independence of the Clark data). *Let $x_1, \dots, x_m \in \overline{\mathbb{Q}} \cap \mathbb{R}^\times$, and suppose that $x_j^2 \neq x_k^2$ for $j \neq k$. Then the m numbers*

$$R(x_j) := \frac{H(ix_j)}{H(-ix_j)}$$

are algebraically independent over $\mathcal{E}_{\text{exp}}$. Separately, the m numbers

$$\Theta(x_j) = e^{ix_j} R(x_j)$$

are algebraically independent over $\mathcal{E}_{\text{exp}}$. In particular each of

$$\{R(n) : n \geq 1\}, \quad \{\Theta(n) : n \geq 1\}$$

is a countably algebraically independent family.

If $\alpha \in \mathbb{Q} \cap \mathbb{T}$, every real solution of $\Theta(x) = \alpha$ is transcendental, apart from the unavoidable solution $x = 0$ when $\alpha = 1$.

Proof. The $2m$ algebraic points $ix_j, -ix_j$ are distinct. Their Joukowski fibres are pairwise disjoint. Theorem 5.1, followed by the same disjoint-support linear projection used in Theorem 26.4, shows that

$$H(ix_1), H(-ix_1), \dots, H(ix_m), H(-ix_m)$$

are algebraically independent over $\mathcal{E}_{\text{exp}}$. Ratios made from disjoint pairs of independent indeterminates remain algebraically independent: indeed, a polynomial relation among the ratios becomes, after clearing the denominator monomial, a nonzero polynomial relation among the original indeterminates. Multiplication by the nonzero coefficients $e^{ix_j} \in \mathcal{E}_{\text{exp}}$ preserves independence.

If a nonzero algebraic real x satisfied $\Theta(x) = \alpha$, then

$$e^{ix} H(ix) - \alpha H(-ix) = 0$$

would be a nontrivial linear relation over $\mathcal{E}_{\text{exp}}$ between two independent values. This is impossible. At $x = 0$, one has $\Theta(0) = 1$, which explains the unique exception. $\square$

Remark 27.9 (Why signs must not be duplicated). The restriction $x_j^2 \neq x_k^2$ is necessary: on the real axis

$$\Theta(-x) = \Theta(x)^{-1}, \quad R(-x) = R(x)^{-1}.$$

Thus a family containing both x and $-x$ has an immediate algebraic relation.

27.4 Recovery of γ from arithmetic scattering data

For $n \geq 1$, put

$$a_n = \frac{n + \sqrt{n^2 + 4}}{2}, \quad R_n = R(n) = \frac{H(in)}{H(-in)} = e^{-in}\Theta(n), \quad T_n = \frac{\pi i R_n + 1}{2 R_n - 1}. \quad (244)$$

Theorem 27.8 implies $R_n \neq 1$, so T_n is well defined. Since $|R_n| = 1$, it is real.

Theorem 27.10 (Scattering recovery and its first correction). *One has*

$$\gamma = \lim_{n \rightarrow \infty} (T_n - \log n), \quad T_n - \log n = \gamma + O\left(\frac{\log n}{n}\right). \quad (245)$$

More precisely, the completely explicit corrected estimator

$$\hat{\gamma}_n = \left(1 - \frac{2(1 + \cos a_n)}{\pi a_n}\right) T_n + \frac{\sin a_n}{a_n} - \log a_n \quad (246)$$

satisfies

$$\hat{\gamma}_n = \gamma + O\left(\frac{\log n}{n^2}\right). \quad (247)$$

Thus γ is recovered from ratios of values generated by Ein at moving algebraic points, even though every finite family of those ratios is algebraically independent over $\mathcal{E}_{\text{exp}}$.

Proof. The identity $a_n - a_n^{-1} = n$ gives $J(ia_n) = in$, hence

$$H(in) = \text{Ein}(ia_n) + \text{Ein}(-i/a_n). \quad (248)$$

Put $L_n = \gamma + \log a_n$ and $b = \pi/2$. On the positive imaginary axis, the connection formula and the standard asymptotic expansion

$$E_1(ia) = \frac{e^{-ia}}{ia} \left(1 - \frac{1}{ia} + O(a^{-2})\right)$$

combine with $\text{Ein}(-i/a) = -i/a + 1/(4a^2) + O(a^{-3})$. Separating real and imaginary parts in (248) gives

$$\Re H(in) = L_n - \frac{\sin a_n}{a_n} + \frac{\cos a_n + 1/4}{a_n^2} + O(a_n^{-3}), \quad (249)$$

$$\Im H(in) = b - \frac{1 + \cos a_n}{a_n} - \frac{\sin a_n}{a_n^2} + O(a_n^{-3}). \quad (250)$$

Because $H(-in) = \overline{H(in)}$, direct algebra gives

$$T_n = b \frac{\Re H(in)}{\Im H(in)}. \quad (251)$$

Equations (249)–(251) first yield $T_n = L_n + O(L_n/a_n)$. Since $a_n = n + O(n^{-1})$, this proves (245).

For the sharper statement, multiply (251) by

$$1 - \frac{1 + \cos a_n}{ba_n}$$

and add $\sin(a_n)/a_n$. The terms of order a_n^{-1} in both the numerator and denominator cancel, leaving

$$\left(1 - \frac{1 + \cos a_n}{ba_n}\right) T_n + \frac{\sin a_n}{a_n} = L_n + O\left(\frac{L_n}{a_n^2}\right).$$

Since $b = \pi/2$, subtraction of $\log a_n$ gives (246) and (247). $\square$

27.5 An exact inverse-resonance trace

Theorem 27.11 (Inverse-resonance trace). *Let $Z(H)$ denote the multiset of all zeros of H . Then*

$$\sum_{\rho \in Z(H)} \frac{-2\Re \rho}{|\rho|^2} = 1 + \frac{\sin 1}{\text{Cin}(1)}. \quad (252)$$

The series converges absolutely, every summand is positive, and its value is transcendental over $\mathcal{E}_{\text{exp}}$, hence also over $\overline{\mathbb{Q}}(e, e^i)$.

Proof. By Theorem 27.7, Θ is a meromorphic inner function of zero mean type. Meromorphic continuation across the real axis excludes a finite singular measure in its inner factorization, and zero mean type excludes an exponential inner factor. Thus Θ is a pure Blaschke product. Its zeros are

$$\zeta_\rho = -i\rho \in \mathbb{C}_+ \quad (\rho \in Z(H)).$$

Logarithmic differentiation of the Blaschke product at zero gives

$$\frac{1}{i} \frac{\Theta'(0)}{\Theta(0)} = \sum_{\rho \in Z(H)} \frac{2\Im \zeta_\rho}{|\zeta_\rho|^2} = \sum_{\rho \in Z(H)} \frac{-2\Re \rho}{|\rho|^2}. \quad (253)$$

The Blaschke condition gives absolute convergence; equivalently, all terms are positive and there are no zeros near the origin.

On the other hand, $\Theta(0) = 1$, $H(0) = 2 \text{Cin}(1)$, and $H'(0) = \sin 1$. Differentiating (240) directly yields

$$\frac{1}{i} \frac{\Theta'(0)}{\Theta(0)} = 1 + 2 \frac{H'(0)}{H(0)} = 1 + \frac{\sin 1}{\text{Cin}(1)}. \quad (254)$$

Together with (253), this proves the trace identity.

Finally $\sin 1 \in \mathcal{E}_{\text{exp}}^\times$, whereas

$$\text{Cin}(1) = \frac{\text{Ein}(i) + \text{Ein}(-i)}{2}$$

is transcendental over $\mathcal{E}_{\text{exp}}$ by Theorem 5.1. Hence the right side of (252) is transcendental over $\mathcal{E}_{\text{exp}}$. $\square$

Remark 27.12 (Logical scope). The stability and Clark results above are unconditional consequences of the Joukowski descent and the multipoint Ein theorem. The recovery formula (245) is an asymptotic encoding of γ , not an irrationality criterion: passing from algebraic independence at each finite set of moving points to an arithmetic statement about the limit requires a genuinely uniform measure. Likewise Proposition 27.5 isolates an exact algebraic witness but does not establish that the witness exists.

28 Reciprocal traces, phase recovery, and height saturation

This section separates two reciprocal functions which have different arithmetic meanings. Put

$$\Psi(z) = \text{Ein}(z) + \text{Ein}(z^{-1}) = H(J(z)), \quad \mathcal{F}_\gamma(z) = E_1(z) + E_1(z^{-1}) = \Psi(z) - 2\gamma, \quad (255)$$

where branches are chosen coherently so that $\text{Log}(z^{-1}) = -\text{Log } z$. The first function has rational Laurent coefficients and is genuinely independent of γ . The second is the Euler-level

translate of the first; the opposite monodromies of its two E_1 terms cancel, so the displayed sum denotes its single-valued analytic continuation on $\mathbb{C}^\times$.

This distinction is also essential for effectivity. The entire function Ein is a Siegel E -function, whereas E_1 has a logarithmic singularity at the origin and is not an E -function. In particular, an algebraic approximation to a zero of $\mathcal{F}_\gamma$ does not put all terms of

$$\text{Log } z + E_1(z) + \text{Ein}(z^{-1})$$

inside the effective theory of E -values. Indeed the connection formula gives the identity

$$\text{Log } z + E_1(z) + \text{Ein}(z^{-1}) = \Psi(z) - \gamma = \gamma + \mathcal{F}_\gamma(z). \quad (256)$$

Thus evaluation at a zero of $\mathcal{F}_\gamma$ recovers γ , but only tautologically. The nontrivial recovery below uses instead the zero divisor and the asymptotic phase of reciprocal functions.

28.1 An exact Hadamard trace at the Euler level

Let $\{u_k\}$ be the zeros of $H(u) - 2\gamma$, repeated according to multiplicity. The index set includes the unique real zero from Theorem 26.7 and both members of every nonreal conjugate pair.

Theorem 28.1 (Euler-level Hadamard trace). *The series $\sum_k |u_k|^{-2}$ converges and*

$$\boxed{\sum_k \frac{1}{u_k^2} = \frac{\sin^2 1 - \sin 1 \, \text{Ci}(1)}{4 \, \text{Ci}(1)^2}}. \quad (257)$$

Consequently, if T denotes the left side, then

$$\text{Ci}(1) = \frac{\sin 1(\sqrt{1+16T} - 1)}{8T}, \quad \gamma = \text{Cin}(1) + \frac{\sin 1(\sqrt{1+16T} - 1)}{8T}. \quad (258)$$

Proof. Set $f(u) = H(u) - 2\gamma$. By Corollary 26.10, f is entire of order one. Hence its zero divisor has exponent of convergence at most one and admits a genus-one Hadamard factorization. Since

$$f(0) = 2 \, \text{Cin}(1) - 2\gamma = -2 \, \text{Ci}(1) \neq 0,$$

logarithmic differentiation twice at the origin gives

$$-\left(\frac{f'}{f}\right)'(0) = \sum_k \frac{1}{u_k^2}. \quad (259)$$

The product formula for H' in Proposition 26.1 gives the exact origin jet

$$H'(0) = \sin 1, \quad H''(0) = -\frac{1}{2} \sin 1.$$

Therefore

$$-\left(\frac{f'}{f}\right)'(0) = \frac{f'(0)^2 - f''(0)f(0)}{f(0)^2} = \frac{\sin^2 1 - \sin 1 \, \text{Ci}(1)}{4 \, \text{Ci}(1)^2},$$

which proves (257). The standard bounds $0 < \text{Ci}(1) < \sin 1$ (which also follow directly from the corresponding alternating series and elementary bounds for γ) show that the trace T is positive. Solving $4TX^2 + \sin(1)X - \sin^2(1) = 0$ for the positive root $X = \text{Ci}(1)$ gives (258). $\square$

Remark 28.2 (What the trace does not prove). Formula (258) is an exact spectral reparametrization, not a transcendence proof: the divisor on the left is itself the divisor of the γ -dependent level $H - 2\gamma$. No arithmetic assertion in this paper relies on a numerical zero count. In particular, an exact floor formula suggested by finite computations is not used; the rigorous bounds remain those of Corollary 26.10.

28.2 A nontrivial phase recovery from the γ -free zero set

The situation is different for zeros of Ψ , because the defining Laurent series then contains no γ . Let $z_{k,0}$ be the upper half-plane zero with $|z_{k,0}| > 1$ constructed in the proof of Theorem 26.9 at level $c = 0$, and write $\rho_k = z_{k,0}$. Put

$$t_k = 2\pi k, \quad q_k = \log t_k.$$

Theorem 28.3 (Second-order zero phase). *As $k \rightarrow \infty$,*

$$\begin{aligned} \rho_k = & -q_k - \frac{1}{2} \log \left((q_k + \gamma)^2 + \frac{\pi^2}{4} \right) \\ & + i \left[t_k + \arctan \left(\frac{2(q_k + \gamma)}{\pi} \right) \right] + O \left(\frac{q_k}{t_k} \right). \end{aligned} \quad (260)$$

In particular the zero set of the γ -free function Ψ determines γ through

$$\boxed{\gamma = \lim_{k \rightarrow \infty} \left\{ \frac{\pi}{2} \tan(\Im \rho_k - 2\pi k) - \log(2\pi k) \right\}}. \quad (261)$$

More precisely, the expression in braces is

$$\gamma + O \left(\frac{(\log k)^3}{k} \right). \quad (262)$$

Every ρ_k is transcendental.

Proof. At level $c = 0$, the fixed-point equation in the proof of Theorem 26.9 is

$$z = (2k + 1)\pi i - \text{Log } z - \text{Log}(\text{Log } z + \gamma). \quad (263)$$

The exact zero ρ_k differs from its model fixed point by $O(t_k^{-1})$. Uniformly in the unit disc used there,

$$\text{Log } z = q_k + \frac{\pi i}{2} + O(q_k/t_k),$$

and hence

$$\begin{aligned} \text{Log}(\text{Log } z + \gamma) = & \frac{1}{2} \log \left((q_k + \gamma)^2 + \frac{\pi^2}{4} \right) \\ & + i \arctan \left(\frac{\pi}{2(q_k + \gamma)} \right) + O(q_k/t_k). \end{aligned}$$

Substitution in (263), followed by

$$\frac{\pi}{2} - \arctan \left(\frac{\pi}{2x} \right) = \arctan \left(\frac{2x}{\pi} \right) \quad (x > 0),$$

proves (260).

Write $\Im \rho_k - t_k = \arctan(2(q_k + \gamma)/\pi) + \varepsilon_k$, where $\varepsilon_k = O(q_k/t_k)$. Since $q_k/t_k = o(q_k^{-1})$, the whole mean-value segment remains a distance asymptotic to q_k^{-1} from the nearest pole of $\tan$. Consequently $\sec^2 = O(q_k^2)$ uniformly there,

$$\frac{\pi}{2} \tan(\Im \rho_k - t_k) = q_k + \gamma + O(q_k^3/t_k).$$

The error tends to zero and proves both (261) and (262). Finally $\Psi(\rho_k) = 0$, so ρ_k lies over the algebraic level $c = 0$; its transcendence follows from Corollary 26.5. $\square$

Theorem 28.3 is analytically nonvacuous: unlike (256), the object whose zeros are used has rational Laurent coefficients. It still gives no irrationality result, because no effective arithmetic control of these transcendental zeros is established here.

28.3 Laurent truncations saturate the height budget

Define

$$\Psi_N(z) = \sum_{m=1}^N \frac{(-1)^{m+1}}{m m!} (z^m + z^{-m}), \quad L_N = \text{lcm}(1, \dots, N), \quad (264)$$

and

$$D_N^* = \text{lcm}_{1 \leq m \leq N} (m m!), \quad Q_N(z) = D_N^* z^N \Psi_N(z). \quad (265)$$

Lemma 28.4 (The denominator scale is intrinsic). *One has*

$$Q_N \in \mathbb{Z}[z], \quad Q_N \text{ primitive}, \quad \deg Q_N = 2N, \quad H(Q_N) = D_N^*,$$

and therefore

$$\log H(Q_N) = N \log N + o(N). \quad (266)$$

Moreover,

$$N! \prod_{p \leq N} p \mid D_N^* \mid N! L_N, \quad (267)$$

so $\log D_N^* = N \log N + o(N)$ as well.

Proof. The coefficients in (265) are $D_N^*/(m m!)$, hence are integral and have maximum absolute value D_N^* , attained at $m = 1$. Their gcd is one: otherwise division by a common prime would give a smaller common denominator than D_N^* . This proves the first assertions.

For the right divisibility in (267), use $m! \mid N!$ and $m \mid L_N$. For the left divisibility, fix a prime $p \leq N$ and take $m = p \lfloor N/p \rfloor$. There is no multiple of p strictly between m and N , so $v_p(m!) = v_p(N!)$ and $v_p(m m!) \geq v_p(N!) + 1$. Taking the least common multiple over m adds at least one copy of every prime $p \leq N$ to $N!$. The prime number theorem and Stirling's formula give (266). $\square$

Theorem 28.5 (Factorial approximation and defining-height saturation). *Let ρ be a fixed simple zero of Ψ with $|\rho| > 1$. For all sufficiently large N , Ψ_N has a zero α_N converging to ρ , and*

$$\alpha_N - \rho = \frac{(-1)^{N+2} \rho^{N+1}}{\Psi'(\rho)(N+1)(N+1)!} \left(1 + O_\rho(N^{-1})\right). \quad (268)$$

Consequently

$$\frac{-\log |\alpha_N - \rho|}{\log H(Q_N)} \longrightarrow 1. \quad (269)$$

Here Q_N is the primitive integer polynomial obtained by minimal denominator clearing; no claim about the minimal-polynomial height of α_N is made.

Proof. On a fixed neighbourhood of ρ , the tail $R_N = \Psi - \Psi_N$ satisfies

$$R_N(z) = \frac{(-1)^{N+2} z^{N+1}}{(N+1)(N+1)!} \left(1 + O_\rho(N^{-1})\right). \quad (270)$$

Indeed the ratio of consecutive exterior terms is $O_\rho(N^{-1})$, while the terms in z^{-m} are exponentially smaller. Rouché's theorem on a small circle about the simple zero ρ gives a unique nearby zero α_N of Ψ_N . The standard perturbation formula yields

$$\alpha_N - \rho = \frac{R_N(\rho)}{\Psi'(\rho)} \left(1 + O_\rho(N^{-1})\right),$$

which is (268). Stirling's formula and Lemma 28.4 give

$$-\log |\alpha_N - \rho| = N \log N + O_\rho(N), \quad \log H(Q_N) = N \log N + o(N),$$

and hence (269). $\square$

Corollary 28.6 (Full primitive integerization destroys the small value). *Under the hypotheses of Theorem 28.5,*

$$Q_N(\rho) = (-1)^{N+3} \frac{D_N^*}{(N+1)(N+1)!} \rho^{2N+1} \left(1 + O_\rho(N^{-1})\right). \quad (271)$$

In particular

$$\log |Q_N(\rho)| = N(1 + 2 \log |\rho|) + o(N) > 0. \quad (272)$$

Also

$$|\Psi(\alpha_N)| = H(Q_N)^{-1+o(1)}. \quad (273)$$

Proof. Since $\Psi(\rho) = 0$, one has $Q_N(\rho) = -D_N^* \rho^N R_N(\rho)$. Substitution of (270) gives (271). Lemma 28.4 and Stirling's formula then give (272). Finally $\Psi_N(\alpha_N) = 0$, so the same tail estimate at α_N and (266) give (273). $\square$

The last corollary closes the naive use of this full primitive clearing polynomial. Taken alone, it does not exclude systematic factorization or a minimal-height compression for the selected root; the dyadic theorem in Section 29 closes precisely that escape on $N = 2^r$. The factorial analytic error is real, but after denominator clearing the full primitive polynomial value grows rather than tends to zero. It also explains why the pointwise measure of Adamczewski–Faverjon cannot simply be compared with $H(Q_N)$: in their theorem the height belongs to the relation polynomial in fixed E -values, whereas here the algebraic evaluation point and its degree move with N .

28.4 The exterior-rank barrier

Let C_m be the monic Chebyshev polynomials from (206) and set

$$H_N(w) = \sum_{m=1}^N \frac{(-1)^{m+1}}{m m!} C_m(w). \quad (274)$$

Theorem 28.7 (The level first appears in the top norm). *Fix a level $\ell \in \mathbb{C}$ and let $w_{1,N}, \dots, w_{N,N}$ be the roots, with multiplicity, of $H_N(w) - \ell$. Then the elementary symmetric functions*

$$e_1(w_{1,N}, \dots, w_{N,N}), \dots, e_{N-1}(w_{1,N}, \dots, w_{N,N})$$

are independent of ℓ , while

$$\prod_{j=1}^N w_{j,N} = N N! (\ell - H_N(0)). \quad (275)$$

Thus the first elementary symmetric invariant which sees the level is the full rank- N norm, and it carries the factorial factor $N!$.

Proof. Since C_N is monic, the leading coefficient of $H_N - \ell$ is $(-1)^{N+1}/(N N!)$. The level ℓ changes only the constant coefficient. Vieta's formulas therefore make $e_1, \dots, e_{N-1}$ level-independent, and give

$$e_N = (-1)^N \frac{H_N(0) - \ell}{(-1)^{N+1}/(N N!)} = N N! (\ell - H_N(0)).$$

$\square$

At the Euler level $\ell = 2\gamma$, since $H_N(0) \rightarrow H(0) = 2 \operatorname{Cin}(1)$, one has

$$\prod_{j=1}^N w_{j,N} = 2 \operatorname{Ci}(1) N N! (1 + o(1)), \quad \log \left| \prod_{j=1}^N w_{j,N} \right| = N \log N - N + O(\log N).$$

Thus, by Newton's identities, the power sums of order below N and all proper exterior traces of the finite zero set forget γ ; the first elementary symmetric invariant which remembers it pays the full determinant cost. A possible route not excluded by the theorem would have to remove the universal $N \log N$ contribution canonically, for example through a zeta-regularized determinant or an arithmetic spectral projector. No such arithmetic regularization is constructed here.

29 Dyadic irreducibility and exact capacity limitations

29.1 Dyadic irreducibility and minimal Weil-height saturation

The defining-height result above leaves open, for a general truncation index, the possibility that the root selected near a zero of Ψ belongs to a factor of much smaller degree or height. On the dyadic subsequence this possibility can be excluded completely. Recall H_N , D_N^* , and Q_N from (274) and (265), and put

$$P_N(w) = D_N^* H_N(w), \quad A_N = \frac{D_N^*}{N N!}. \quad (276)$$

Thus, with $b_{m,N} = D_N^*/(m m!)$,

$$P_N(w) = \sum_{m=1}^N (-1)^{m+1} b_{m,N} C_m(w), \quad (277)$$

$$Q_N(z) = z^N P_N(z + z^{-1}) = \sum_{m=1}^N (-1)^{m+1} b_{m,N} (z^{N+m} + z^{N-m}). \quad (278)$$

Theorem 29.1 (Dyadic Eisenstein and reciprocal irreducibility). *Let $N = 2^r \geq 2$. Then A_N is an odd integer and*

$$\boxed{P_N(w) \equiv -A_N(w^N + 2) \pmod{4}.} \quad (279)$$

Consequently P_N is a primitive 2-Eisenstein polynomial and is irreducible over $\mathbb{Q}$ of degree N . Moreover, Q_N is primitive and irreducible over $\mathbb{Q}$ of degree $2N$.

Proof. Legendre's formula gives

$$v_2(N!) = N - s_2(N) = N - 1,$$

where $s_2(N)$ is the sum of the binary digits of N . Hence

$$v_2(N N!) = N + r - 1. \quad (280)$$

If $1 \leq m < N$, then $v_2(m) \leq r - 1$ and $v_2(m!) \leq m - 1$, so

$$v_2(m m!) \leq m + r - 2 \leq N + r - 3.$$

It follows that the maximum defining $v_2(D_N^*)$ is attained at $m = N$, that A_N is odd, and that

$$4 \mid b_{m,N} \quad (1 \leq m < N). \quad (281)$$

The identity $C_{2m} = C_m^2 - 2$, beginning with $C_2(w) = w^2 - 2$, gives by induction

$$C_{2^r}(w) \equiv w^{2^r} + 2 \pmod{4}. \quad (282)$$

Since N is even, the coefficient of C_N in (277) is $-A_N$. Equations (281) and (282) prove (279). Thus the leading coefficient of P_N is odd, all its remaining coefficients are even, and its constant coefficient is congruent to $-2A_N$ modulo 4. The polynomial Q_N is primitive by Lemma 28.4; hence P_N is primitive as well, because any common divisor of the coefficients of P_N would divide every coefficient of $z^N P_N(z + z^{-1}) = Q_N$. Eisenstein's criterion now proves the irreducibility of P_N .

It remains to check that the reciprocal substitution does not split the polynomial. Since $C_m(2) = 2$ and $C_m(-2) = 2(-1)^m$,

$$P_N(2) = 2D_N^* \sum_{m=1}^N \frac{(-1)^{m+1}}{m m!} > 0, \quad (283)$$

$$P_N(-2) = -2D_N^* \sum_{m=1}^N \frac{1}{m m!} < 0. \quad (284)$$

The first sign follows by pairing consecutive terms of the decreasing alternating sum, since N is even.

Let θ be a root of P_N , let $K = \mathbb{Q}(\theta)$, and write $a_N = \text{lc}(P_N) = -A_N$. Because P_N is irreducible of degree N ,

$$N_{K/\mathbb{Q}}(\theta^2 - 4) = \frac{P_N(2)P_N(-2)}{a_N^2} < 0. \quad (285)$$

A square in K has norm equal to a square in $\mathbb{Q}$, so $\theta^2 - 4$ is not a square in K . Therefore $X^2 - \theta X + 1$ is irreducible over K . If α is one of its roots, then $\theta = \alpha + \alpha^{-1}$ and consequently

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] = 2[K : \mathbb{Q}] = 2N.$$

The number α is a zero of Q_N , whose degree is $2N$. Thus Q_N is irreducible; its primitivity was already established in Lemma 28.4. $\square$

Theorem 29.2 (Dyadic minimal-height saturation). *Let ρ be a fixed simple zero of Ψ with $|\rho| > 1$, and, for all sufficiently large N , let α_N be the zero of Ψ_N converging to ρ in Theorem 28.5. Along $N = 2^r \rightarrow \infty$,*

$$[\mathbb{Q}(\alpha_N) : \mathbb{Q}] = 2N, \quad h(\alpha_N) = \frac{1}{2} \log N + O(1). \quad (286)$$

More precisely,

$$\frac{-\log |\alpha_N - \rho|}{[\mathbb{Q}(\alpha_N) : \mathbb{Q}] h(\alpha_N)} \longrightarrow 1, \quad (287)$$

$$\frac{-\log |\Psi(\alpha_N)|}{[\mathbb{Q}(\alpha_N) : \mathbb{Q}] h(\alpha_N)} \longrightarrow 1. \quad (288)$$

Here h is the absolute logarithmic Weil height. In particular, Q_N , up to sign, is the primitive integer minimal polynomial of α_N ; the full clearing polynomial is not concealing a factor of smaller degree or height on the dyadic subsequence.

Proof. By Theorem 29.1, Q_N is the primitive irreducible integer polynomial of α_N , up to sign. The coefficient–Mahler inequalities for a polynomial of degree $2N$, together with Lemma 28.4, give

$$4^{-N} D_N^* \leq M(Q_N) \leq \|Q_N\|_2 \leq \sqrt{2N+1} D_N^*. \quad (289)$$

Indeed, the lower bound follows from $H(Q_N) \leq 2^{2N} M(Q_N)$, while the upper bound is Landau’s inequality and $H(Q_N) = D_N^*$. Since $\log D_N^* = N \log N + o(N)$, it follows that

$$\log M(Q_N) = N \log N + O(N). \quad (290)$$

The degree assertion in (286) follows from Theorem 29.1, and the standard height identity gives

$$2N h(\alpha_N) = \log M(Q_N).$$

This proves the height estimate in (286).

The root asymptotic (268) gives

$$-\log |\alpha_N - \rho| = N \log N + O_\rho(N). \quad (291)$$

Moreover $\Psi_N(\alpha_N) = 0$, so the uniform tail estimate used in the proof of Theorem 28.5 yields

$$\Psi(\alpha_N) = \frac{(-1)^{N+2} \alpha_N^{N+1}}{(N+1)(N+1)!} \left(1 + O_\rho(N^{-1})\right).$$

Since $\alpha_N \rightarrow \rho$,

$$-\log |\Psi(\alpha_N)| = N \log N + O_\rho(N). \quad (292)$$

Dividing (291) and (292) by $2N h(\alpha_N) = N \log N + O(N)$ proves (287) and (288). $\square$

Corollary 29.3 (The precise moving-point threshold). *For every real $c < 1$, infinitely many algebraic numbers in the dyadic family above satisfy*

$$\log |\Psi(\alpha_N)| < -c [\mathbb{Q}(\alpha_N) : \mathbb{Q}] h(\alpha_N). \quad (293)$$

Equivalently, no estimate of the form

$$-\log |\Psi(\alpha)| \leq (c + o(1)) [\mathbb{Q}(\alpha) : \mathbb{Q}] h(\alpha), \quad c < 1,$$

can hold uniformly on any moving family containing these dyadic points. For every $c > 1$, the reverse strict inequality to (293) holds eventually along this family, so this sequence alone cannot contradict a bound with coefficient greater than one.

Proof. This is exactly the limit in (288). $\square$

Remark 29.4 (Scope of the obstruction). The corollary identifies the normalized threshold for the term $[\mathbb{Q}(\alpha) : \mathbb{Q}] h(\alpha)$ alone. It does not rule out estimates with additional terms of comparable size, such as $[\mathbb{Q}(\alpha) : \mathbb{Q}] \log [\mathbb{Q}(\alpha) : \mathbb{Q}]$, nor does it exclude nonlinear auxiliary constructions or non-dyadic phenomena. What it closes is the specific escape through a smaller irreducible factor or smaller minimal Weil height for the selected truncation root.

29.2 Two exact limitations of capacity normalization

The critical constants appearing in the reciprocal and Hankel constructions are stable under two natural operations. The statements below are deliberately limited to polynomial pullbacks of the standard adelic set and to changes of basis of the full Hankel determinant. They do not constitute a no-go theorem for weighted, place-dependent, or nonlinear auxiliary constructions. This is the boundary at which the adelic-capacity mechanism used in, for example, Dimitrov's Schinzel–Zassenhaus argument ceases to improve the present normalization [18]; we use the standard local capacity transformation law in the form reviewed by Bost and Chambert-Loir [6].

We normalize the absolute values of $\mathbb{Q}$ by $|p|_p = p^{-1}$ and the usual absolute value at infinity. At the archimedean place put

$$\mathcal{E}_\infty = [-2, 2],$$

and, at every finite place, let

$$\mathcal{E}_p = \{z \in \mathbb{A}_{\mathbb{C}_p}^{1,\text{an}} : |z|_p \leq 1\}$$

be the full closed Berkovich unit disk. Capacities below are logarithmic capacities relative to infinity.

Proposition 29.5 (Polynomial pullbacks preserve critical adelic capacity). *One has*

$$\text{cap}_\infty(\mathcal{E}_\infty) = 1, \quad \text{cap}_p(\mathcal{E}_p) = 1 \quad \text{for every prime } p.$$

Consequently the global capacity of the adelic set $\mathcal{E} = (\mathcal{E}_v)_{v \leq \infty}$ is

$$\prod_{v \leq \infty} \text{cap}_v(\mathcal{E}_v) = 1.$$

More generally, let

$$R(X) = a_d X^d + \cdots + a_0 \in \mathbb{Q}[X], \quad d \geq 1, \quad a_d \neq 0.$$

Then, at every place v ,

$$\text{cap}_v(R^{-1}(\mathcal{E}_v)) = \left(\frac{\text{cap}_v(\mathcal{E}_v)}{|a_d|_v} \right)^{1/d}, \quad (294)$$

and hence

$$\boxed{\prod_{v \leq \infty} \text{cap}_v(R^{-1}(\mathcal{E}_v)) = 1.} \quad (295)$$

Thus rational affine changes, rational polynomial pullbacks, and in particular rational Chebyshev pullbacks cannot turn this specific critical adelic capacity into a strict inequality below one.

Proof. The classical identity $\text{cap}([-2, 2]) = 1$ gives the archimedean assertion, and the closed Berkovich disk of radius one has nonarchimedean capacity one. The local polynomial transformation law for logarithmic capacity gives (294). Multiplying it over all places yields

$$\prod_v \text{cap}_v(R^{-1}(\mathcal{E}_v)) = \left(\frac{\prod_v \text{cap}_v(\mathcal{E}_v)}{\prod_v |a_d|_v} \right)^{1/d}.$$

The first product in the numerator is one, while the product formula for $a_d \in \mathbb{Q}^\times$ gives $\prod_v |a_d|_v = 1$. This proves (295). $\square$

Remark 29.6 (Why the finite-place set is not $\mathbb{Z}_p$). The full Berkovich unit disk in Proposition 29.5 must not be replaced silently by the classical compact subset $\mathbb{Z}_p \subset \mathbb{Q}_p$. With the standard logarithmic kernel,

$$\text{cap}_p(\mathbb{Z}_p) = p^{-1/(p-1)}, \quad (296)$$

not one. Indeed, if μ_p is normalized Haar measure on $\mathbb{Z}_p$, then

$$\int_{\mathbb{Z}_p^2} v_p(x-y) d\mu_p(x) d\mu_p(y) = \sum_{k \geq 1} \Pr(v_p(x-y) \geq k) = \sum_{k \geq 1} p^{-k} = \frac{1}{p-1}.$$

The corresponding equilibrium energy is $\log(p)/(p-1)$, which gives (296). In particular, the literal family with finite components $\mathbb{Z}_p$ does not have global capacity one.

We next make the critical cancellation in Theorem 19.1 integral. For $n \geq 1$, put

$$\mathcal{B}_n = n! \prod_{j=1}^n \binom{2j}{j} \binom{2j-1}{j}, \quad \ell_n = \text{lcm}(1, \dots, n), \quad \mathcal{I}_n = \ell_n \mathcal{B}_n. \quad (297)$$

Proposition 29.7 (Integral Hankel saturation and full-determinant rigidity). *For every $n \geq 1$ and every $x \in \mathbb{C}$,*

$$\mathcal{I}_n D_n(x) = (-1)^n (2\ell_n H_n - \ell_n x). \quad (298)$$

In particular, the two coefficients on the right side are integers. One has

$$\mathcal{B}_n^{1/n^2} \longrightarrow 4, \quad \mathcal{I}_n^{1/n^2} \longrightarrow 4. \quad (299)$$

For every fixed $x \in \mathbb{C}$,

$$|D_n(x)|^{1/n^2} \longrightarrow \frac{1}{4}, \quad \boxed{|\mathcal{I}_n D_n(x)|^{1/n^2} \longrightarrow 1}. \quad (300)$$

Let

$$M_n(x) = (H_{i+j} - x)_{0 \leq i, j \leq n}.$$

If $U_n, V_n \in \text{GL}_{n+1}(\mathbb{Z})$, then

$$\det(U_n M_n(x) V_n) = \pm D_n(x). \quad (301)$$

Hence unimodular row and column lattice reduction cannot alter the capacity constant of the full determinant.

Proof. Theorem 19.1 gives

$$D_n(x) = (-1)^n \frac{2H_n - x}{\mathcal{B}_n}.$$

Multiplication by $\mathcal{I}_n = \ell_n \mathcal{B}_n$ proves (298); the integrality follows from $\ell_n H_n \in \mathbb{Z}$. The Stirling estimate already used in the proof of Theorem 19.1 gives

$$\log \mathcal{B}_n = (\log 4)n^2 + O(n \log n).$$

The elementary Chebyshev bound $\log \ell_n = O(n)$ therefore proves (299). Since $H_n = \log n + O(1)$, for fixed x one has

$$\log |2H_n - x| = O(\log \log n)$$

for all sufficiently large n . Formula (300) now follows either directly or together with Theorem 19.1. Finally,

$$\det(U_n M_n(x) V_n) = \det(U_n) \det(V_n) D_n(x),$$

and both integer determinants are ± 1 , proving (301). $\square$

Remark 29.8 (Exact scope of the limitation). Proposition 29.7 concerns the covolume, or equivalently the full determinant, of this particular Hankel lattice. It does not exclude a short individual row or linear form produced by lattice reduction, a useful proper minor, a change of the underlying lattice, an x -dependent auxiliary construction, or a nonlinear combination of the moments. Likewise, Proposition 29.5 does not exclude weighted capacity, place-dependent changes of the local compact sets, proper local subsets, rational maps with poles, or nonpolynomial pullbacks. The proved obstruction is precisely that rational polynomial pullback and unimodular full-determinant normalization preserve the existing critical equality; no broader capacity or lattice no-go is asserted.

30 The operator carrier: normal form, determinants, and scope

We first isolate the universal operator-theoretic content of the construction. Throughout this section, inner products are linear in the second variable.

Proposition 30.1 (Rank-one similarity collapse). *Let E be a complex Hilbert space, let $0 \neq \psi, h \in E$, and put*

$$C = |\psi\rangle\langle h|, \quad \alpha = \langle h, \psi \rangle \neq 0.$$

Let S be the unilateral shift on $\ell^2(\mathbb{N}_0)$ and define

$$A = S^* \otimes C, \quad B = S \otimes C.$$

There is an $R \in \mathrm{GL}(E)$ and an orthogonal rank-one projection P_1 such that

$$RCR^{-1} = \alpha P_1.$$

Consequently the pair (A, B) is simultaneously similar to

$$((\alpha S^*) \oplus 0, (\alpha S) \oplus 0).$$

Moreover, if $P_0 = |e_0\rangle\langle e_0|$, then

$$[A, B] = P_0 \otimes C^2, \quad \mathrm{Tr}[A, B] = \alpha^2,$$

and hence

$$\det_F(e^A e^B e^{-A} e^{-B}) = e^{\alpha^2}. \quad (302)$$

Proof. The identity $C^2 = \alpha C$ shows that $P = C/\alpha$ is a bounded rank-one idempotent. The decomposition $E = \mathrm{ran} P \dot{+} \ker P$ is topological, so a bounded invertible change of coordinates sends P to an orthogonal rank-one projection. Tensoring this change of coordinates with the identity on ℓ^2 gives the asserted simultaneous normal form. Finally, $S^*S = I$, $SS^* = I - P_0$, and $\mathrm{tr}(C^2) = \alpha^2$. The last assertion is the Helton–Howe–Pincus formula; see, for example, [38]. $\square$

For the Euler specialization, let

$$H = L^2(\mathbb{R}_+, e^{-x} dx), \quad Xf(x) = xf(x), \quad A = X + I,$$

take $E = H \oplus \mathbb{C}$, and put

$$\psi = (-A^{-1}1, 1), \quad h_\gamma = (A \log X 1, 0).$$

Both vectors belong to E , and direct integration gives

$$\langle h_\gamma, \psi \rangle = - \int_0^\infty e^{-x} \log x \, dx = \gamma.$$

Thus $C = |\psi\rangle\langle h_\gamma|$ has $\alpha = \gamma$, and (302) becomes e^{γ^2} . The proposition also makes the limitation precise: the simultaneous similarity class sees only the scalar γ and is blind to the zero-mode norm $2 - \delta$ and to the remaining Hilbert-complex geometry.

There is, however, a natural relative Fredholm determinant on the δ -side. With the preceding H and A , put $u = A^{-1/2}1$ and let

$$Q : \mathcal{D}(A^{1/2}) \oplus \mathbb{C} \longrightarrow H, \quad Q(f, c) = A^{1/2}f + cu.$$

Proposition 30.2 (Zero-mode perturbation determinant). *The operator Q is a closed surjective Fredholm operator of index one,*

$$QQ^* = A + |u\rangle\langle u|, \quad \ker Q = \mathbb{C}\psi, \quad \psi = (-A^{-1}1, 1).$$

For $z \in \mathbb{C} \setminus [1, \infty)$ define

$$\Delta_Q(z) = \det_F \left((QQ^* - z)(A - z)^{-1} \right).$$

Here the product denotes its bounded extension from the natural common domain. Then

$$\begin{aligned} \Delta_Q(z) &= 1 + \langle u, (A - z)^{-1}u \rangle \\ &= 1 + \int_0^\infty \frac{e^{-x}}{(1+x)(1+x-z)} \, dx. \end{aligned} \tag{303}$$

Writing $\delta = eE_1(1)$ and taking the principal branch of E_1 , this is

$$\Delta_Q(z) = 1 + \frac{e^{1-z}E_1(1-z) - \delta}{z} \quad (z \neq 0),$$

with a removable value at zero given by

$$\Delta_Q(0) = 2 - \delta = \|\psi\|^2. \tag{304}$$

Proof. Surjectivity follows by taking $(f, c) = (A^{-1/2}g, 0)$ for $g \in H$; the kernel formula is immediate. Direct computation gives $QQ^* = A + |u\rangle\langle u|$. The rank-one Fredholm determinant lemma then yields (303). Finally,

$$\int_0^\infty \frac{e^{-x}}{(1+x)^2} \, dx = 1 - \delta,$$

which proves (304) and the index assertion. $\square$

Let $\mathcal{H} = \ell^2(\mathbb{N}_0) \otimes E$, let $\pi : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})/\mathcal{L}^1(\mathcal{H})$ be the quotient map, and put $a = \pi(e^A)$, $b = \pi(e^B)$. These elements commute. If $\xi = \{a, b\} \in K_2^{\text{alg}}(\mathcal{B}(\mathcal{H})/\mathcal{L}^1(\mathcal{H}))$, then Brown's determinant invariant gives [38, 31]

$$\det(\partial_2 \xi) = \det_F(e^A e^B e^{-A} e^{-B}) = e^{\alpha^2}. \tag{305}$$

In particular, when $e^{\alpha^2} \neq 1$, both $\partial_2 \xi \in K_1(\mathcal{B}(\mathcal{H}), \mathcal{L}^1(\mathcal{H}))$ and ξ are nontrivial. Notice that (305) is an algebraic K_2 statement detected on a relative K_1 boundary; it is not a “relative K_2 ” class.

Scope. The preceding statements do not produce an arithmetic relation between e^{γ^2} and δ . Indeed, replacing h by any covector with $\langle h, \psi \rangle = \alpha$ realizes the arbitrary scalar e^{α^2} , while Proposition 30.1 erases δ completely. Nor does (305) yield a new topological C^* -algebra K -class: after passage to the Calkin algebra, the two displayed invertibles are exponentials and hence have trivial individual topological K_1 classes.

Under the Hardy-space identification, the active normal form is the one-mode Toeplitz pair with symbols $e^{\alpha z^{-1}}$ and $e^{\alpha z}$. For $\alpha \neq 0$, the first symbol has an essential singularity at the origin, so meromorphic tame-symbol formulas do not apply. Finally, no Wodzicki-residue interpretation follows from this model: no classical pseudodifferential calculus or spectral triple has been specified, the relevant commutator is finite rank and hence is annihilated by singular traces, and a multiplicative residue determinant sends a multiplicative commutator to 1 whenever all factors lie in its multiplicative domain. Thus (302) is an ordinary Fredholm determinant anomaly, not a noncommutative-residue anomaly.

31 Frobenius framing and a depth-one rapid-decay target

The additive normalization freedom visible in (256) and in the Kummer tower is often expressed by the unipotent matrices

$$S_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

Inside an unframed Tannakian category, adding further tensors already fixed by the motivic Galois group cannot remove this direction. A Frobenius operator with distinct graded eigenvalues does remove the corresponding centralizer, but this fact has to be interpreted carefully: after forgetting the filtration and comparison data, the same separation of slopes also makes the local extension split. We begin by recording both sides of this linear algebra.

Lemma 31.1 (Frobenius-framed rigidity). *Let K be a field of characteristic zero and let V be a two-dimensional K -realization with a K -linearized Frobenius operator (for example, an f th iterate of a semilinear Frobenius for which $\sigma^f = \text{id}_K$) which, in a basis compatible with its two graded pieces, has matrix*

$$\Phi_p = \begin{pmatrix} 1 & c_p \\ 0 & q_p \end{pmatrix}, \quad q_p \neq 1. \quad (306)$$

Then the unipotent subgroup $U = \{S_t : t \in K\}$ has trivial intersection with the centralizer of Φ_p :

$$U \cap Z_{\text{GL}(V)}(\Phi_p) = \{1\}. \quad (307)$$

In particular this applies when $q_p = p^f$ or p^{-f} , including $f = 1$.

Proof. Direct multiplication gives

$$S_t \Phi_p - \Phi_p S_t = \begin{pmatrix} 0 & (q_p - 1)t \\ 0 & 0 \end{pmatrix}.$$

Since $q_p \neq 1$, commutation is equivalent to $t = 0$. □

For a raw σ -semilinear Frobenius, the corresponding centralizer condition is $q_p t = \sigma(t)$ rather than $(q_p - 1)t = 0$. The associated inhomogeneous equation has the opposite-looking consequence.

Theorem 31.2 (Automatic splitting of a bare Frobenius extension). *Let K be a complete non-archimedean field, let $\sigma: K \rightarrow K$ be an isometric automorphism, and let $q \in K^\sigma$ satisfy $|q| \neq 1$. For every $c \in K$, consider the σ -semilinear operator whose matrix is*

$$A = \begin{pmatrix} 1 & c \\ 0 & q \end{pmatrix}.$$

There is a unique $t \in K$ such that

$$qt - \sigma(t) = c. \quad (308)$$

For this value of t one has

$$S_t^{-1} A \sigma(S_t) = \begin{pmatrix} 1 & 0 \\ 0 & q \end{pmatrix}. \quad (309)$$

Thus the underlying unfiltered semilinear Frobenius extension is split, and its upper-unipotent splitting is unique.

Proof. If $|q| < 1$, the convergent series

$$t = - \sum_{n \geq 0} q^n \sigma^{-n-1}(c)$$

satisfies (308). If $|q| > 1$, the corresponding solution is

$$t = \sum_{n \geq 0} q^{-n-1} \sigma^n(c).$$

Convergence follows because σ is isometric. If t_1 and t_2 are two solutions, their difference u satisfies $qu = \sigma(u)$; taking absolute values gives $|q||u| = |u|$, and hence $u = 0$. Finally,

$$S_t^{-1} A \sigma(S_t) = \begin{pmatrix} 1 & c + \sigma(t) - qt \\ 0 & q \end{pmatrix},$$

which proves (309). □

Theorem 31.2 does not contradict Lemma 31.1. The homogeneous centralizer equation has only the zero solution, while the inhomogeneous splitting equation has exactly one solution. Consequently a marked Frobenius basis may remove a gauge ambiguity, but the bare Frobenius isocrystal does not by itself retain the global extension class.

Proposition 31.3 (A Kummer local-to-global counterexample). *Let*

$$M_2 = [\mathbb{Z} \xrightarrow{1 \mapsto 2} \mathbb{G}_m]$$

be the Kummer 1-motive over $\mathbb{Q}$. Its canonical sequence

$$0 \longrightarrow [0 \rightarrow \mathbb{G}_m] \longrightarrow M_2 \longrightarrow [\mathbb{Z} \rightarrow 0] \longrightarrow 0 \quad (310)$$

is non-split in the isogeny category. Nevertheless, M_2 has good reduction at every prime $p \neq 2$, and at each such prime its underlying rational crystalline F -isocrystal is split. Hence the implication

$$\left. \begin{array}{l} \text{the unfiltered rational } F\text{-isocrystal is split} \\ \text{at every good prime} \end{array} \right\} \implies \text{the global motive is split}$$

is false.

Proof. A splitting of (310) up to isogeny would make the point $2 \in \mathbb{G}_m(\mathbb{Q})$ torsion, equivalently would force $2^m = 1$ for some $m \geq 1$. Thus the global extension is non-split. For $p \neq 2$, the point 2 extends to a unit over $\mathbb{Z}_p$, so the motive has good reduction. Its reduction $\bar{2} \in \mathbb{F}_p^\times$ has finite order. The reduced Kummer 1-motive therefore splits up to isogeny, and applying the rational crystalline realization gives a split sequence with graded slopes 0 and 1, up to the convention for signs. This is the root-of-unity case of the Kummer calculation in [37, Example 6.3]; the surrounding slope formalism may be compared with [32]. $\square$

The qualifier *unfiltered* in the proposition is essential. Neither the p -adic Galois realization nor the filtered Frobenius module is asserted to split. The relative position of the slope splitting and the Hodge–de Rham filtration can retain the Kummer extension; in the preceding example its coordinate is represented, subject to conventions, by $\log_p 2$. Accordingly, a viable arithmetic enhancement of the Euler extension must use filtered-Frobenius, p -adic integration, or syntomic comparison data, not the bare F -isocrystal alone.

The rapid-decay pairing itself is well established: Bloch and Esnault construct it for irregular connections on curves, while Hien develops the surface and arbitrary-dimensional theories [7, 26, 27]. The classical arithmetic comparison is also instructive. Huber and Wüstholz determine the algebraic linear relations among periods of classical 1-motives [29]. For 1-motives over number fields with good reduction, Mohajer and Sebbar construct canonical rational structures and a p -adic analytic-subgroup framework at depths one and two [37]. On the irregular side, Snodgrass constructs rational rapid-decay realizations and E -periods in the E -operator setting under his stated hypotheses [54]. None of these results supplies the numerical-to-motivic implication needed below.

Let M_γ denote the Euler–Mascheroni exponential motive in the normalization of Fresán and Jossen. They prove that it is a non-split extension

$$0 \longrightarrow \mathbb{Q}(0) \longrightarrow M_\gamma \longrightarrow \mathbb{Q}(-1) \longrightarrow 0, \quad (311)$$

that $\mathrm{Hom}(M_\gamma, \mathbb{Q}(0)) = 0$, and that its complex extension coordinate is γ [24, Sec. 12.8].

The relevant motivic separation from classical periods is already known; it is not an additional mixed-extension conjecture.

Theorem 31.4 (Fresán–Jossen classical separation). *Let N be any classical motive over $\mathbb{Q}$, put $N^+ = N \oplus M_\gamma$, and denote their motivic Galois groups by G_N and G_{N^+} . Then the natural faithfully flat morphism*

$$G_{N^+} \longrightarrow G_N$$

has a subgroup isomorphic to $\mathbb{G}_a$ in its kernel. In particular,

$$\dim G_{N^+} > \dim G_N. \quad (312)$$

The statement applies, in particular, to every finite direct sum of Tate and Kummer motives.

Proof. Fresán and Jossen show that the perverse realization of a classical motive has trivial fundamental group, whereas adjoining M_γ introduces the full unipotent group $\mathbb{G}_a$. Their comparison diagram embeds this copy of $\mathbb{G}_a$ into G_{N^+} and maps it trivially to G_N . This is precisely the proof of [24, Cor. 12.8.12]; see also their Proposition 12.8.7 and Corollary 12.8.8. $\square$

Theorem 31.4 settles the motivic part of the Euler–Kummer separation problem, object by object for every finite classical family. What remains open is the comparison step that turns a numerical relation among complex period coordinates into a motivic relation. The minimal form needed for γ is the following degree-one fullness statement.

Conjecture 31.5 (Euler rapid-decay comparison fullness). *Assume that M_γ is equipped with its rational rapid-decay realization and with coherent filtered-Frobenius or syntomic comparison data at the good places. If*

$$\lambda\gamma + \beta = 0, \quad \lambda, \beta \in \overline{\mathbb{Q}}, \quad \lambda \neq 0, \quad (313)$$

then this numerical relation is induced by a morphism in the exponential-motivic category; equivalently in the present rank-two setting, it forces a splitting of (311).

This is deliberately a rank-two, degree-one specialization of the exponential-period philosophy. It is not a consequence of the existence of rapid-decay homology, a rational structure, or bare Frobenius realizations. The filtered or syntomic data are possible tools for proving the comparison assertion, not substitutes for it.

Proposition 31.6 (Conditional transcendence). *Conjecture 31.5 implies that γ is transcendental.*

Proof. If γ were algebraic, then (313) would hold with $\lambda = 1$ and $\beta = -\gamma$. The conjecture would force (311) to split, contradicting the non-splitting proved by Fresán and Jossen [24, Prop. 12.8.7]. $\square$

Conjecture 31.7 (Euler–Baker linear period theorem). *Let $\alpha_1, \dots, \alpha_r \in \overline{\mathbb{Q}}^\times$ be multiplicatively independent, and fix branches of their logarithms. If*

$$\lambda\gamma + \beta + \nu(2\pi i) + \sum_{j=1}^r \mu_j \operatorname{Log} \alpha_j = 0, \quad \lambda, \beta, \nu, \mu_j \in \overline{\mathbb{Q}}, \quad (314)$$

then

$$\lambda = \beta = \nu = \mu_1 = \dots = \mu_r = 0.$$

For clarity, let N_α be a finite classical motive containing the relevant Tate and Kummer objects, so that its period matrix contains $1, 2\pi i$, and the chosen $\operatorname{Log} \alpha_j$.

Proposition 31.8 (The exact relative comparison gap). *Assume degree-one comparison fullness for $N_\alpha \oplus M_\gamma$: every $\overline{\mathbb{Q}}$ -linear relation among the displayed comparison-matrix entries is induced by motivic functoriality. Then Conjecture 31.7 holds for $\alpha_1, \dots, \alpha_r$.*

Proof. Under comparison fullness, a relation (314) is motivic. The copy of $\mathbb{G}_a$ in the kernel furnished by Theorem 31.4 fixes all period coordinates coming from N_α and translates the Euler extension coordinate. Invariance of the relation therefore forces $\lambda = 0$. The remaining relation

$$\beta + \nu(2\pi i) + \sum_{j=1}^r \mu_j \operatorname{Log} \alpha_j = 0$$

is classical. The Baker–Lindemann theorem [4], together with the assumed multiplicative independence, gives $\beta = \nu = \mu_1 = \dots = \mu_r = 0$. $\square$

Thus no additional theorem asserting motivic independence of the Euler extension from the span of Kummer extensions is missing: that separation is already contained in Theorem 31.4. The genuine missing input for the Euler–Baker statement is the relative *numerical-to-motivic comparison theorem* used in Proposition 31.8.

Remark 31.9 (An adelic formulation of the missing estimate). Let K now be a number field. The local Dwork scales $p^{-1/(p-1)}$ should not be identified numerically across different primes. A viable slope strategy would first have to place coherent filtered-Frobenius or syntomic norms on one rapid-decay extension and then form an adelic degree

$$\widehat{\deg}(Ks) = - \sum_{w \in M_K} n_w \log \|s\|_w.$$

It would additionally require a theorem showing that an algebraic complex period relation produces a global section of impossible adelic degree unless the extension splits. The bare slope decomposition cannot provide this: Proposition 31.3 shows that it can split at every good place while the global motive remains non-split. Neither the required coherence of the filtered local norms nor the global slope inequality is presently proved.

The conclusion is diagnostic. Frobenius centralizer rigidity is a correct uniqueness statement, but the rank-two scalar semilinear model of Theorem 31.2 splits automatically when $|q| \neq 1$. Fresán and Jossen already prove that the Euler $\mathbb{G}_a$ -direction survives after adjoining every fixed classical motive. The remaining depth-one problem is therefore exactly a comparison/fullness theorem transferring a complex rapid-decay relation to the motivic category. Its minimal rank-two case would prove the transcendence of γ ; its finite relative Kummer form would prove the Euler–Baker linear period theorem.

32 Corrections, closure ledger, and remaining problems

The proof audit changes the status of several statements in the preceding draft. The multipoint theorem now includes the Laurent-denominator argument needed to pass from a rational differential field to a finite Laurent sum. The Padé recurrence is proved for every degree from the Laguerre recurrence and the Padé Casoratian, rather than inferred from finite symbolic checks. The Laguerre–Hankel identity is proved for every n by the Geronimus–Markov formula. The reciprocal-Gamma calculation uses only square-free mixed derivatives in distinct variables; it makes no assertion that all ordinary higher derivatives vanish.

Three former interpretations are explicitly withdrawn. A formal shift of a differential equation is not an arithmetic impossibility theorem unless it preserves the Stieltjes boundary condition. The reciprocal Euler equation has infinitely many off-circle complex zeros, so its single unit-circle orbit is not the whole zero set. Finally, a membership equivalence between two formal classes is not a reduction of the transcendence problem unless the new membership question is independently decidable.

Two further effectivity corrections are recorded in the present version. First, E_1 is not a Siegel E -function: its logarithmic singularity at the origin contains the same connection constant which one is trying to control. The exact recovery identity (256) is therefore not an effective E -value representation of γ . Second, the height in the Adamczewski–Favrejon measure is the height of a relation polynomial for fixed functions at a fixed algebraic point, not the height of a moving algebraic approximation point. A former comparison of a putative $\kappa < 1$ requirement with a Dirichlet exponent is consequently withdrawn. The correct moving-point difficulty is quantified instead by Theorem 28.5.

Version 6.2 makes four additional cross-field corrections. The classical compact set $\mathbb{Z}_p$ has capacity $p^{-1/(p-1)}$, not one; the capacity-one local object used here is the full closed Berkovich unit disk. The auxiliary primitive $\mathcal{K}$ used at the Euler point is entire and holonomic but is not an E -function; its algebraic-point values nevertheless belong to the ring of finite E -values by the explicit finite Ein-value formula (239). The Hermite–Biehler construction

yields a de Branges space and discrete Clark spectra, but no unproved regularity of a canonical-system Hamiltonian is asserted. Finally, in the proved rank-two scalar semilinear model, $|q| \neq 1$ makes the underlying bare extension split locally. Thus raw good-prime splitting cannot recover a global extension class. Conversely, Fresán–Jossen already prove the required motivic $\mathbb{G}_a$ -separation from every fixed classical motive; the remaining gap is numerical-to-motivic comparison fullness, not a new mixed-Ext¹ independence theorem.

Version 6.3 records a further five-document audit. The Bell–Gould prime-power lift and its coefficientwise CRT synthesis survive as exact divisibility theorems, but neither implies small primitive linear forms. A proposed compactifier estimate is false already at its first index, and a quoted finite denominator bound lacks the recurrence, primitive denominator, and strict numerical inequality needed for certification; neither is used here. In a separate compact double-integral family, the formerly stated 16^{-n} asymptotic misses a factor n^{-1} , and the partial-fraction formula omits higher-order poles. The corrected family and its unavoidable positive logarithmic contamination are stated only in their proved range. Crossing the surviving multipoint theorem with the classical polyexponential differential chain yields the resonant all-order theorem, the all-order Gamma-jet split, and the exact $2N - 1$ rank of each finite Jacobi spectral tower. These deductions have complete proofs here, but the one-point nonresonant polyexponential phenomenon is classical and no historical novelty is claimed for it. A proposed multiplicative-dependence criterion for general algebraic arguments is false and is not used.

Table 1: Cross-field closure ledger. Here *closure* means that the stated finite classification or obstruction is proved, not that a named long-standing problem is solved.

Imported area	Exact closure in this paper	What remains open
Differential algebra and toric geometry	Full finite relation ideal for exponential and Ein coordinates; all-order resonant one-level polyexponential tower; pure-exponential base change; common-shift defect one and the exact residual dichotomy	The functions and the one-point nonresonant phenomenon are classical; exact priority of the multipoint resonant toric formulation still requires specialist review.
Graph, matroid, and Grassmannian geometry	Cycle-space ideals; every $\mathbb{Q}$ -representable algebraic matroid; exact Plücker ideal for explicit Ein-matrices	These are arithmetic realizations of classical generic constructions, not new classifications of matroids or Grassmannians.
Classical special functions and moment theory	Countable independence of the integral and quadratic E_1 families; all-algebraic-level four-value exclusion; all-order positive Hausdorff systems with strict total positivity, independent Hankel/Favard coordinates, and exact finite spectral-tower rank $2N - 1$; all-order Gamma-jet/tail split	These constructions do not isolate γ , δ , individual zeta values, or individual tail integrals; exact priority of the arithmetic realizations remains to be audited.
p -adic and umbral congruences	Exact all-prime radius phase, denominator law, p -curvature, Borel logarithm, prime-power Bell–Gould annihilators, and arbitrary-modulus CRT synthesis	Individual irrationality of p -adic Euler sums and δ , as well as analytic smallness of CRT representatives, remain open.

Imported area	Exact closure in this paper	What remains open
Orthogonal polynomials and isomonodromy	All-degree Kummer determinant, Riccati/ P_V boundary, positive mixed determinants, finite recovery of $\mathbb{Q}(\gamma, \delta)$, Nevai limits, and arcsine spectral distribution	The classical normalizations produce no small linear forms for δ .
Statistical mechanics	Monomer–dimer representation and a complex vertical zero strip	The determinant is not real-rooted, and no Lee–Yang theorem is obtained.
Entire-function and Clark geometry	Finite Ein-witness rigidity; Joukowski descent, rightmost-fibre stability, simple transcendental zero sets, meromorphic inner quotient, algebraically independent scattering samples, two γ -recovery limits, and exact S_2, S_3 trace inversion	The recoveries do not provide the uniform moving-point arithmetic measure needed to decide γ .
Reciprocal spectral arithmetic	Exact Euler-level Hadamard and inverse-resonance traces, γ -free phase recovery, top-norm barrier, and dyadic irreducibility with actual minimal-Weil-height saturation	No arithmetic regularized determinant or uniform moving-point E -value measure is available; non-dyadic and nonlinear auxiliaries remain open.
Adelic capacity and Hankel lattices	Rational polynomial pullback preserves global capacity one; primitive Hankel clearing saturates at one; the scalar one-pole Borel–Laguerre family has an exact factorial wall; the compact Sondow variant has strictly positive logarithmic contamination	Weighted or place-dependent capacity, coupled determinants, general rational maps with poles, short vectors, and nonlinear auxiliaries are not excluded.
Operator theory	Exact Fredholm determinant and relative determinant, plus similarity collapse	The carrier forgets the coupling needed to relate e^{γ^2} to δ ; no residue or topological K -theory solution follows.
Frobenius and exponential motives	The rank-two scalar bare-Frobenius model with $ q \neq 1$ splits uniquely; local splitting does not imply global splitting; the Euler $\mathbb{G}_a$ -direction is motivically separated from every fixed classical motive	A filtered-Frobenius or syntomic numerical-to-motivic comparison/fullness theorem remains open.

The remaining high-value targets are therefore sharply delimited:

1. determine whether either γ or δ is irrational;
2. produce an actual relation or fixed-degree super-small sequence in a detector such as (80) or (81);
3. prove a genuinely nonvacuous intersection theorem between finite-point E -values and arithmetic-Gevrey values;
4. find a weighted, place-dependent, or nonlinear normalization outside the rational polynomial pullbacks and full-determinant transformations proved critical in Propositions 29.5 and 29.7;
5. decide the arithmetic nature of the unique real Euler-level point w_* in Theorem 26.7. Its algebraicity would already imply the algebraic independence of γ and δ .

6. construct an arithmetic zeta-regularized determinant or spectral projector which removes the universal $N \log N$ term in Theorem 28.7 without removing the reciprocal phase;
7. prove a mixed E -value/Baker measure uniform in the algebraic sample point for the Clark scattering sequence of Theorem 27.8;
8. prove the Euler rapid-decay splitting principle 31.5; Fresán–Jossen already provide the required motivic separation from every fixed classical family, so the missing input is precisely numerical-to-motivic comparison fullness.

Acknowledgement of scope

This document is a revised research draft, not a peer-reviewed publication. The results labelled as applications or syntheses rely on the cited theorems, and their exact priority should be checked by a specialist literature review before submission. The negative statements are intentionally local to the specified formal classes and normalizations.

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